

Statistical Data Analysis and Supplement Information

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1 Dependencies, functions and data

1.1 Dependencies

```
### Packages
library(emmeans)
library(ggstance)
library(tidyverse)
library(patchwork)
library(data.table)
library(kableExtra)
library(pals)
library(here)
library(lme4)
library(lmerTest)
library(ggh4x)
library(GGally)
library(broom.mixed)
library(flowCore)
library(flowViz)
# library(bemovi)
# library(LEEF.measurement.bemovi)
library(survey)
library(lavaan)
library(lavaan.survey)
library(effectsize)
library(r2glmm)

#### Regarding lavaan.survey
# lavaan.survey is an extension of lavaan that can deal with data independencies.
# It is no longer on cran, but it is a trustworthy package (there is also
# a publication for it). The source code is here
# https://github.com/daob/lavaan.survey.

options(dplyr.summarise.inform = FALSE)
```

1.2 Functions

```
## other useful functions
source(file = here("R/1_Functions/OtherFunctions.R"))
## Permutation entropy function
source(file = here("R/1_Functions/PermutationEntropy.R"))
```

1.3 Data

```
# Time series
dd_list <-
  readRDS(here(
    "Data/2_TimeSeriesData/Transformed/DensityBiomassTimeSeries.RDS"
  ))

# Forecast results
load(here(
```

```

    "Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"
  ))
load(here(
  "Data/3_ForecastsResults/3_MergedResults/seg.complete.forecasts.RData"
))
load(here(
  "Data/3_ForecastsResults/3_MergedResults/RMSEBackT.complete.forecasts.RData"
))
load(here(
  "Data/3_ForecastsResults/3_MergedResults/inSample.complete.forecasts.RData"
))

```

2 Implementation of the experiment

2.1 Organisms

```

SpeciesTab <- data.frame(
  Species = c(
    "\\textit{Chlamydomonas}\\n\\textit{reinhardtii}",
    "\\textit{Cryptomonas} sp.",
    "\\textit{Cosmarium}\\n\\textit{botrytis}",
    "\\textit{Desmodesmus}\\n\\textit{armatus}",
    "\\textit{Monoraphidium}\\n\\textit{obtusum}",
    "\\textit{Staurastrum}\\n\\textit{gracile}",
    "\\textit{Staurastrum}\\n\\textit{polytrichum}",
    "\\textit{Colpidium}\\n\\textit{striatum}",
    "\\textit{Dexiostoma}\\n\\textit{campylum}",
    "\\textit{Loxocephalus} sp.",
    "\\textit{Tetrahymena}\\n\\textit{thermophila}",
    "\\textit{Coleps} sp.",
    "\\textit{Paramecium}\\n\\textit{caudatum}",
    "\\textit{Stylonychia}\\n\\textit{mytilus}",
    "\\textit{Stylonychia} sp.",
    "\\textit{Euplotes}\\n\\textit{daidaleos}",
    "\\textit{Paramecium}\\n\\textit{bursaria}",
    "\\textit{Didinium}\\n\\textit{nasutum}"
  ),
  FunctionalGrp = c(
    "Edible\\nalgae",
    "Edible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Inedible\\nalgae",
    "Bacterivores",
    "Bacterivores",
    "Bacterivores",
    "Bacterivores",
    "Omnivores",
    "Omnivores",
    "Omnivores",

```

```

"Omnivores",
"Mixotrophs",
"Mixotrophs",
"Predator"
),
Origin = c(
  "SAG\\textsuperscript{a}",
  "SAG",
  "SAG",
  "SAG",
  "SAG",
  "CCAP\\textsuperscript{b}",
  "SAG",
  "Carolina Biological\\nSupply",
  "CCAP",
  "Carolina Biological\\nSupply",
  "CCAP",
  "Pond\\nUniversity Zurich",
  "Carolina Biological\\nSupply",
  "Pond\\nUniversity Zurich",
  "Pond\\nUniversity Zurich",
  "Sciento",
  "SAG",
  "Sciento"
),
StrainNumber = c(
  "11-32b",
  "26.80",
  "136.80",
  "276-4d",
  "55.81",
  "679/3",
  "20.98",
  NA,
  "1611/1",
  NA,
  "1630/1U",
  NA,
  NA,
  NA,
  NA,
  "P229",
  "27.96",
  "P220"
),
Prey = c(
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,

```

```

"Bacteria",
"Bacteria",
"Bacteria",
"Bacteria",
"\\textit{Cryptomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Cryptomonas},\\nBacteria",
"\\textit{Chlamydomonas},\\nBacteria",
"\\textit{Paramecium}\\n\\textit{caudatum}"
),
indistFrom = c(
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  NA,
  "\\textit{Tetrahymena}",
  NA,
  "\\textit{Dexiostoma}",
  NA,
  NA,
  "\\textit{Stylonychia} sp.",
  "\\textit{Stylonychia mytilus}",
  NA,
  NA,
  NA
)
)

options(knitr.kable.NA = '')
SpeciesTab %>%
  mutate_all(linebreak, align="l") %>%
  kbl(format = "latex",
      col.names = linebreak(c("Species", "Functional\\ngroup",
                              "Origin", "Strain\\nnumber", "Prey", "Indistin-\\nguishable\\nfrom"),
                              align = "c"),
      caption = "Algae and ciliates used in the experiment.",
      escape = F,
      booktabs = T,
      linesep = "",
      label = "SpeciesTab") %>%
  kable_styling(font_size = 7, latex_options = "striped") %>%
  footnote(alphabet = c("SAG: Culture Collection of Algae at Göttingen University",
                        "CCAP: Culture Collection of Algae and Protozoa"))

```

Table 1: Algae and ciliates used in the experiment.

Species	Functional group	Origin	Strain number	Prey	Indistinguishable from
<i>Chlamydomonas reinhardtii</i>	Edible algae	SAG ^a	11-32b		
<i>Cryptomonas</i> sp.	Edible algae	SAG	26.80		
<i>Cosmarium botrytis</i>	Inedible algae	SAG	136.80		
<i>Desmodesmus armatus</i>	Inedible algae	SAG	276-4d		
<i>Monoraphidium obtusum</i>	Inedible algae	SAG	55.81		
<i>Staurastrum gracile</i>	Inedible algae	CCAP ^b	679/3		
<i>Staurastrum polytrichum</i>	Inedible algae	SAG	20.98		
<i>Colpidium striatum</i>	Bacterivores	Carolina Biological Supply		Bacteria	
<i>Dexiostoma campylum</i>	Bacterivores	CCAP	1611/1	Bacteria	<i>Tetrahymena</i>
<i>Loxocephalus</i> sp.	Bacterivores	Carolina Biological Supply		Bacteria	
<i>Tetrahymena thermophila</i>	Bacterivores	CCAP	1630/1U	Bacteria	<i>Dexiostoma</i>
<i>Coleps</i> sp.	Omnivores	Pond University Zurich		<i>Cryptomonas</i> , Bacteria	
<i>Paramecium caudatum</i>	Omnivores	Carolina Biological Supply		<i>Chlamydomonas</i> , Bacteria	
<i>Stylonychia mytilus</i>	Omnivores	Pond University Zurich		<i>Chlamydomonas</i> , Bacteria	<i>Stylonychia</i> sp.
<i>Stylonychia</i> sp.	Omnivores	Pond University Zurich		<i>Chlamydomonas</i> , Bacteria	<i>Stylonychia mytilus</i>
<i>Euplotes daidaleos</i>	Mixotrophs	Sciento	P229	<i>Cryptomonas</i> , Bacteria	
<i>Paramecium bursaria</i>	Mixotrophs	SAG	27.96	<i>Chlamydomonas</i> , Bacteria	
<i>Didinium nasutum</i>	Predator	Sciento	P220	<i>Paramecium caudatum</i>	

^a SAG: Culture Collection of Algae at Göttingen University^b CCAP: Culture Collection of Algae and Protozoa

2.2 Experimental design

```
# plot the planned change of stressor levels over the course of the experiment
stressor_levels <-
  read.csv(here("Data/1_ExperimentDesignInfo/LightDeclineSchedule.csv"))

# change to long format
stressor_levels_long <-
  gather(stressor_levels, level, value = value, constant:declining)

stressor_levels_long %>%
  mutate(level = ifelse(level == "declining", "Decreasing", "Constant")) %>%
  ggplot(aes(day, value, col = level)) +
  scale_color_manual(values = c("#C4961A", "#293352"),
    aesthetics = c("color", "fill")) +
  geom_point() +
  labs(x = "Sampling day", y = "Light level (%)", col = "Light conditions") +
  theme_bw() +
  theme(legend.position = "bottom")
```

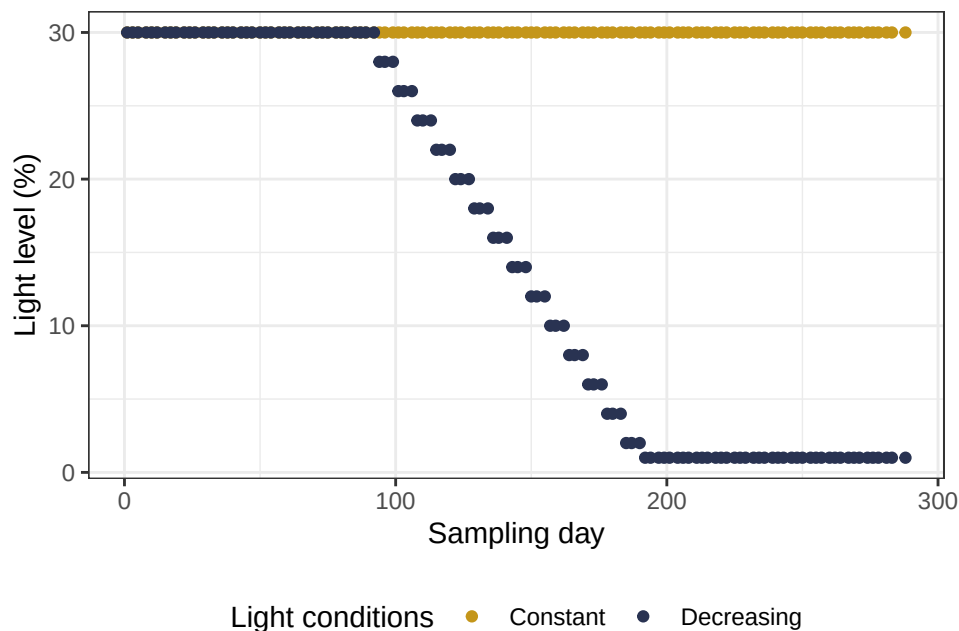


Figure 1: Light levels over the course of the experiment. After 3 months of constant environmental conditions, bottles either remained at the initial light level (30% light) or experienced a gradual decline in light for three months. Then, bottles remained three more months at the final light levels (30 or 1% light intensity). Each point represents a sampling day.

```
PlannedRichTab <- data.frame(
  Tot = c(7, 10, 14),
  Edible = c(2, 2, 2),
  Inedible = c(1, 2, 3),
  Bacterivores = c(1, 2, 3),
  Omni = c(1, 2, 3),
  Mixo = c(1, 2, 2),
```

Table 2: Species richness per functional group in the four diversity levels. Numbers in brackets are the number of available species.

Total (18)	Edible algae (2)	Inedible algae (5)	Bacterivores (4)	Omnivores (4)	Mixotrophs (2)	Predator (1)
7	2	1	1	1	1	1
10	2	2	2	2	2	1
14	2	3	3	3	2	1

```

    Pred = 1
  )

PlannedRichTab %>%
  kbl(
    format = "latex",
    col.names = linebreak(
      c(
        "Total\n(18)",
        "Edible algae\n(2)",
        "Inedible algae\n(5)",
        "Bacterivores\n(4)",
        "Omnivores\n(4)",
        "Mixotrophs\n(2)",
        "Predator\n(1)"
      )
    ),
    align = "c"
  ),
  caption = "Species richness per functional group in the four diversity
levels. Numbers in brackets are the number of available species. ",
  escape = F,
  align = "c",
  booktabs = T,
  linesep = "",
  label = "PlannedRichTab"
)

design <-
  read.csv(here("Data/1_ExperimentDesignInfo/ExperimentalDesign.csv")) %>%
  mutate(lightTreatment = ifelse(light_treatment == "constant",
                                "Constant", "Decreasing")) %>%
  dplyr::select(bottle, composition, richness, lightTreatment, incubator)

design %>%
  kbl(
    booktabs = T,
    col.names = c(
      "Bottle",
      "Composition",
      "Planned richness",
      "Light conditions",
      "Incubator"
    )
  ),

```

Table 3: The summarized experimental design.

Bottle	Composition	Planned richness	Light conditions	Incubator
b_01	c_08	10	Constant	A
b_02	c_02	7	Constant	A
b_03	c_11	14	Constant	A
b_04	c_05	7	Constant	A
b_05	c_13	14	Decreasing	B
b_06	c_11	14	Decreasing	B
b_07	c_06	10	Decreasing	B
b_08	c_01	7	Decreasing	B
b_09	c_02	7	Decreasing	C
b_10	c_07	10	Decreasing	C
b_11	c_08	10	Decreasing	C
b_12	c_14	14	Decreasing	C
b_13	c_06	10	Constant	D
b_14	c_03	7	Constant	D
b_15	c_14	14	Constant	D
b_16	c_15	14	Constant	D
b_17	c_05	7	Decreasing	E
b_18	c_04	7	Decreasing	E
b_19	c_12	14	Decreasing	E
b_20	c_09	10	Decreasing	E
b_21	c_07	10	Constant	F
b_22	c_10	10	Constant	F
b_23	c_04	7	Constant	F
b_24	c_13	14	Constant	F
b_25	c_03	7	Decreasing	G
b_26	c_15	14	Decreasing	G
b_27	c_10	10	Decreasing	G
b_28	c_09	10	Constant	H
b_29	c_12	14	Constant	H
b_30	c_01	7	Constant	H

```
caption = "The summarized experimental design.",
# escape = F,
align = "lclcl",
linesep = "",
label = "designTab"
) %>%
kable_styling(latex_options = "striped", font_size = 8)
```

2.3 Selection of compositions

```
comps <-
  read_csv(here("Data/1_ExperimentDesignInfo/Compositions.csv"))
rich <- comps$richness

comps2 <- comps %>%
  dplyr::select(-richness) %>%
  t() %>%
  as.data.frame()

colnames(comps2) <- gsub("_", "\\_\\_\\_", comps2[2, 1:15])

comps2 <- comps2 %>%
```

```

slice_tail(n = 18) %>%
mutate_if(is.character, as.numeric)

comps2 <- cbind(Species = rownames(comps2), comps2)
rownames(comps2) <- NULL

comps2 <- rbind(comps2, c("Planned richness", rich))

comps2 <- comps2 %>%
  mutate(
    Species = case_when(
      Species == "Chlamydomonas_reinhardtii" ~ "\\textit{Chlamydomonas}\\n\\textit{reinhardtii}",
      Species == "Cryptomonas_sp" ~ "\\textit{Cryptomonas} sp.\\n ",
      Species == "Monoraphidium_obtusum" ~ "\\textit{Monoraphidium}\\n\\textit{obtusum}",
      Species == "Cosmarium_botrytis" ~ "\\textit{Cosmarium}\\n\\textit{botrytis}",
      Species == "Stauroastrum_gracile" ~ "\\textit{Stauroastrum}\\n\\textit{gracile}",
      Species == "Stauroastrum_polytrichum" ~ "\\textit{Stauroastrum}\\n\\textit{polytrichum}",
      Species == "Desmodesmus_armatus" ~ "\\textit{Desmodesmus}\\n\\textit{armatus}",
      Species == "Tetrahymena_thermophila" ~ "\\textit{Tetrahymena}\\n\\textit{thermophila}",
      Species == "Colpidium_striatum" ~ "\\textit{Colpidium}\\n\\textit{striatum}",
      Species == "Loxocephalus_sp" ~ "\\textit{Loxocephalus} sp.\\n ",
      Species == "Dexiostoma_campylum" ~ "\\textit{Dexiostoma}\\n\\textit{campylum}",
      Species == "Paramecium_caudatum" ~ "\\textit{Paramecium}\\n\\textit{caudatum}",
      Species == "Stylonychia_mytilus" ~ "\\textit{Stylonychia}\\n\\textit{mytilus}",
      Species == "Stylonychia_sp" ~ "\\textit{Stylonychia} sp.\\n ",
      Species == "Coleps_sp" ~ "\\textit{Coleps} sp.\\n ",
      Species == "Paramecium_bursaria" ~ "\\textit{Paramecium}\\n\\textit{bursaria}",
      Species == "Euplotes_daidaleos" ~ "\\textit{Euplotes}\\n\\textit{daidaleos}",
      Species == "Didinium_nasutum" ~ "\\textit{Didinium}\\n\\textit{nasutum}",
      Species == "Planned richness" ~ "Planned\\nrichness"
    )
  )

comps2 %>%
  mutate_all(linebreak, align = "l") %>%
  kbl(
    booktabs = T,
    caption = "The 15 community compositions used in the experiment.
A 1 indicates that a species was used in a specific composition,
while a 0 indicates that it was not.",
    escape = F,
    align = c("l", rep("c", 15)),
    linesep = "",
    label = "CompsTab"
  ) %>%
  kable_styling(latex_options = "striped", font_size = 7) %>%
  row_spec(19, bold = T) %>%
  add_header_above(header = c(" " = 1, "Community composition" = 15)) %>%
  column_spec(2:15, width = "4mm") %>%
  row_spec(18, hline_after = T)

```

Table 4: The 15 community compositions used in the experiment. A 1 indicates that a species was used in a specific composition, while a 0 indicates that is was not.

Species	Community composition														
	c_01	c_02	c_03	c_04	c_05	c_06	c_07	c_08	c_09	c_10	c_11	c_12	c_13	c_14	c_15
<i>Chlamydomonas reinhardtii</i>	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
<i>Cryptomonas</i> sp.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
<i>Monoraphidium obtusum</i>	0	0	1	0	0	1	0	0	1	0	0	1	1	0	1
<i>Cosmarium botrytis</i>	0	0	0	1	0	0	0	1	0	1	1	1	1	0	0
<i>Staurastrum gracile</i>	0	1	0	0	0	0	1	0	1	0	1	1	0	1	0
<i>Staurastrum polytrichum</i>	1	0	0	0	0	0	1	0	0	1	0	0	1	1	1
<i>Desmodesmus armatus</i>	0	0	0	0	1	1	0	1	0	0	1	0	0	1	1
<i>Tetrahymena thermophila</i>	1	1	0	0	0	0	0	1	1	0	1	0	1	0	0
<i>Colpidium striatum</i>	0	0	0	0	1	0	1	0	1	1	1	1	1	1	1
<i>Loxocephalus</i> sp.	0	0	1	0	0	1	1	1	0	0	1	1	1	1	1
<i>Dexiostoma campylum</i>	0	0	0	1	0	1	0	0	0	1	0	1	0	1	1
<i>Paramecium caudatum</i>	0	1	0	0	0	1	1	1	0	0	1	1	1	1	1
<i>Stylonychia mytilus</i>	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0
<i>Stylonychia</i> sp.	0	0	1	0	0	0	1	0	0	1	1	1	0	0	1
<i>Coleps</i> sp.	0	0	0	0	1	0	0	1	1	1	1	1	1	1	1
<i>Paramecium bursaria</i>	1	0	0	0	1	0	1	1	1	0	1	1	1	1	1
<i>Euplotes daidaleos</i>	0	1	1	1	0	1	0	0	0	1	1	1	1	1	1
<i>Didinium nasutum</i>	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Planned richness	7	7	7	7	7	10	10	10	10	10	14	14	14	14	14

3 Sampling of the experiment

3.1 Algae

```
flowcam.selection.wide <-  
  read.csv(here("Data/1_ExperimentDesignInfo/FlowCamFiltering.csv"))  
  
flowcam.selection.wide %>%  
  mutate(species = case_when(species == "Chlamydomonas" ~ "Chlamydomonas reinhardtii",  
                             species == "ChlamydomonasClumps" ~ "Chlamydomonas clumps",  
                             species == "Cosmarium" ~ "Cosmarium botrytis",  
                             species == "Debris" ~ "Debris",  
                             species == "Desmodesmus" ~ "Desmodesmus armatus",  
                             species == "DesmodesmusClumps" ~ "Digested algae",  
                             species == "DigestedAlgae" ~ "Digested algae",  
                             species == "DividingChlamydomonas" ~ "Dividing Chlamydomonas",  
                             species == "Monoraphidium" ~ "Monoraphidium obtusum",  
                             species == "Staurastrum1" ~ "Staurastrum gracile",  
                             species == "Staurastrum2" ~ "Staurastrum polytrichum")) %>%  
  ggplot(aes(unfiltered, filtered, col = species)) +  
  geom_point(alpha = 0.7) +  
  geom_abline(intercept = log10(1), slope = 1) +  
  labs(x = expression(paste(  
    "Density in unfiltered sample (cells ", mL ^ {-1}, ")")  
  )),  
  y = expression(paste(  
    "Density in filtered sample (cells ", mL ^ {-1}, ")")  
  )),  
  col = "Taxon") +  
  theme_bw() +  
  scale_color_manual(values = unname(kelly()[2:12])) +  
  scale_x_log10() +  
  scale_y_log10()
```

3.2 Bacteria

```
bacteria.wide <-  
  read.csv(here("Data/1_ExperimentDesignInfo/FlowcytometerFiltering.csv"))  
  
bacteria.wide %>%  
  ggplot(aes(unfiltered, filtered)) +  
  geom_point(alpha = 0.7) +  
  geom_abline(intercept = log10(1), slope = 1) +  
  labs(x = expression(paste(  
    "Density in unfiltered sample (cells ", mL ^ {-1}, ")")  
  )),  
  y = expression(paste(  
    "Density in filtered sample (cells ", mL ^ {-1}, ")")  
  )),  
  col = "Taxon") +  
  theme_bw() +  
  scale_color_manual(values = unname(kelly()[2:12])) +  
  scale_x_log10() +  
  scale_y_log10()
```

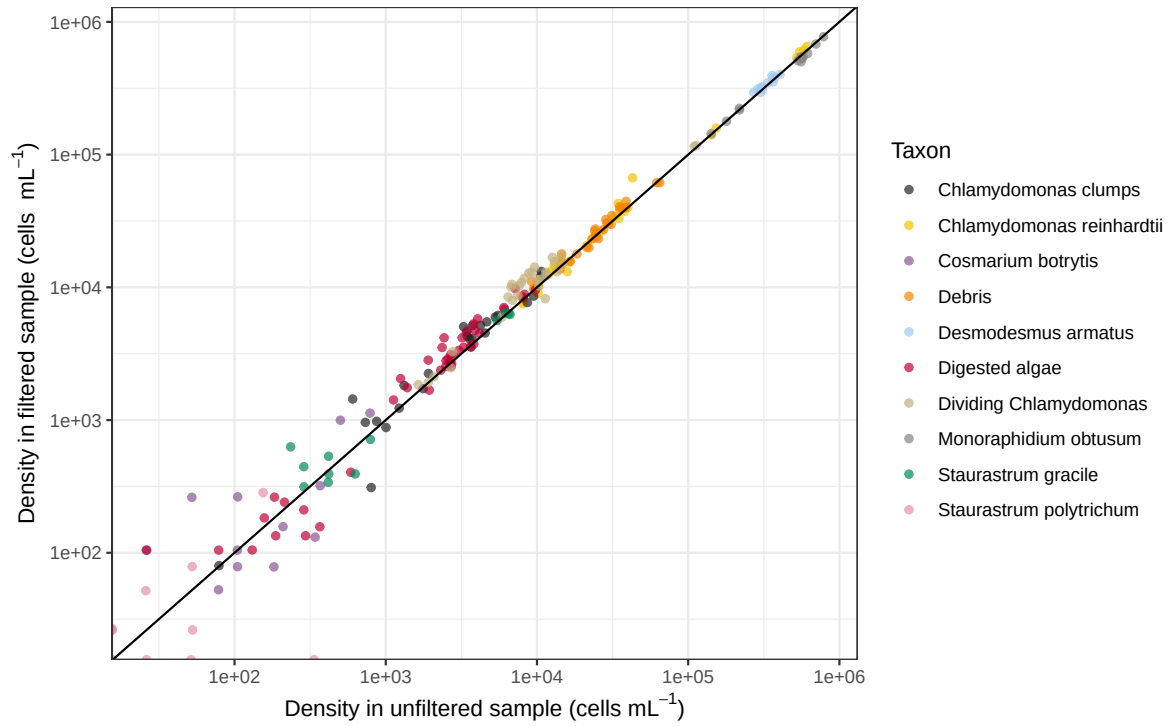


Figure 2: Comparison of filtered and unfiltered algae samples. On one sampling day, we made two flowcam measurements for each of the 30 samples, one from an unfiltered sample and one from a sample that had been filtered through a 200 μ m mesh. The black line is the identity line.

```

    "Density in filtered sample (cells mL-1 ^ {-1}
      -1
    }, ")",
  ))) +
  theme_bw() +
  scale_x_log10() +
  scale_y_log10()

```

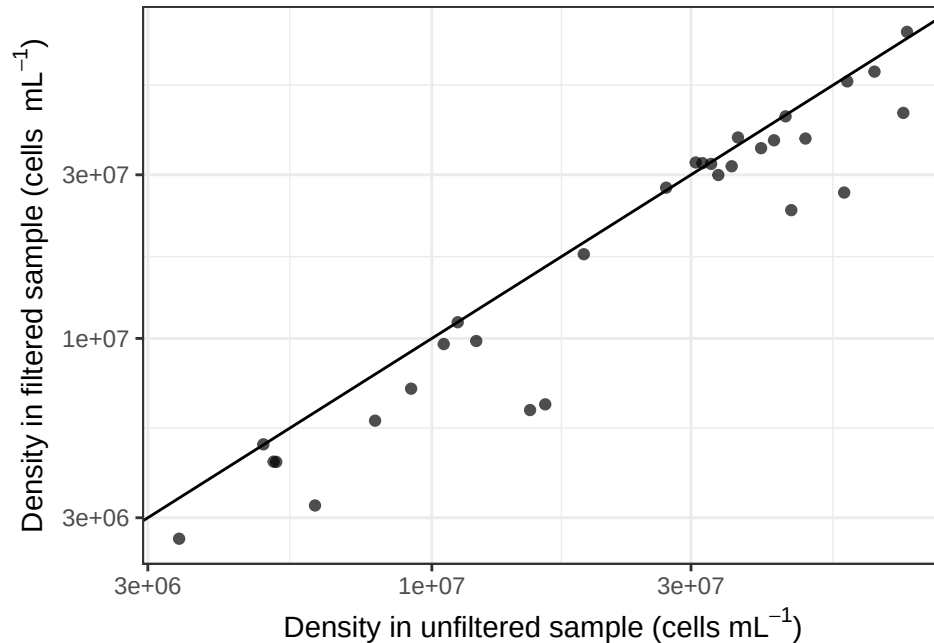


Figure 3: Comparison of filtered and unfiltered bacteria samples. On one sampling day, we made two flowcytometer measurements for each of the 30 samples: one from an unfiltered sample and one from a sample that had been filtered through 200 μm . The black line is the identity line.

```

# load data and gate coordinates
gates_coordinates <- read.csv(here("Data/1_ExperimentDesignInfo/gates_coordinates.csv"))

# load the example sample data
flowFrame <- readRDS(here("Data/1_ExperimentDesignInfo/flowFrame.rds"))
flow_df <- flowFrame@exprs
set.seed(435)
idx <- sample(1:nrow(flow_df), size = round(0.1*nrow(flow_df)), replace = F)
flowFrame@exprs <- flow_df[idx,]

# Apply gate
polyGate_bacteria <- as.matrix(gates_coordinates[1:4, 1:2])
colnames(polyGate_bacteria) <- c("FL1-H", "FL3-H")
bacteria_gate <- flowCore::polygonGate(filterId = "Bacteria", .gate = polyGate_bacteria)

# plot the gate for one sample:
flowViz::xyplot(`FL3-H` ~ `FL1-H`, data = flowFrame, filter = bacteria_gate, smooth=FALSE, xlab="log10(

```

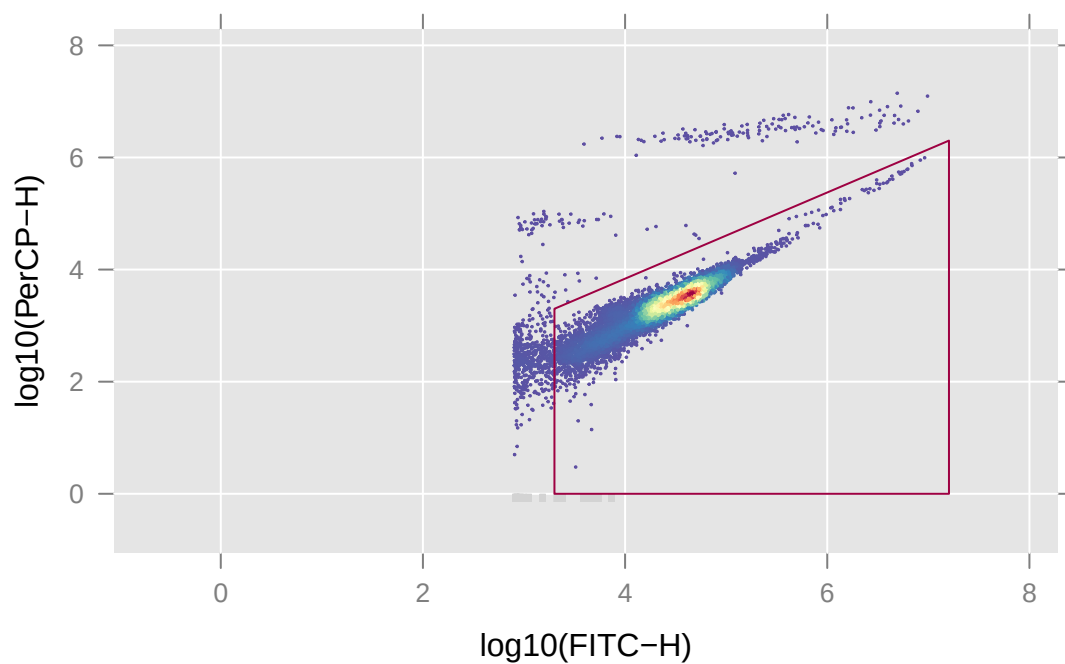


Figure 4: Example of a sample measured on the flowcytometer to illustrate the gating strategy. Bacteria were gated on the plot of FL-1 (i.e. FITC-H) versus FL-3 (i.e. PerCP-H). In this example, the bacteria population contained 178,063 cells. A random subsample of 10% of the measured particles is displayed to avoid over-plotting.

4 Quality controls and time series processing

4.1 Quality controls and missing data percentages

```
tab1.1 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  group_by(method, bottle) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%
  rename(Grouping = bottle)

tab1.2 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  group_by(method, light_treatment) %>%
  dplyr::mutate(
    light_treatment = ifelse(
      light_treatment == "constant",
      "Constant light",
      "Decreasing light"
    )
  ) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%
  rename(Grouping = light_treatment)

tab1.3 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  group_by(method, richness) %>%
  dplyr::mutate(richness = ifelse(
    richness == 7,
    "Planned richness: 07",
    ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
  )) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs) %>%
  rename(Grouping = richness) %>%
  arrange(Grouping)

tab1.4 <- dd_list$densities %>%
  dplyr::filter(method != "Manual count") %>%
  mutate(Grouping = "Experiment") %>%
  group_by(Grouping, method) %>%
  summarize(n_NAs = sum(is.na(density)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  pivot_wider(names_from = method, values_from = perc_NAs)
```

```

tab1 <- rbind(tab1.1, tab1.2, tab1.3, tab1.4)

####

tab2.1 <- dd_list$o2 %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  group_by(bottle) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = bottle, Oxygen = perc_NAs)

tab2.2 <- dd_list$o2 %>%
  group_by(light_treatment) %>%
  dplyr::mutate(
    light_treatment = ifelse(
      light_treatment == "constant",
      "Constant light",
      "Decreasing light"
    )
  ) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = light_treatment, Oxygen = perc_NAs)

tab2.3 <- dd_list$o2 %>%
  group_by(richness) %>%
  dplyr::mutate(richness = ifelse(
    richness == 7,
    "Planned richness: 07",
    ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
  )) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  dplyr::select(-n_NAs) %>%
  rename(Grouping = richness, Oxygen = perc_NAs) %>%
  arrange(Grouping)

tab2.4 <- dd_list$o2 %>%
  mutate(Grouping = "Experiment") %>%
  group_by(Grouping) %>%
  summarize(n_NAs = sum(is.na(percent_o2)),
            perc_NAs = round(n_NAs / n() * 100, 2)) %>%
  rename(Oxygen = perc_NAs) %>%
  dplyr::select(-n_NAs)

tab2 <- rbind(tab2.1, tab2.2, tab2.3, tab2.4)

####

tab3.1 <- dd_list$water_chem %>%

```

```

mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
group_by(bottle) %>%
summarize(n_NAs = sum(is.na(concentration)),
           perc_NAs = round(n_NAs / n() * 100, 2)) %>%
dplyr::select(-n_NAs) %>%
rename(Grouping = bottle, Water_chemistry = perc_NAs)

tab3.2 <- dd_list$water_chem %>%
group_by(light_treatment) %>%
dplyr::mutate(
  light_treatment = ifelse(
    light_treatment == "constant",
    "Constant light",
    "Decreasing light"
  )
) %>%
summarize(n_NAs = sum(is.na(concentration)),
           perc_NAs = round(n_NAs / n() * 100, 2)) %>%
dplyr::select(-n_NAs) %>%
rename(Grouping = light_treatment, Water_chemistry = perc_NAs)

tab3.3 <- dd_list$water_chem %>%
group_by(richness) %>%
dplyr::mutate(richness = ifelse(
  richness == 7,
  "Planned richness: 07",
  ifelse(richness == 10, "Planned richness: 10", "Planned richness: 14")
)) %>%
summarize(n_NAs = sum(is.na(concentration)),
           perc_NAs = round(n_NAs / n() * 100, 2)) %>%
dplyr::select(-n_NAs) %>%
rename(Grouping = richness, Water_chemistry = perc_NAs) %>%
arrange(Grouping)

tab3.4 <- dd_list$water_chem %>%
mutate(Grouping = "Experiment") %>%
group_by(Grouping) %>%
summarize(n_NAs = sum(is.na(concentration)),
           perc_NAs = round(n_NAs / n() * 100, 2)) %>%
rename(Water_chemistry = perc_NAs) %>%
dplyr::select(-n_NAs)

tab3 <- rbind(tab3.1, tab3.2, tab3.3, tab3.4)

###

tab <- full_join(full_join(tab1, tab2), tab3)

# kableExtra::kable(tab)

tab %>%
  # mutate_all(linebreak, align="l") %>%
  kbl(

```

```

"latex",
booktabs = T,
col.names = c(
  "Grouping",
  "FlowCam",
  "Flowcytometer",
  "Video",
  "Oxygen",
  "TOC/TN analyzer"
),
caption = "Percentage of missing data for different
sampling methods and grouping variables.",
escape = F,
align = "lrrrrr",
linesep = "",
label = "NAtab"
) %>%
kable_styling(latex_options = "striped", font_size = 8) %>%
row_spec(36, bold = T) %>%
add_header_above(header = c(
  " " = 1,
  "Percentage of missing data for different sampling methods" = 5
)) %>%
# column_spec(2:5, width = "4mm") %>%
row_spec(30, hline_after = T) %>%
row_spec(32, hline_after = T) %>%
row_spec(35, hline_after = T)

```

4.2 The recorded time series and their processing

```

stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == T) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%

```

Table 5: Percentage of missing data for different sampling methods and grouping variables.

Grouping	Percentage of missing data for different sampling methods				
	FlowCam	Flowcytometer	Video	Oxygen	TOC/TN analyzer
Bottle 01	0.81	4.84	0.81	2.42	0.81
Bottle 02	0.81	4.84	0.81	1.61	0.81
Bottle 03	0.81	4.84	0.81	1.61	0.81
Bottle 04	0.81	4.84	0.81	1.61	0.81
Bottle 05	0.81	4.84	0.81	1.61	0.81
Bottle 06	0.81	4.84	0.81	1.61	0.81
Bottle 07	0.81	4.84	0.81	1.61	1.21
Bottle 08	0.81	4.84	0.81	1.61	0.81
Bottle 09	0.81	4.84	0.81	1.61	1.01
Bottle 10	0.81	4.84	0.81	1.61	1.01
Bottle 11	0.81	4.84	0.81	1.61	1.01
Bottle 12	0.81	4.84	0.81	1.61	1.01
Bottle 13	0.81	4.84	0.81	1.61	1.01
Bottle 14	0.81	4.84	0.81	1.61	1.41
Bottle 15	0.81	4.84	0.81	1.61	0.81
Bottle 16	0.81	4.84	0.81	1.61	0.81
Bottle 17	0.81	4.84	0.81	1.61	1.21
Bottle 18	0.81	4.84	1.21	1.61	0.81
Bottle 19	0.81	4.84	0.81	1.61	0.81
Bottle 20	0.81	4.84	0.81	1.61	0.81
Bottle 21	0.81	4.84	0.81	1.61	0.81
Bottle 22	0.81	4.84	0.81	1.61	0.81
Bottle 23	0.81	4.84	1.61	1.61	0.81
Bottle 24	0.81	4.84	0.81	1.61	0.81
Bottle 25	0.81	4.84	1.21	1.61	0.81
Bottle 26	0.81	4.84	0.81	1.61	0.81
Bottle 27	0.81	4.84	0.81	1.61	1.21
Bottle 28	0.81	4.84	0.81	1.61	1.01
Bottle 29	0.81	4.84	0.81	2.42	1.01
Bottle 30	0.81	4.84	0.81	1.61	1.01
Constant light	0.81	4.84	0.83	1.72	0.90
Decreasing light	0.81	4.84	0.85	1.61	0.94
Planned richness: 07	0.81	4.84	0.97	1.61	0.95
Planned richness: 10	0.81	4.84	0.81	1.69	0.97
Planned richness: 14	0.81	4.84	0.81	1.69	0.85
Experiment	0.81	4.84	0.84	1.67	0.92

```

ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
    # legend.key.size = unit(3, "line"),
    # strip.text = element_text(colour = "white"),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
    {
      -1
    }, " "))),
    x = "Sampling day") +
  guides(col = guide_legend(ncol = 3, byrow = TRUE))

stripColdd2 <- dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == F) %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip2 <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd2$stripCol))

dd_list$densities %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ),

```

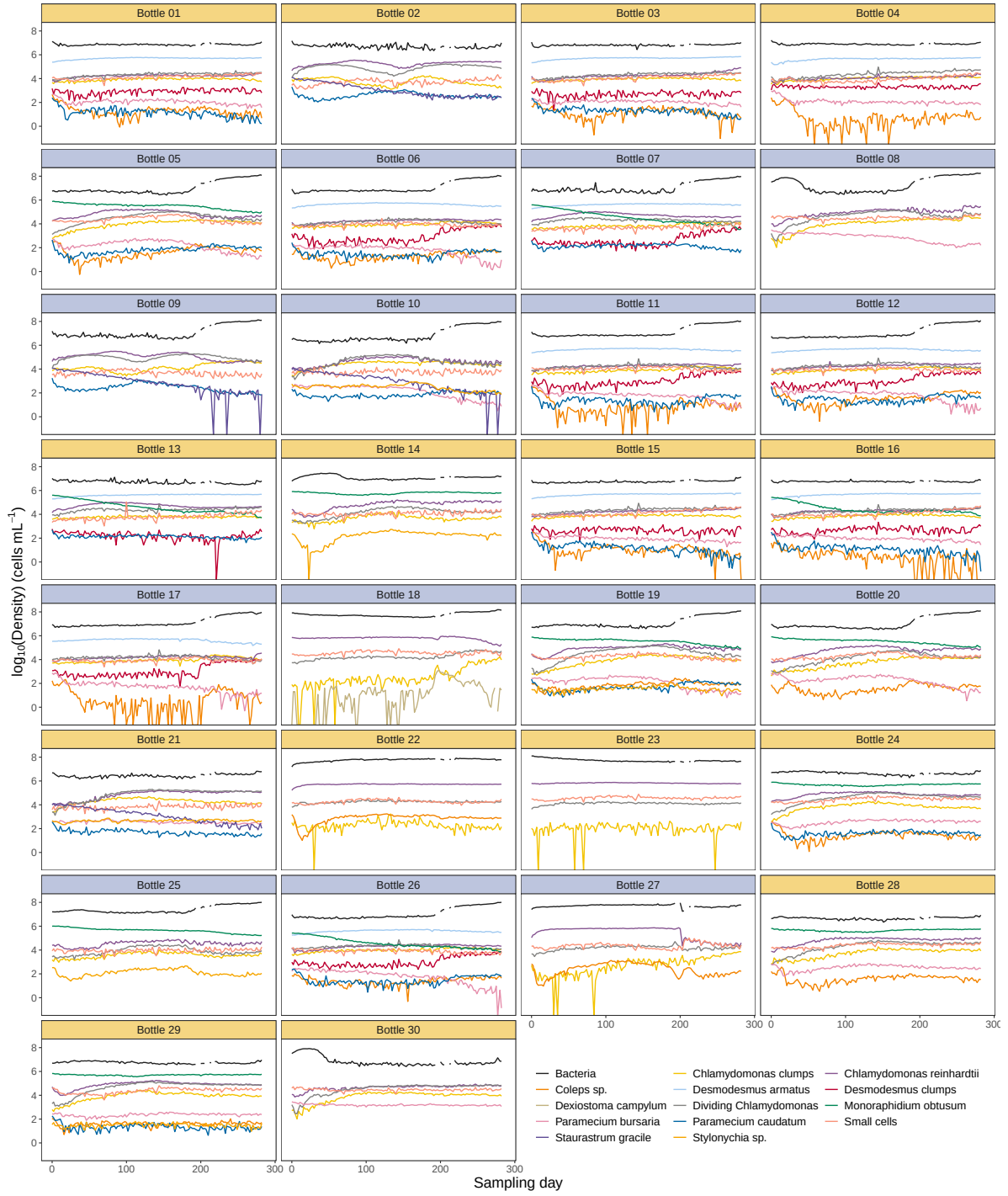


Figure 5: The *untransformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

ForecastTarget == F) %>%
mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
ggplot(aes(day, log10(density), col = species)) +
geom_line(linewidth = 0.6) +
facet_wrap2( ~ bottle, ncol = 4, strip = strip2) +
theme_bw() +
theme(
  legend.position = c(0.75, 0.07),
  panel.grid = element_blank(),
  legend.text = element_text(size = 12),
  axis.text = element_text(size = 11),
  axis.title = element_text(size = 16),
  strip.text = element_text(size = 12),
  # legend.key.size = unit(3, "line"),
  # strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
{
  -1
}, ")")),
  x = "Sampling day") +
guides(col = guide_legend(ncol = 3, byrow = TRUE))

dd_list$biomass %>%
mutate(
  bottle = gsub("b_", "Bottle ", bottle),
  composition = gsub("c_", "Composition ", composition)
) %>%
ggplot(aes(day, log10(biomass), col = composition)) +
geom_line(linewidth = 0.6) +
facet_wrap( ~ light_treatment) +
# facet_wrap2(~bottle, ncol = 4, strip = strip2)+
theme_bw() +
theme(
  legend.position = "bottom",
  panel.grid = element_blank(),
  legend.text = element_text(size = 12),
  axis.text = element_text(size = 11),
  axis.title = element_text(size = 16),
  strip.text = element_text(size = 12),
  # legend.key.size = unit(3, "line"),
  # strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "",
  y = expression(paste("log"[10], "(Total biomass) (", g ~ mL ^ {
    -1
  }, ")")),
  x = "Sampling day")

o2_both <-
read_csv(here("Data/2_TimeSeriesData/Transformed/OxygenBothSensors.csv"))
o2_both <-

```

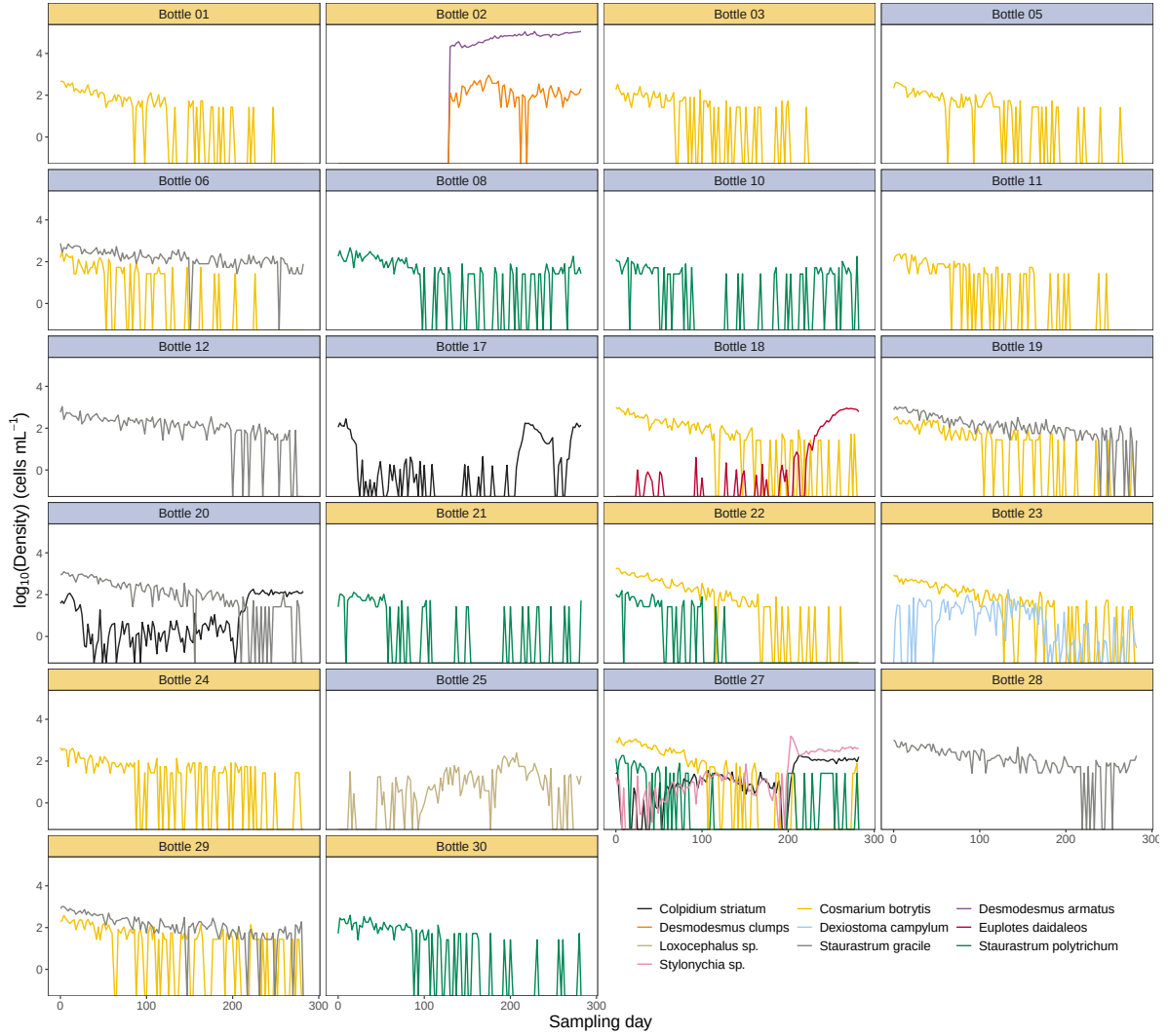


Figure 6: The *untransformed* biotic density time series that were *not* used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by species or species clump. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

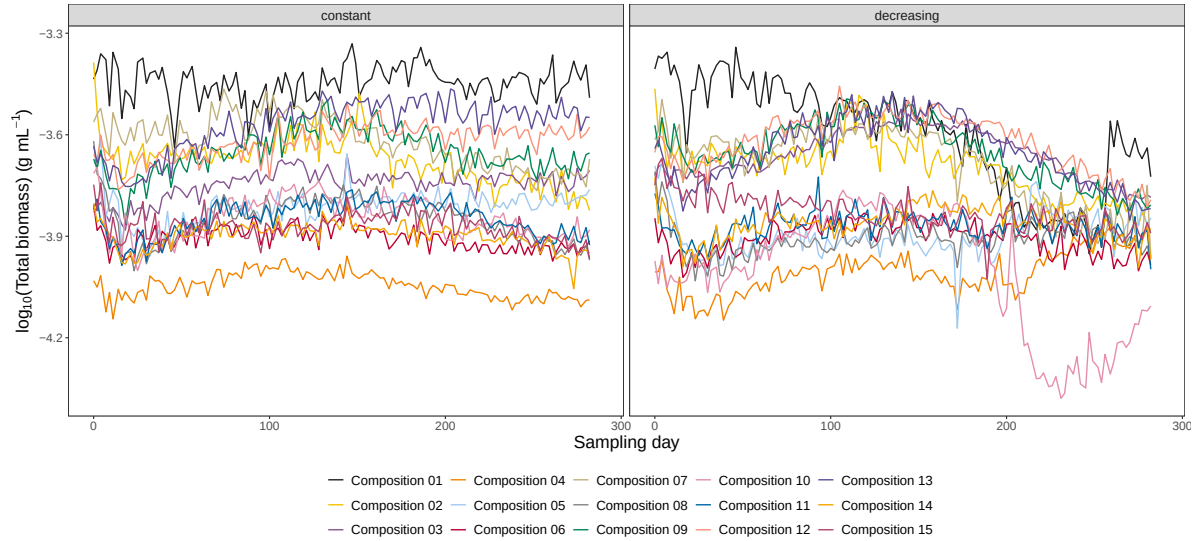


Figure 7: The *untransformed* total biomass time series of the communities, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

```
full_join(
  o2_both %>% mutate(
    sensor = ifelse(sensor == 4, "Sensor at 400 mL height",
                    "Sensor at 900 mL height")
  ),
  dd_list$o2 %>% mutate(sensor = "Average of the two sensors")
)

o2_both %>%
  mutate(
    bottle = gsub("b_", "Bottle ", bottle),
    composition = gsub("c_", "Composition ", composition)
  ) %>%
  ggplot(aes(day, percent_o2, col = sensor)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle,
    ncol = 4,
    strip = strip,
    scales = "free_y") +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
  ) +
  scale_color_brewer(palette = "Dark2") +
  labs(col = "",
    y = "Oxygen concentration (%)",
```

```

x = "Sampling day")

dd_list$water_chem %>%
  mutate(type = case_when(type=="IC" ~ "IC: Inorganic Carbon",
                           type=="TC" ~ "DC: Dissolved Carbon",
                           type=="TN" ~ "DN: Dissolved Nitrogen",
                           type=="TOC" ~ "DOC: Dissolved Organic Carbon"),
         bottle = gsub("b_", "Bottle ", bottle),
         composition = gsub("c_", "Composition ", composition)) %>%
  ggplot(aes(day, concentration, col=type))+
  geom_line(linewidth=0.6)+
  facet_wrap2(~bottle, ncol = 4, strip = strip, scales = "free_y")+
  theme_bw()+
  theme(legend.position = c(0.75,0.055),
        panel.grid=element_blank(),
        legend.text = element_text(size=12),
        axis.text = element_text(size=11),
        axis.title = element_text(size=16),
        strip.text = element_text(size=12),
        ) +
  scale_color_brewer(palette = "Dark2") +
  labs(col="",
       y = expression(paste("Concentration (", mg~L-1, "))),
       x = "Sampling day")

```

```

dd_list$densities.int.detrend.complete %>%
  dplyr::filter(!(species %in% c("Algae", "Debris", "Didinium nasutum",
                                "Digested algae", "Small unidentified",
                                "Colpidium vacuoles")),
               ForecastTarget==T) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, density, col=species))+
  geom_line(linewidth=0.6)+
  facet_wrap2(~bottle, ncol = 4, strip = strip)+
  theme_bw()+
  theme(legend.position = c(0.75,0.055),
        panel.grid=element_blank(),
        legend.text = element_text(size=12),
        axis.text = element_text(size=11),
        axis.title = element_text(size=16),
        strip.text = element_text(size=12),
        # legend.key.size = unit(3, "line"),
        # strip.text = element_text(colour = "white"),
        ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col="", y = "Standardized density",
       x = "Sampling day") +
  guides(col=guide_legend(ncol=3, byrow=TRUE))

```

```

dd_list$biomass.int.detrend.complete %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle),
         composition = gsub("c_", "Composition ", composition)) %>%
  ggplot(aes(day, biomass, col=composition))+
  geom_line(linewidth=0.6)+

```

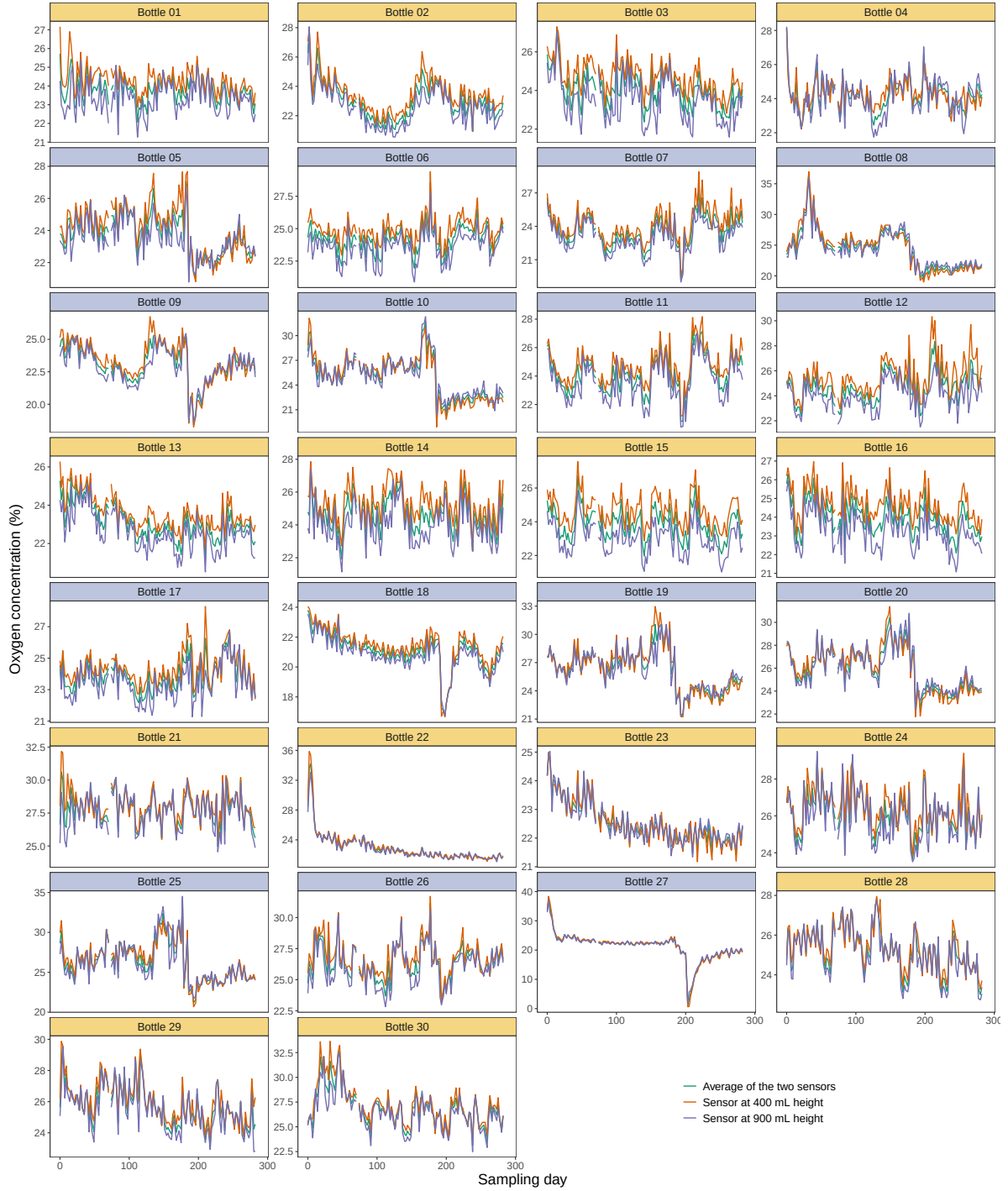


Figure 8: The *untransformed* oxygen concentration time series, shown in the different bottles (i.e. the panels). The time series are coloured by sensor height (with the average of the two time series also shown). The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.



Figure 9: The *untransformed* carbon and nitrogen concentration time series, shown in the different bottles (i.e. the panels). The time series are coloured by the type of carbon and nitrogen concentration. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.



Figure 10: The *transformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

facet_wrap(~light_treatment)+
# facet_wrap2(~bottle, ncol = 4, strip = strip2)+
theme_bw()+
theme(legend.position = "bottom",
      panel.grid=element_blank(),
      legend.text = element_text(size=12),
      axis.text = element_text(size=11),
      axis.title = element_text(size=16),
      strip.text = element_text(size=12),
      # legend.key.size = unit(3,"line"),
      # strip.text = element_text(colour = "white"),
      ) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col="",
     y = "Standardized total biomass",
     x = "Sampling day")

```

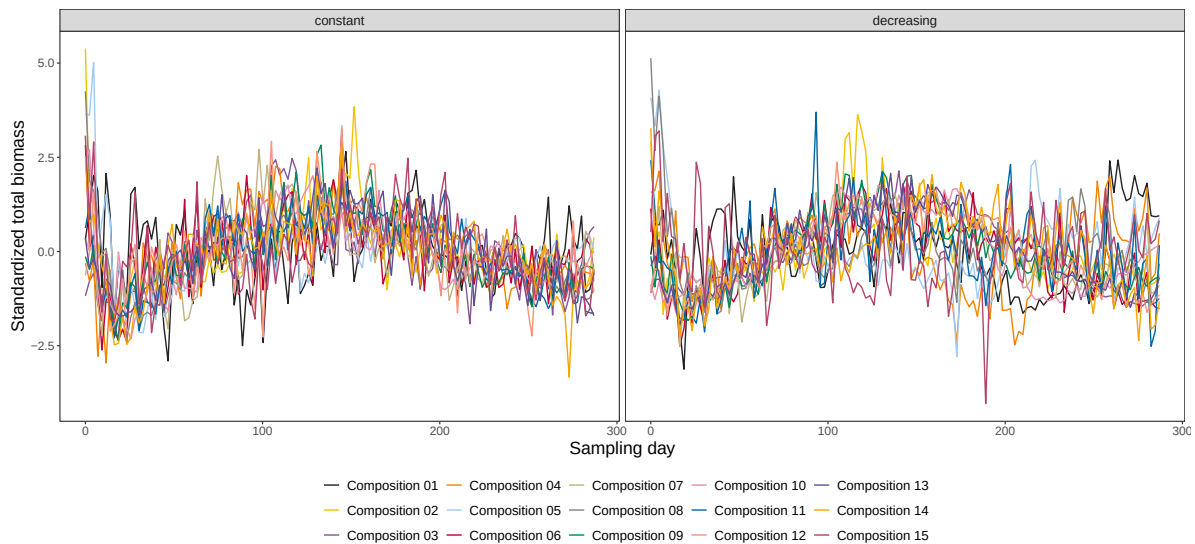


Figure 11: The *transformed* total biomass time series of the communities, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

```

dd_list$o2.int.detrend.complete %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle),
         composition = gsub("c_", "Composition ", composition)) %>%
  ggplot(aes(day, percent_o2, col=composition))+
  geom_line(linewidth=0.6)+
  facet_wrap(~light_treatment)+
# facet_wrap2(~bottle, ncol = 4, strip = strip2)+
theme_bw()+
theme(legend.position = "bottom",
      panel.grid=element_blank(),
      legend.text = element_text(size=12),
      axis.text = element_text(size=11),
      axis.title = element_text(size=16),
      strip.text = element_text(size=12),
      # legend.key.size = unit(3,"line"),

```

```

    # strip.text = element_text(colour = "white"),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col="",
       y = "Standardized Oxygen concentration",
       x = "Sampling day")

```

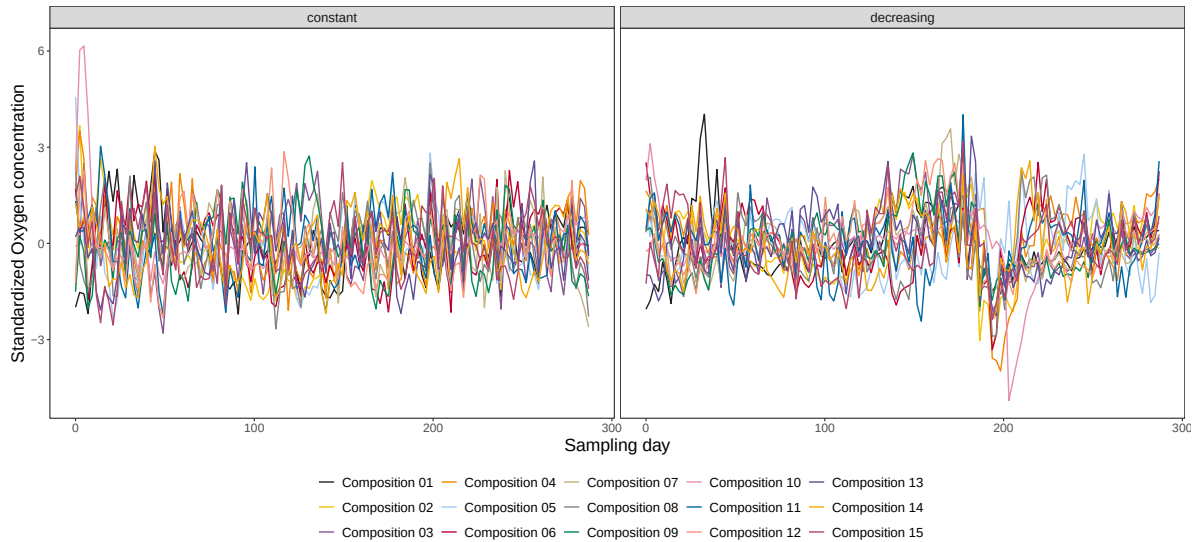


Figure 12: The *transformed* oxygen concentration time series, shown in the different light conditions (i.e. the panels). The time series are coloured by community composition.

5 Calculation of realized richness, biomass and time series properties

5.1 The realized species richness

```

dd.rich.test <- dd_list$densities %>%
  dplyr::filter(day == 287) %>%
  group_by(
    bottle,
    light_treatment,
    mean_richness,
    median_richness,
    composition,
    realized_richness
  ) %>%
  dplyr::summarise(n = n())

# t.test.light <-
#   t.test(
#     mean_richness ~ light_treatment,
#     paired = TRUE,
#     alternative = "two.sided",
#     data = dd.rich.test

```

```

# )
# t.test.light

lowerfun <- function(data, mapping,...){
  ggplot(data = data, mapping = mapping)+
    geom_abline(slope = 1, intercept = 0)+
    geom_smooth(method = "lm", ...) +
    geom_point(...) +
    theme_bw()
}

my_dens <- function(data, mapping, ...) {
  ggplot(data = data, mapping = mapping) +
    geom_density(..., alpha=0.5) +
    theme_bw()
}

my_hist <- function(data, mapping, ...) {
  ggplot(data = data, mapping = mapping) +
    geom_histogram(..., alpha=0.3, position="identity") +
    theme_bw()
}

## realized species richness

numb.species <- dd_list$densities %>%
  dplyr::filter(
    species %in% c(
      "Bacteria",
      "Chlamydomonas reinhardtii",
      "Coleps sp.",
      "Cosmarium botrytis",
      "Desmodesmus armatus",
      "Paramecium bursaria",
      "Paramecium caudatum",
      "Stauroastrum gracile",
      "Monoraphidium obtusum",
      "Stauroastrum polytrichum",
      "Stylonychia sp.",
      "Colpidium striatum",
      "Dexiostoma campylum",
      "Euplotes daidaleos",
      "Loxoccephallus sp."
    )
  ) %>%
  group_by(bottle, richness, composition, light_treatment, ForecastTarget) %>%
  dplyr::summarize(
    realized_richness = length(unique(species)),
    # persisting.species = list(unique(species)),
    mean_richness = unique(mean_richness),
    median_richness = unique(median_richness)
  )

colnames(numb.species) <-

```

```

c(
  "bottle",
  "Planned richness",
  "composition",
  "light_treatment",
  "ForecastTarget",
  "Minimal richness",
  "Mean richness",
  "Median richness"
)

cor.numb.species <- numb.species %>%
  ggpairs(
    mapping = aes(col = light_treatment, fill = light_treatment),
    columns = c(
      "Planned richness",
      "Minimal richness",
      "Mean richness",
      "Median richness"
    ),
    lower = list(continuous = lowerfun, alpha = 0.5),
    diag = list(continuous = my_hist),
    legend = 1
  ) +
  scale_color_manual(values = c("#C4961A", "#293352"),
    aesthetics = c("color", "fill"))

cor.numb.species +
  labs(col = "Light conditions", fill = "Light conditions") +
  theme(legend.position = "bottom")

dd.rich.test <- dd_list$densities %>%
  dplyr::filter(day==287) %>%
  group_by(bottle, light_treatment, mean_richness, median_richness, composition,
    realized_richness) %>%
  dplyr::summarise(n=n()) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle),
    composition = gsub("c_", "Composition ", composition))

rich.test.1 <- dd.rich.test %>%
  ggplot(aes(light_treatment, mean_richness, group=composition))+
  geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
  geom_line(col="black", alpha=0.3,
    position=position_dodge(width = 0.3))+
  geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
    na.rm = T, alpha=1, shape=21)+
  theme_bw()+
  theme(legend.position = "bottom")+
  scale_fill_manual(values = unname(kelly()[2:16]))+
  scale_y_continuous(limits = c(0,8)) +
  scale_color_manual(values = c("#C4961A", "#293352"))+
  labs(x="Light conditions", y="Mean richness",
    fill="") +
  guides(col=guide_legend(ncol = 1,byrow=TRUE))

```

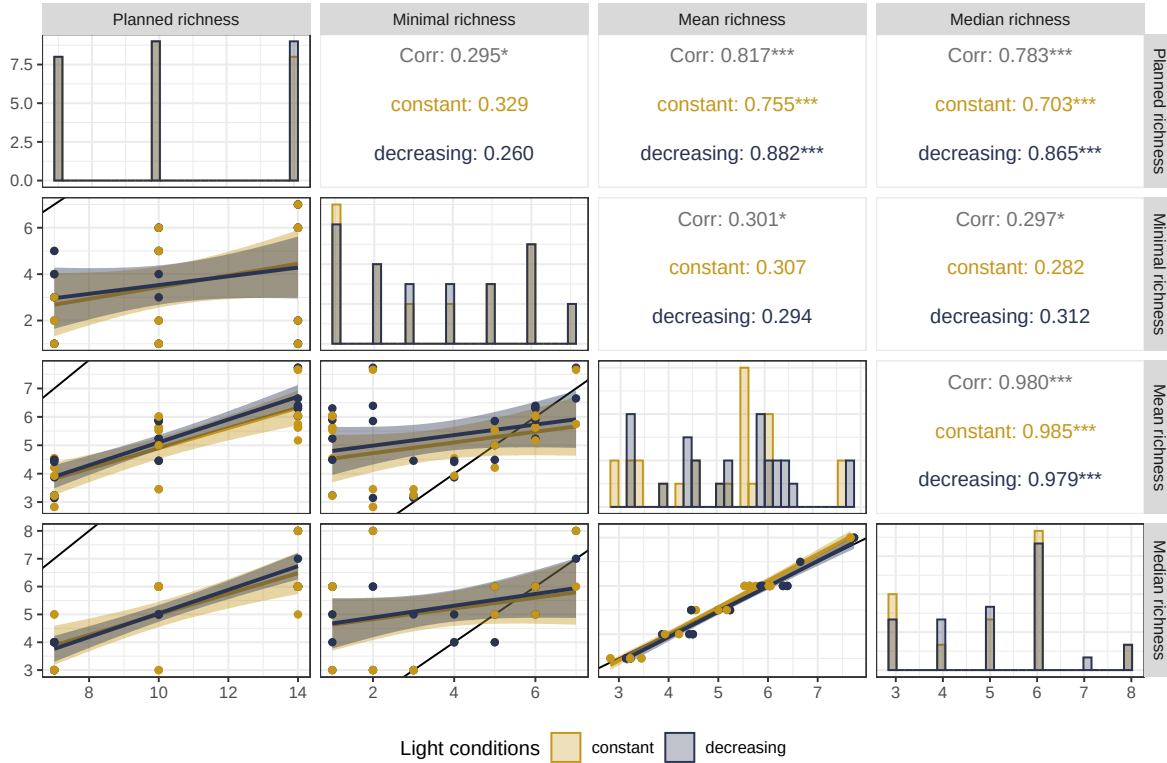


Figure 13: Comparison of the different species richness metrics (i.e. the planned and the realized median, mean and minimal richness), colour-coded for the light conditions. The panels in the diagonal show the bar-plots of the richness variables. The panels below the diagonal show the pairwise scatter-plots as indicated by the facet labels (the black line is the identity line, the coloured lines and the corresponding shaded areas show the fit and the 95% CI of a linear regression). The panels above the diagonal showing the corresponding correlation estimates.

```

rich.test.2 <- dd.rich.test %>%
  ggplot(aes(light_treatment, median_richness, group=composition))+
  geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
  geom_line(col="black", alpha=0.3,
            position=position_dodge(width = 0.3))+
  geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
             na.rm = T, alpha=1, shape=21)+
  theme_bw()+
  theme(legend.position = "bottom")+
  scale_fill_manual(values = unname(kelly()[2:16]))+
  scale_y_continuous(limits = c(0,8)) +
  scale_color_manual(values = c("#C4961A", "#293352"))+
  labs(x="Light conditions", y="Median richness",
       fill="") +
  guides(col=guide_legend(ncol = 1,byrow=TRUE))

rich.test.3 <- dd.rich.test %>%
  ggplot(aes(light_treatment, realized_richness, group=composition))+
  geom_boxplot(aes(col=light_treatment, group=light_treatment), show.legend = F)+
  geom_line(col="black", alpha=0.3,
            position=position_dodge(width = 0.3))+
  geom_point(aes(fill=composition), position=position_dodge(width = 0.3), size=3,
             na.rm = T, alpha=1, shape=21)+
  theme_bw()+
  theme(legend.position = "bottom")+
  scale_fill_manual(values = unname(kelly()[2:16]))+
  scale_y_continuous(limits = c(0,8)) +
  scale_color_manual(values = c("#C4961A", "#293352"))+
  labs(x="Light conditions", y="Minimal richness",
       fill="") +
  guides(col=guide_legend(ncol = 1,byrow=TRUE))

rich.test.1 + rich.test.2 + rich.test.3 + plot_layout(guides = "collect") +
  plot_annotation(tag_levels = "a") &
  theme(legend.position = 'bottom')

```

6 Forecasting

6.1 Correlation of forecasts across methods

```

complete.forecasts.wide <- complete.forecasts %>%
  dplyr::select(
    bottle,
    richness,
    composition,
    incubator,
    realized_richness,
    light_treatment,
    Target,
    RMSE,
    forecast.method,

```

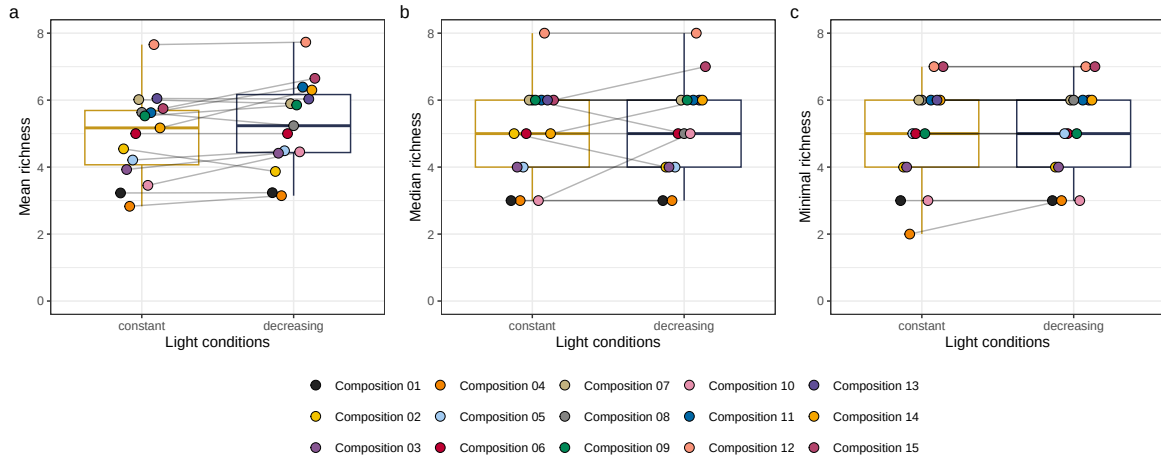


Figure 14: Scatter- and box-plot to investigate the effects of the light conditions on the realized species richness measures. (a) Mean richness; (b) Median richness; (c) Minimal richness. The points are coloured based on species composition and bottles with the same starting species composition are connected by a line. The boxplots are coloured based on light condition (orange = constant, blue = decreasing).

```
mean_richness,
median_richness
) %>%
pivot_wider(names_from = forecast.method, values_from = RMSE)

cor.plot.complete <- complete.forecasts.wide %>%
ggpairs(
  mapping = aes(col = light_treatment, fill = light_treatment),
  columns = c("ARIMA", "Multiview EDM", "RNN", "Random Forest", "Simplex EDM"),
  lower = list(continuous = lowerfun, alpha = 0.5),
  diag = list(continuous = my_dens),
  legend = 1
) +
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill"))

cor.plot.complete +
  labs(col = "Light conditions", fill = "Light conditions") +
  theme(legend.position = "bottom")
```

7 Additional figures and tables

7.1 Effects of experimental conditions on forecasts

```
##### -----
##### Species forecast model regressions
##### -----

formula <-
```

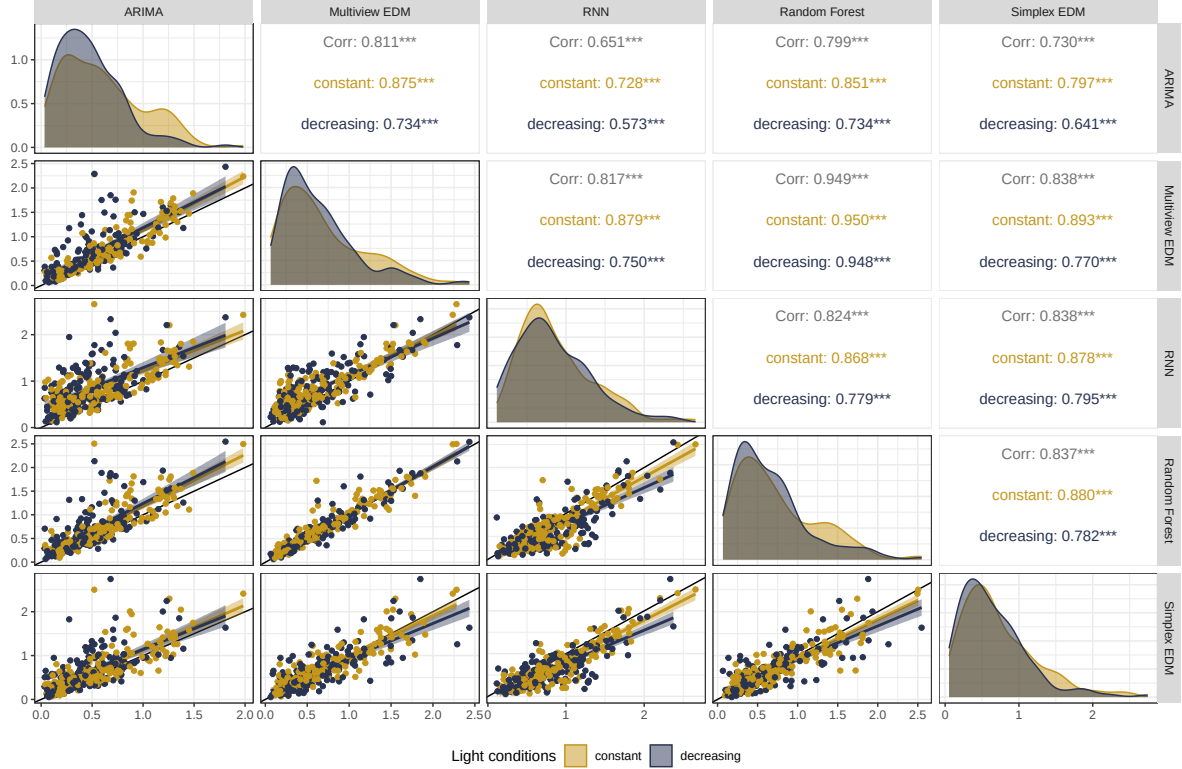


Figure 15: Comparison of forecast errors values (i.e. RMSE) produced by the different forecast methods (i.e. ARIMA, Multiview EDM, RNN, Random Forest and Simplex EDM), colour-coded for the light conditions. The panels in the diagonal show the density plots of the forecast errors. The panels below the diagonal show the pairwise scatter-plots as indicated by the facet labels (the black line is the identity line, the coloured lines and the corresponding shaded areas show the fit and the 95% CI of a linear regression). The panels above the diagonal showing the corresponding correlation estimates.

```

as.formula(
  RMSE ~ median_richness_center * light_treatment +
    (1 | Target) + (1 | bottle) + (1 | incubator)
)

m.simplex <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Simplex EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.multiview <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Multiview EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.arima <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "ARIMA",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.rnn <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "RNN",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

m.rf <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Random Forest",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

```

```

)

m.species.list <- list(
  `Simplex EDM` = m.simplex,
  `Multiview EDM` = m.multiview,
  `ARIMA` = m.arima,
  `RNN` = m.rnn,
  `Random Forest` = m.rf
)

##### -----
##### Total biomass forecast model regressions
##### -----

formula <- as.formula(RMSE ~ median_richness_center * light_treatment +
                      (1 | composition) + (1 | incubator))
formula.no.int <- as.formula(RMSE ~ median_richness_center + light_treatment +
                             (1 | composition) + (1 | incubator))

m.simplex.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM", Target == "Total_biomass")
)

m.multiview.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM", Target == "Total_biomass")
)

m.arima.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA", Target == "Total_biomass")
)

m.rnn.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN", Target == "Total_biomass")
)

m.rf.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest", Target == "Total_biomass")
)

## no interaction

m.simplex.bio.no.int <- lmer(
  formula.no.int,

```

```

complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM", Target == "Total_biomass")
)

m.multiview.bio.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM", Target == "Total_biomass")
)

m.arima.bio.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA", Target == "Total_biomass")
)

m.rnn.bio.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN", Target == "Total_biomass")
)

m.rf.bio.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest", Target == "Total_biomass")
)

m.bio.list <- list(
  `Simplex EDM` = m.simplex.bio.no.int,
  `Multiview EDM` = m.multiview.bio.no.int,
  `ARIMA` = m.arima.bio.no.int,
  `RNN` = m.rnn.bio,
  `Random Forest` = m.rf.bio.no.int
)

##### -----
##### Mean oxygen forecast model regressions
##### -----

formula <- as.formula(RMSE ~ median_richness_center * light_treatment +
  (1 | composition) + (1 | incubator))
formula.no.int <- as.formula(RMSE ~ median_richness_center + light_treatment +
  (1 | composition) + (1 | incubator))

m.simplex.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM", Target == "Mean_oxygen")
)

```

```

m.multiview.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM", Target == "Mean_oxygen")
)

m.arima.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA", Target == "Mean_oxygen")
)

m.rnn.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN", Target == "Mean_oxygen")
)

m.rf.oxy <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest", Target == "Mean_oxygen")
)

## no interaction
m.simplex.oxy.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM", Target == "Mean_oxygen")
)

m.multiview.oxy.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Multiview EDM", Target == "Mean_oxygen")
)

m.arima.oxy.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "ARIMA", Target == "Mean_oxygen")
)

m.rnn.oxy.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "RNN", Target == "Mean_oxygen")
)

m.rf.oxy.no.int <- lmer(
  formula.no.int,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Random Forest", Target == "Mean_oxygen")
)

```

```

)

m.oxy.list <- list(
  `Simplex EDM` = m.simplex.oxy.no.int,
  `Multiview EDM` = m.multiview.oxy.no.int,
  `ARIMA` = m.arima.oxy.no.int,
  `RNN` = m.rnn.oxy.no.int,
  `Random Forest` = m.rf.oxy.no.int
)

##### -----
##### Total biomass vs Mean Species forecast model regressions
##### -----

average.complete.forecasts2 <- complete.forecasts %>%
  dplyr::filter(!(Target %in% c("Mean_oxygen")) %>%
    mutate(aggregatedMeasure = ifelse(Target %in% c("Total_biomass", "Mean_oxygen"), T, F)) %>%
    group_by(
      bottle,
      richness,
      composition,
      incubator,
      realized_richness,
      mean_richness,
      median_richness,
      light_treatment,
      forecast.method,
      median_richness_center,
      mean_richness_center,
      aggregatedMeasure
    ) %>%
    summarise(meanRMSE = mean(RMSE), sdRMSE = sd(RMSE))

formula <- as.formula(
  meanRMSE ~ median_richness_center * light_treatment +
    median_richness_center * aggregatedMeasure +
    light_treatment * aggregatedMeasure +
    (1 | bottle) + (1 | incubator)
)
formula.no.int <- as.formula(
  meanRMSE ~ median_richness_center + light_treatment +
    aggregatedMeasure +
    (1 | bottle) + (1 | incubator)
)

m.simplex.aggr.bio <- lmer(
  formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

```

```

m.multiview.aggr.bio <- lmer(
  formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Multiview EDM")
)

m.arima.aggr.bio <- lmer(formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "ARIMA"))

m.rnn.aggr.bio <- lmer(formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "RNN"))

m.rf.aggr.bio <- lmer(
  formula,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Random Forest")
)

# adpated models with dropped interactions if insignificant

m.simplex.aggr.bio.no.int <- lmer(
  formula.no.int,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

m.multiview.aggr.bio.no.int <- lmer(
  meanRMSE ~ median_richness_center * light_treatment +
    aggregatedMeasure + (1 |
      bottle) + (1 | incubator),
  average.complete.forecasts2 %>% dplyr::filter(forecast.method == "Multiview EDM")
)

m.arima.aggr.bio.no.int <- lmer(
  formula.no.int,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "ARIMA")
)

m.rnn.aggr.bio.no.int <- lmer(
  meanRMSE ~ median_richness_center * light_treatment +
    aggregatedMeasure + (1 |
      bottle) + (1 | incubator),
  average.complete.forecasts2 %>% dplyr::filter(forecast.method == "RNN")
)

m.rf.aggr.bio.no.int <- lmer(
  formula.no.int,
  average.complete.forecasts2 %>%
    dplyr::filter(forecast.method == "Random Forest")
)

```

```

m.aggr.bio.list <- list(
  `Simplex EDM` = m.simplex.aggr.bio.no.int,
  `Multiview EDM` = m.multiview.aggr.bio.no.int,
  `ARIMA` = m.arima.aggr.bio.no.int,
  `RNN` = m.rnn.aggr.bio.no.int,
  `Random Forest` = m.rf.aggr.bio.no.int
)

##### -----
##### Mean oxygen vs Mean Species forecast model regressions
##### -----

average.complete.forecasts.oxy <- complete.forecasts %>%
  dplyr::filter(!(Target %in% c("Total_biomass")) %>%
    mutate(aggregatedMeasure = ifelse(Target %in% c("Mean_oxygen", "Total_biomass"), T, F)) %>%
    group_by(
      bottle,
      richness,
      composition,
      incubator,
      realized_richness,
      mean_richness,
      median_richness,
      light_treatment,
      forecast.method,
      median_richness_center,
      mean_richness_center,
      aggregatedMeasure
    ) %>%
    summarise(meanRMSE = mean(RMSE), sdRMSE = sd(RMSE))

formula <- as.formula(
  meanRMSE ~ median_richness_center * light_treatment +
    median_richness_center * aggregatedMeasure +
    light_treatment * aggregatedMeasure +
    (1 | bottle) + (1 | incubator)
)

m.simplex.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

m.multiview.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Multiview EDM")
)

```

```

m.arima.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "ARIMA")
)

m.rnn.aggr.oxy <- lmer(formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "RNN"))

m.rf.aggr.oxy <- lmer(
  formula,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Random Forest")
)

# adapted models with dropped interactions if insignificant

formula.adapt <- as.formula(
  meanRMSE ~ median_richness_center + light_treatment + aggregatedMeasure +
    (1 | bottle) + (1 | incubator)
)

m.simplex.aggr.oxy.adapt <- lmer(
  formula.adapt,
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Simplex EDM")
)

m.multiview.aggr.oxy.adapt <- lmer(
  meanRMSE ~ median_richness_center + light_treatment +
    aggregatedMeasure + median_richness_center:aggregatedMeasure +
    (1 | bottle) + (1 | incubator),
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "Multiview EDM")
)

m.arima.aggr.oxy.adapt <- lmer(
  meanRMSE ~ median_richness_center * light_treatment + aggregatedMeasure +
    (1 | bottle) + (1 | incubator),
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "ARIMA")
)

m.rnn.aggr.oxy.adapt <- lmer(
  meanRMSE ~ median_richness_center * light_treatment + aggregatedMeasure +
    (1 | bottle) + (1 | incubator),
  average.complete.forecasts.oxy %>%
    dplyr::filter(forecast.method == "RNN")
)

m.rf.aggr.oxy.adapt <- lmer(

```

```

meanRMSE ~ median_richness_center * light_treatment + aggregatedMeasure +
  (1 | bottle) + (1 | incubator),
average.complete.forecasts.oxy %>%
  dplyr::filter(forecast.method == "Random Forest")
)

```

```

m.aggr.oxy.list <- list(
  `Simplex EDM` = m.simplex.aggr.oxy.adapt,
  `Multiview EDM` = m.multiview.aggr.oxy.adapt,
  `ARIMA` = m.arima.aggr.oxy.adapt,
  `RNN` = m.rnn.aggr.oxy.adapt,
  `Random Forest` = m.rf.aggr.oxy.adapt
)

```

```

formula <-
  as.formula(
    RMSE ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator)
  )
m.simplex <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Simplex EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)
eta_squared(m.simplex)
r2beta(m.simplex, method = "kr") # kenward roger
r2beta(m.simplex, method = "nsj") # nakagawa similar

```

```

complete.forecasts.split.c <- complete.forecasts %>%
  dplyr::filter(
    forecast.method == "Simplex EDM",
    light_treatment == "constant",
    Target != "Total_biomass",
    Target != "Mean_oxygen"
  ) %>%
  mutate(log10median_richness = log10(median_richness))

```

```

complete.forecasts.split.d <- complete.forecasts %>%
  dplyr::filter(
    forecast.method == "Simplex EDM",
    light_treatment == "decreasing",
    Target != "Total_biomass",
    Target != "Mean_oxygen"
  ) %>%
  mutate(log10median_richness = log10(median_richness))

```

```

formula.split <- as.formula(RMSE ~ median_richness_center +
  (1 |

```

```

                                Target) + (1 | bottle) + (1 | incubator))

formula.split2 <- as.formula(RMSE ~ log10median_richness +
                                (1 |
                                Target) + (1 | bottle) + (1 | incubator))

m.simplex.constant <- lmer(formula.split2, complete.forecasts.split.c)
m.simplex.decreasing <- lmer(formula.split2, complete.forecasts.split.d)

newdat.m.simplex.c <- expand.grid(
  log10median_richness = seq(
    min(complete.forecasts.split.c$log10median_richness),
    max(complete.forecasts.split.c$log10median_richness),
    length.out = 100
  ),
  light_treatment = unique(complete.forecasts.split.c$light_treatment)
)

newdat.m.simplex.d <- expand.grid(
  log10median_richness = seq(
    min(complete.forecasts.split.d$log10median_richness),
    max(complete.forecasts.split.d$log10median_richness),
    length.out = 100
  ),
  light_treatment = unique(complete.forecasts.split.d$light_treatment)
)

newdat.m.simplex.c <- conf_interval(m.simplex.constant,
                                   newdat.m.simplex.c,
                                   "RMSE",
                                   "Forecast error (RMSE)")
newdat.m.simplex.d <- conf_interval(m.simplex.decreasing,
                                   newdat.m.simplex.d,
                                   "RMSE",
                                   "Forecast error (RMSE)")

#### -----
#### Table for simplex species forecasts regression (Main text)
#### -----

r.simplex <- rand(m.simplex)

spec.ab.rmse.tab <- data.frame(
  Response = "Forecast error\nof target\nabundances",
  Covariate = c(
    "Intercept",
    "Richness",
    "Light (decreasing)",
    "Richness:Light",
    "Target (taxa)",
    "Bottle",
    "Incubator"
  ),
  Type = c(

```

```

    rep("Fixed\\neffect", 4),
    "Random\\nintercept",
    "Random\\nintercept",
    "Random\\nintercept"
  ),
  Estimate = c(fixef(m.simplex), as.data.frame(VarCorr(m.simplex))$sdcor[1:3]),
  DF = c(round(summary(m.simplex)$coef[, 3, drop = FALSE], 0), r.simplex$Df[2:4]),
  SE = c(summary(m.simplex)$coef[, 2, drop = FALSE], NA, NA, NA),
  Test = c(rep("\\textit{t}", 4), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
  Test.value = c(summary(m.simplex)$coef[, 4, drop = FALSE], r.simplex$LRT[2:4]),
  p.value = c(summary(m.simplex)$coef[, 5, drop = FALSE], r.simplex$`Pr(>Chisq)`[2:4])
)

spec.ab.rmse.tab.anova.tab <- data.frame(
  Response = "Forecast error\\nof target\\nabundances",
  Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
  Type = "Sum of\\nsquares",
  Estimate = anova(m.simplex)[, 1],
  DF = paste(anova(m.simplex)[, 3], round(anova(m.simplex)[, 4], 1), sep = ", "),
  SE = NA,
  Test = "\\textit{F}",
  Test.value = anova(m.simplex)[, 5],
  p.value = anova(m.simplex)[, 6]
)

spec.ab.rmse.tab <- rbind(spec.ab.rmse.tab, spec.ab.rmse.tab.anova.tab)

#### -----
#### Table for species forecasts regression (Supplementary): all forecasts
#### -----

newdat.taxa <- data.frame()
tab.taxa <- data.frame()

df <- complete.forecasts %>%
  dplyr::filter(!(Target == c("Total_biomass", "Mean_oxygen")))

newdat.t <- expand.grid(
  median_richness_center = seq(
    min(df$median_richness_center),
    max(df$median_richness_center),
    length.out = 100
  ),
  light_treatment = unique(df$light_treatment)
)

for (m.name in names(m.species.list)) {
  m <- m.species.list[[m.name]]
  newdat.t2 <- conf_interval(m, newdat.t, "RMSE", "Forecast error (RMSE)")
  newdat.t2$forecast.method <- m.name
}

```

```

r <- rand(m)

tab1 <- data.frame(
  Response = "Forecast\nerror of\ntaxa\nabundances",
  forecast.method = m.name,
  Covariate = c(
    "Intercept",
    "Richness",
    "Light (decreasing)",
    "Richness:Light",
    "Target (taxa)",
    "Bottle",
    "Incubator"
  ),
  Type = c(
    rep("Fixed\neffect", 4),
    "Random\nintercept",
    "Random\nintercept",
    "Random\nintercept"
  ),
  Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:3]),
  DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:4]),
  SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA, NA),
  Test = c(rep("\\textit{t}", 4), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
  Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:4]),
  p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:4])
)

tab2 <- data.frame(
  Response = "Forecast\nerror of\ntaxa\nabundances",
  forecast.method = m.name,
  Covariate = c("Richness", "Light (decreasing)", "Richness:Light"),
  Type = "Sum of\nsquares",
  Estimate = anova(m)[, 1],
  DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
  SE = NA,
  Test = "\\textit{F}",
  Test.value = anova(m)[, 5],
  p.value = anova(m)[, 6]
)

tab1 <- rbind(tab1, tab2)
newdat.taxa <- rbind(newdat.taxa, newdat.t2)
tab.taxa <- rbind(tab.taxa, tab1)
}

```

```

#### -----
#### Table for species forecasts regression (Supplementary): split analysis
#### -----

```

```

conf.int.c <- confint(m.simplex.constant)
conf.int.c.coef <- conf.int.c[5:6, ]
conf.int.c.coef <- round(conf.int.c.coef, 3)
conf.int.c.coef <- apply(conf.int.c.coef, 1, paste, collapse = " - ")
coefs.c <- coef(summary(m.simplex.constant))

split.tab.1 <-
  data.frame(
    Light = "Constant",
    Covariate = c("Intercept", "log10(Richness)"),
    Estimate = fixef(m.simplex.constant),
    CI = conf.int.c.coef,
    SS = c(NA, anova(m.simplex.constant)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex.constant)[, 3], round(anova(m.simplex.constant)[, 4], 1), sep = ", "
    )),
    F.value = c(NA, anova(m.simplex.constant)[, 5]),
    p.value = c(NA, anova(m.simplex.constant)[, 6])
  )

conf.int.d <- confint(m.simplex.decreasing)
conf.int.d.coef <- conf.int.d[5:6, ]
conf.int.d.coef <- round(conf.int.d.coef, 3)
conf.int.d.coef <- apply(conf.int.d.coef, 1, paste, collapse = " - ")
coefs.d <- coef(summary(m.simplex.decreasing))

split.tab.2 <-
  data.frame(
    Light = "Decreasing",
    Covariate = c("Intercept", "log10(Richness)"),
    Estimate = fixef(m.simplex.decreasing),
    CI = conf.int.d.coef,
    SS = c(NA, anova(m.simplex.decreasing)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex.decreasing)[, 3], round(anova(m.simplex.decreasing)[, 4], 1), sep = ", "
    )),
    F.value = c(NA, anova(m.simplex.decreasing)[, 5]),
    p.value = c(NA, anova(m.simplex.decreasing)[, 6])
  )

split.tab <- rbind(split.tab.1, split.tab.2)

#### -----
#### Table for simplex community biomass regression (Main text)
#### -----

r.simplex.bio <- rand(m.simplex.bio.no.int)

bio.rmse.tab <- data.frame(
  Response = "Forecast error\nof community\nbiomass",
  Covariate = c(
    "Intercept",

```

```

    "Richness",
    "Light (decreasing)",
    "Composition",
    "Incubator"
  ),
  Type = c(
    rep("Fixed\\neffect", 3),
    "Random\\nintercept",
    "Random\\nintercept"
  ),
  Estimate = c(fixef(m.simplex.bio.no.int), as.data.frame(VarCorr(
    m.simplex.bio.no.int
  ))$sdcor[1:2]),
  DF = c(round(summary(
    m.simplex.bio.no.int
  )$coef[, 3, drop = FALSE], 0), r.simplex.bio$Df[2:3]),
  SE = c(summary(m.simplex.bio.no.int)$coef[, 2, drop = FALSE], NA, NA),
  Test = c(rep("\\textit{t}", 3), "$\\chi^2$", "$\\chi^2$"),
  Test.value = c(summary(m.simplex.bio.no.int)$coef[, 4, drop = FALSE], r.simplex.bio$LRT[2:3]),
  p.value = c(
    summary(m.simplex.bio.no.int)$coef[, 5, drop = FALSE],
    r.simplex.bio$`Pr(>Chisq)`[2:3]
  )
)

bio.rmse.anova.tab <- data.frame(
  Response = "Forecast error\\nof community\\nbiomass",
  Covariate = c("Richness", "Light (decreasing)"),
  Type = "Sum of\\nsquares",
  Estimate = anova(m.simplex.bio.no.int)[, 1],
  DF = paste(anova(m.simplex.bio.no.int)[, 3], round(anova(
    m.simplex.bio.no.int
  )[, 4], 1), sep = ", "),
  SE = NA,
  Test = "\\textit{F}",
  Test.value = anova(m.simplex.bio.no.int)[, 5],
  p.value = anova(m.simplex.bio.no.int)[, 6]
)

bio.rmse.tab <- rbind(bio.rmse.tab, bio.rmse.anova.tab)

#### -----
#### Table for community biomass regression (Supplementary): all forecasts
#### -----

newdat.bio <- data.frame()
tab.bio <- data.frame()

df <- complete.forecasts %>%
  dplyr::filter(Target == "Total_biomass")

newdat.b <- expand.grid(

```

```

median_richness_center = seq(
  min(df$median_richness_center),
  max(df$median_richness_center),
  length.out = 100
),
light_treatment = unique(df$light_treatment)
)

for (m.name in names(m.bio.list)) {
  m <- m.bio.list[[m.name]]
  newdat.b2 <- conf_interval(m, newdat.b, "RMSE", "Forecast error (RMSE)")
  newdat.b2$forecast.method <- m.name

  r <- rand(m)

  if (length(fixef(m)) == 3) {
    tab1 <- data.frame(
      Response = "Forecast\nerror of\ncommunity\nbiomass",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Composition",
        "Incubator"
      ),
      Type = c(
        rep("Fixed\neffect", 3),
        "Random\nintercept",
        "Random\nintercept"
      ),
      Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
      DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
      SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
      Test = c(NA, rep("\textit{F}", 2), "$\\chi^2$", "$\\chi^2$"),
      Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
      p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
    )
  } else {
    tab1 <- data.frame(
      Response = "Forecast\nerror of\ncommunity\nbiomass",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Richness:Light",
        "Composition",
        "Incubator"
      ),
      Type = c(
        rep("Fixed\neffect", 4),
        "Random\nintercept",

```

```

      "Random\\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\\textit{F}", 3), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
}

newdat.bio <- rbind(newdat.bio, newdat.b2)
tab.bio <- rbind(tab.bio, tab1)
}

#### -----
#### Table for mean oxygen regression (Supplementary): all forecasts
#### -----

newdat.oxy <- data.frame()
tab.oxy <- data.frame()

df <- complete.forecasts %>%
  dplyr::filter(Target == "Mean_oxygen")

newdat.o <- expand.grid(
  median_richness_center = seq(
    min(df$median_richness_center),
    max(df$median_richness_center),
    length.out = 100
  ),
  light_treatment = unique(df$light_treatment)
)

for (m.name in names(m.oxy.list)) {
  m <- m.oxy.list[[m.name]]
  newdat.o2 <- conf_interval(m, newdat.o, "RMSE", "Forecast error (RMSE)")
  newdat.o2$forecast.method <- m.name

  r <- rand(m)

  if (length(fixef(m)) == 3) {
    tab1 <- data.frame(
      Response = "Forecast\\nerror\\nof mean\\noxygen",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Composition",
        "Incubator"
      ),
    ),
  }
}

```

```

    Type = c(
      rep("Fixed\\neffect", 3),
      "Random\\nintercept",
      "Random\\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\\textit{F}", 2), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
} else {
  tab1 <- data.frame(
    Response = "Forecast\\nerror\\nof mean\\noxygen",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light",
      "Composition",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\\neffect", 4),
      "Random\\nintercept",
      "Random\\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\\textit{F}", 3), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
}

newdat.oxy <- rbind(newdat.oxy, newdat.o2)
tab.oxy <- rbind(tab.oxy, tab1)
}

#### -----
#### Table for aggregate analysis: Biomass vs Species forecasts: all forecasts
#### -----

newdat.aggr.bio <- data.frame()
tab.aggr.bio <- data.frame()
em.tab.bio <- data.frame()

newdat.a.b <- with(

```

```

average.complete.forecasts2,
expand.grid(
  median_richness_center = seq(
    min(median_richness_center),
    max(median_richness_center),
    length.out = 100
  ),
  light_treatment = unique(light_treatment),
  aggregatedMeasure = unique(aggregatedMeasure)
)
)

for (m.name in names(m.aggr.bio.list)) {
  m <- m.aggr.bio.list[[m.name]]
  newdat.a.b2 <- conf_interval(m, newdat.a.b, "meanRMSE", "Mean forecast error (RMSE)")
  newdat.a.b2$forecast.method <- m.name

  em <- emmeans(m, c("aggregatedMeasure"), by = "light_treatment")
  em <- as.data.frame(pairs(em))
  em$forecast.method = m.name
  em.tab.bio <- rbind(em.tab.bio, em)

  r <- rand(m)

  if (length(fixef(m)) == 4) {
    tab1 <- data.frame(
      Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof community\nbiomass",
      forecast.method = m.name,
      Covariate = c(
        "Intercept",
        "Richness",
        "Light (decreasing)",
        "Measure (biomass)",
        "Bottle",
        "Incubator"
      ),
    ),
    Type = c(
      rep("Fixed\neffect", 4),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\textit{F}", 3), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
} else {
  tab1 <- data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof community\nbiomass",
    forecast.method = m.name,
    Covariate = c(

```

```

    "Intercept",
    "Richness",
    "Light (decreasing)",
    "Measure (biomass)",
    "Richness:Light",
    "Bottle",
    "Incubator"
  ),
  Type = c(
    rep("Fixed\\neffect", 5),
    "Random\\nintercept",
    "Random\\nintercept"
  ),
  Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
  DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
  SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
  Test = c(NA, rep("\\textit{F}", 4), "$\\chi^2$", "$\\chi^2$"),
  Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
  p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
)
}

newdat.aggr.bio <- rbind(newdat.aggr.bio, newdat.a.b2)
tab.aggr.bio <- rbind(tab.aggr.bio, tab1)
}

#### -----
#### Table for aggregate analysis: Oxygen vs Species forecasts: all forecasts
#### -----

newdat.aggr.oxy <- data.frame()
tab.aggr.oxy <- data.frame()
em.tab.oxy <- data.frame()

newdat.ox <- with(
  average.complete.forecasts.oxy,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment),
    aggregatedMeasure = unique(aggregatedMeasure)
  )
)

for (m.name in names(m.aggr.oxy.list)) {
  m <- m.aggr.oxy.list[[m.name]]
  newdat.ox2 <- conf_interval(m, newdat.ox, "meanRMSE", "Mean forecast error (RMSE)")
  newdat.ox2$forecast.method <- m.name
}

```

```

em <- emmeans(m, c("aggregatedMeasure"), by = "light_treatment")
em <- as.data.frame(pairs(em))
em$forecast.method = m.name
em.tab.oxy <- rbind(em.tab.oxy, em)

r <- rand(m)

if (length(fixef(m)) == 4) {
  tab1 <- data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof mean\noxygen\nnc",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Measure (biomass)",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 4),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\textit{F}", 3), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
} else if (m.name == "Multiview EDM") {
  tab1 <- data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof mean\noxygen\nnc",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Measure (biomass)",
      "Richness:Measure",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 5),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\textit{F}", 4), "$\\chi^2$", "$\\chi^2$"),

```

```

    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
} else {
  tab1 <- data.frame(
    Response = "Mean forecast\nerror of\nabundances\n \nVersus\n \nForecast error\nof mean\noxygen\nnc",
    forecast.method = m.name,
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Measure (biomass)",
      "Richness:Light",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 5),
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:2]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:3]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA),
    Test = c(NA, rep("\textit{F}", 4), "$\\chi^2$", "$\\chi^2$"),
    Test.value = c(NA, anova(m)[, 5], r$LRT[2:3]),
    p.value = c(NA, anova(m)[, 6], r$`Pr(>Chisq)`[2:3])
  )
}
newdat.aggr.oxy <- rbind(newdat.aggr.oxy, newdat.ox2)
tab.aggr.oxy <- rbind(tab.aggr.oxy, tab1)
}

#### -----
#### Table for marginal means: aggregated measures vs non-aggregated
#### -----

em.tab.oxy <- em.tab.oxy %>%
  mutate(contrast = "Species - Oxygen") %>%
  dplyr::select(contrast,
    forecast.method,
    light_treatment,
    estimate,
    SE,
    df,
    t.ratio,
    p.value)

em.tab.bio <- em.tab.bio %>%
  mutate(contrast = "Species - Biomass") %>%
  dplyr::select(contrast,

```

```

        forecast.method,
        light_treatment,
        estimate,
        SE,
        df,
        t.ratio,
        p.value)

em.tab <- rbind(em.tab.bio, em.tab.oxy)

#### -----
#### Table for marginal means: light vs no light depending on richness
#### -----

emmeans.comparison <- function(model, char.model, richness.value) {
  em.m <- emmeans(
    model,
    specs = c("median_richness_center", "light_treatment"),
    at = list(
      light_treatment = c("decreasing", "constant"),
      median_richness_center = richness.value
    )
  )
  test <- pairs(em.m)
  test <- as.data.frame(test)
  test$model <- char.model
  test$contrast <- richness.value
  # test$richness <- richness.value
  return(test)
}

mods <- list(m.simplex, m.multiview, m.arima, m.rf, m.rnn)
mod.names <- c("Simplex EDM", "Multiview EDM", "ARIMA", "Random Forest", "RNN")

em.tab.rich <- data.frame()
for (val in c(-2.5, 2.5)) {
  for (i in 1:length(mods)) {
    mod <- mods[[i]]
    mod.name <- mod.names[i]
    em.mod <- emmeans.comparison(mod, mod.name, val)
    em.tab.rich <- rbind(em.tab.rich, em.mod)
  }
}

main.tab <- rbind(spec.ab.rmse.tab, bio.rmse.tab)

rownames(main.tab) <- NULL

main.tab$p.value <- ifelse(main.tab$p.value < 0.0001, "<0.0001", sprintf("%.4f", round(main.tab$p.value

main.tab <- main.tab %>%
  dplyr::mutate(

```

```

    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
main.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of linear mixed models investigating the effects of
species richness and light on forecast errors of taxa and community biomass.
This table is an extended version of Table 1 in the main text and includes
both the regression tables and the analysis of variance tables of the models
(top part of the table: Type III ANOVA; bottom part of the table Type II ANOVA).
The models included median centered richness, light treatment and their
interaction as fixed effects. In the first model, random intercepts were
included for the bottles, incubators and the targets (i.e. taxa). In the
second model, random intercepts were included for composition and incubators.
Forecast error is based on forecasts computed with Simplex EDM. The estimates
for the random intercepts are standard deviations.",
    align = c("lllrrrcrr"),
    escape = F,
    label = "MainTab"
  ) %>%
  collapse_rows(
    columns = c(1, 3),
    latex_hline = 'custom',
    custom_latex_hline = c(1, 3),
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

rownames(split.tab) <- NULL

split.tab$p.value <- ifelse(split.tab$p.value < 0.0001, "<0.0001",
  sprintf("%.4f", round(split.tab$p.value, 4)))

split.tab <- split.tab %>%
  dplyr::mutate(

```

Table 6: Results of linear mixed models investigating the effects of species richness and light on forecast errors of taxa and community biomass. This table is an extended version of Table 1 in the main text and includes both the regression tables and the analysis of variance tables of the models (top part of the table: Type III ANOVA; bottom part of the table Type II ANOVA). The models included median centered richness, light treatment and their interaction as fixed effects. In the first model, random intercepts were included for the bottles, incubators and the targets (i.e. taxa). In the second model, random intercepts were included for composition and incubators. Forecast error is based on forecasts computed with Simplex EDM. The estimates for the random intercepts are standard deviations.

Response	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Forecast error of target abundances	Intercept	Fixed effect	0.685	16	0.093	<i>t</i>	7.371	<0.0001
	Richness		0.056	22	0.033	<i>t</i>	1.695	0.1044
	Light (decreasing)		-0.095	6	0.086	<i>t</i>	-1.109	0.3092
	Richness:Light		-0.102	20	0.046	<i>t</i>	-2.246	0.0363
	Target (taxa)	Random intercept	0.074	1	-	χ^2	47.396	<0.0001
	Bottle		0.250	1	-	χ^2	0.410	0.5219
	Incubator		0.085	1	-	χ^2	1.198	0.2737
	Richness	Sum of squares	0.008	1, 21.1	-	<i>F</i>	0.046	0.8328
	Light (decreasing)		0.225	1, 6.1	-	<i>F</i>	1.230	0.3092
	Richness:Light		0.922	1, 19.7	-	<i>F</i>	5.046	0.0363
Forecast error of community biomass	Intercept	Fixed effect	0.657	20	0.108	<i>t</i>	6.065	<0.0001
	Richness		0.015	20	0.062	<i>t</i>	0.241	0.8118
	Light (decreasing)		-0.017	14	0.099	<i>t</i>	-0.167	0.8698
	Composition	Random intercept	0.302	1	-	χ^2	5.108	0.0238
	Incubator		0.001	1	-	χ^2	0.000	1.0000
	Richness	Sum of squares	0.004	1, 19.8	-	<i>F</i>	0.058	0.8118
	Light (decreasing)		0.002	1, 14.1	-	<i>F</i>	0.028	0.8698

```

Estimate = round(Estimate, 3),
SS = round(SS, 3),
F.value = round(F.value, 3)
)

options(knitr.kable.NA = '')
split.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Light",
      "Covariate",
      "Estimate",
      "95\\% CI",
      "SS",
      "DF",
      "\\textit{F}-value",
      "\\textit{p}-value"
    ),
    caption = "Relation between log10-transformed median richness and the
forecast errors of taxa abundances, with the model carried out separately
for the constant and the decreasing light conditions (i.e. split analysis).
The table includes the estimated coefficients and corresponding 95\\%

```

Table 7: Relation between log10-transformed median richness and the forecast errors of taxa abundances, with the model carried out separately for the constant and the decreasing light conditions (i.e. split analysis). The table includes the estimated coefficients and corresponding 95% confidence intervals, as well as the analysis of variance (Type II) table. The random intercepts were included for the bottles, incubators and the targets (i.e. taxa), but are not shown in the table. Note that the richness variable was log10-transformed to meet the model assumptions. This was not necessary in the full model (i.e. the non-partitioned model). Forecast error is based on forecasts computed with Simplex EDM.

Light	Covariate	Estimate	95% CI	SS	DF	F-value	p-value
Constant	Intercept	0.158	-0.371 - 0.697	0.768	1, 12.6	4.118	0.0642
	log10(Richness)	0.716	0.033 - 1.39				
Decreasing	Intercept	1.169	0.609 - 1.753	0.617	1, 12.5	4.064	0.0659
	log10(Richness)	-0.782	-1.576 - -0.023				

confidence intervals, as well as the analysis of variance (Type II) table. The random intercepts were included for the bottles, incubators and the targets (i.e. taxa), but are not shown in the table. Note that the richness variable was log10-transformed to meet the model assumptions. This was not necessary in the full model (i.e. the non-partitioned model). Forecast error is based on forecasts computed with Simplex EDM.",

```
align = c("llccrrr"),
escape = F,
label = "SplitTab"
) %>%
collapse_rows(
  columns = c(1),
  latex_hline = 'custom',
  custom_latex_hline = c(1),
  valign = "middle"
) %>%
kable_styling(font_size = 7)
```

```
complete.forecasts %>%
dplyr::filter(!(Target %in% c("Total_biomass", "Mean_oxygen"))) %>%
ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
geom_ribbon(
  data = newdat.taxa,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.taxa,
  linewidth = 1.3,
  aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
facet_grid( ~ forecast.method) +
```

```

labs(x = "Median realized species richness", y = "Taxa abundance\nforecast error (RMSE)", col =
      "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                    aesthetics = c("color", "fill")) +

theme_bw() +
theme(legend.position = "bottom")

```

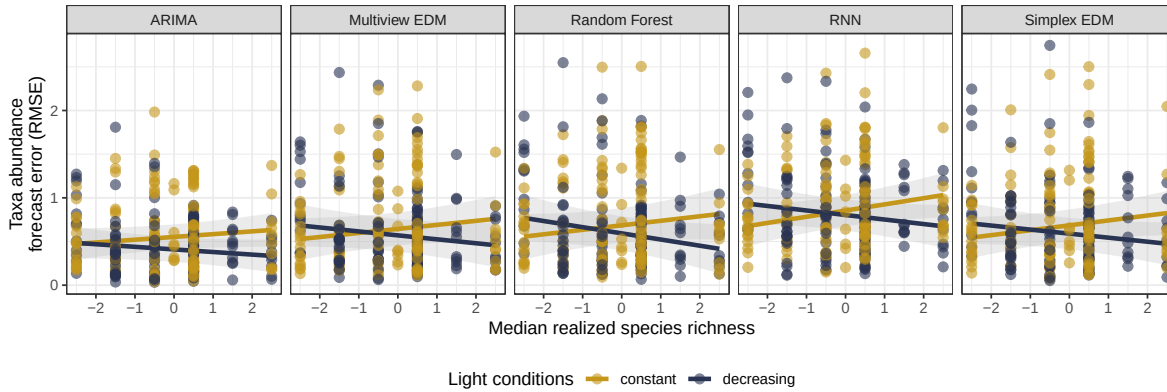


Figure 16: Effects of species richness and light on the forecast error of taxa abundances. Forecasts are for 30 bottles (yellow: constant light, blue: decreasing light) with varying number of taya per bottle (see Figure 1 in the main text). The panels show the results for five different forecasting methods, i.e. ARIMA (Auto-Regressive Integrated Moving Average), Multiview EDM (Empirical Dynamic Modeling), Random Forest, RNN (Recurrent Neural Networks), and Simplex EDM. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Composition and incubator were included as random intercepts in the linear mixed model.

```

rownames(tab.taxa) <- NULL

tab.taxa$p.value <- ifelse(tab.taxa$p.value < 0.0001, "<0.0001", sprintf("%.4f", round(tab.taxa$p.value

tab.taxa <- tab.taxa %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.taxa %>%
  dplyr::filter(Test != "\\textit{F}") %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Method",
      "Covariate",

```

```

      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of the regressions investigating the effects of species
richness and light on the forecast errors of the taxa abundances (for
different forecast methods). The estimates for the random intercepts are
standard deviations.",
    align = c("llllrrrcrr"),
    escape = F,
    label = "TaxaFullTab"
  ) %>%
collapse_rows(
  columns = c(1, 2, 4),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 4),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

rownames(em.tab.rich) <- NULL

em.tab.rich$p.value <- ifelse(em.tab.rich$p.value < 0.0001, "<0.0001",
                             sprintf("%.4f", round(em.tab.rich$p.value, 4)))

em.tab.rich <- em.tab.rich %>%
  dplyr::mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),
    df = round(df, 3),
    t.ratio = round(t.ratio, 3),
  ) %>%
  rename(richness=contrast)

em.tab.rich$contrast <- "Const. light\nvs.\nDecr. light"

em.tab.rich <- em.tab.rich %>%
  dplyr::select(contrast, richness, model, estimate, SE, df, t.ratio, p.value)

options(knitr.kable.NA = '')
em.tab.rich %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Contrast",
      "Centered median\nrichness",
      "Method",

```

Table 8: Results of the regressions investigating the effects of species richness and light on the forecast errors of the taxa abundances (for different forecast methods). The estimates for the random intercepts are standard deviations.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	<i>p</i> -value
Forecast error of taxa abundances	Simplex EDM	Intercept	Fixed effect	0.685	16	0.093	<i>t</i>	7.371	<0.0001
		Richness		0.056	22	0.033	<i>t</i>	1.695	0.1044
		Light (decreasing)		-0.095	6	0.086	<i>t</i>	-1.109	0.3092
		Richness:Light		-0.102	20	0.046	<i>t</i>	-2.246	0.0363
		Target (taxa)	Random intercept	0.074	1	-	χ^2	47.396	<0.0001
		Bottle		0.250	1	-	χ^2	0.410	0.5219
		Incubator		0.085	1	-	χ^2	1.198	0.2737
	Multiview EDM	Intercept	Fixed effect	0.644	16	0.097	<i>t</i>	6.649	<0.0001
		Richness		0.046	22	0.033	<i>t</i>	1.387	0.1792
		Light (decreasing)		-0.074	6	0.090	<i>t</i>	-0.821	0.4436
		Richness:Light		-0.091	20	0.046	<i>t</i>	-1.991	0.0602
		Target (taxa)	Random intercept	0.090	1	-	χ^2	59.107	<0.0001
		Bottle		0.260	1	-	χ^2	1.086	0.2973
		Incubator		0.091	1	-	χ^2	1.385	0.2393
	ARIMA	Intercept	Fixed effect	0.552	16	0.073	<i>t</i>	7.531	<0.0001
		Richness		0.032	24	0.024	<i>t</i>	1.335	0.1945
		Light (decreasing)		-0.143	6	0.053	<i>t</i>	-2.675	0.0365
		Richness:Light		-0.061	22	0.033	<i>t</i>	-1.872	0.0747
		Target (taxa)	Random intercept	0.055	1	-	χ^2	60.559	<0.0001
		Bottle		0.225	1	-	χ^2	0.534	0.4648
		Incubator		0.042	1	-	χ^2	0.388	0.5333
	RNN	Intercept	Fixed effect	0.854	17	0.092	<i>t</i>	9.309	<0.0001
		Richness		0.071	23	0.033	<i>t</i>	2.166	0.0409
		Light (decreasing)		-0.051	6	0.071	<i>t</i>	-0.717	0.5004
		Richness:Light		-0.123	21	0.045	<i>t</i>	-2.710	0.0131
		Target (taxa)	Random intercept	0.095	1	-	χ^2	73.686	<0.0001
		Bottle		0.275	1	-	χ^2	1.531	0.2159
		Incubator		0.049	1	-	χ^2	0.207	0.6495
	Random Forest	Intercept	Fixed effect	0.685	15	0.102	<i>t</i>	6.751	<0.0001
		Richness		0.052	22	0.039	<i>t</i>	1.311	0.2034
		Light (decreasing)		-0.092	6	0.099	<i>t</i>	-0.931	0.3880
		Richness:Light		-0.122	20	0.055	<i>t</i>	-2.242	0.0362
		Target (taxa)	Random intercept	0.136	1	-	χ^2	51.747	<0.0001
		Bottle		0.261	1	-	χ^2	3.753	0.0527
		Incubator		0.092	1	-	χ^2	0.831	0.3619

Table 9: Contrast table comparing the achieved forecast errors of taxa abundances between the two light levels (decreasing and constant) at the highest and lowest achieved (centered) median species richness. The estimate is the difference between the achieved RMSE in the decreasing light setting and the one achieved in the constant light setting.

Contrast	Centered median richness	Method	Estimate	SE	DF	<i>t</i> -value	<i>p</i> -value
Const. light vs. Decr. light	-2.5	Simplex EDM	0.161	0.137	24.225	1.180	0.2496
		Multiview EDM	0.155	0.139	22.903	1.117	0.2757
		ARIMA	0.011	0.093	27.560	0.114	0.9098
		Random Forest	0.214	0.159	23.909	1.349	0.1899
		RNN	0.256	0.126	27.791	2.030	0.0520
	2.5	Simplex EDM	-0.351	0.151	21.970	-2.333	0.0292
		Multiview EDM	-0.302	0.154	22.010	-1.961	0.0626
		ARIMA	-0.297	0.104	22.572	-2.848	0.0092
		Random Forest	-0.398	0.181	23.186	-2.205	0.0376
		RNN	-0.357	0.143	23.096	-2.497	0.0201

```

"Estimate",
"SE",
"DF",
"\\textit{t}-value",
"\\textit{p}-value"
),
caption = "Contrast table comparing the achieved forecast errors of taxa
abundances between the two light levels (decreasing and constant) at the
highest and lowest achieved (centered) median species richness. The estimate
is the difference between the achieved RMSE in the decreasing light setting
and the one achieved in the constant light setting.",
align = c("cclrrrrr"),
escape = F,
label = "emTabRich"
) %>%
collapse_rows(
  columns = c(1,2),
  latex_hline = 'custom',
  custom_latex_hline = c(1,2),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```

complete.forecasts %>%
dplyr::filter(Target %in% c("Total_biomass")) %>%
ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
geom_ribbon(
  data = newdat.bio,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
)

```

```

) +
geom_line(data = newdat.bio,
          linewidth = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
facet_grid( ~ forecast.method) +
labs(x = "Median realized species richness", y = "Forecast error of community biomass (RMSE)", col =
      "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                  aesthetics = c("color", "fill")) +
theme_bw() +
theme(legend.position = "bottom")

```

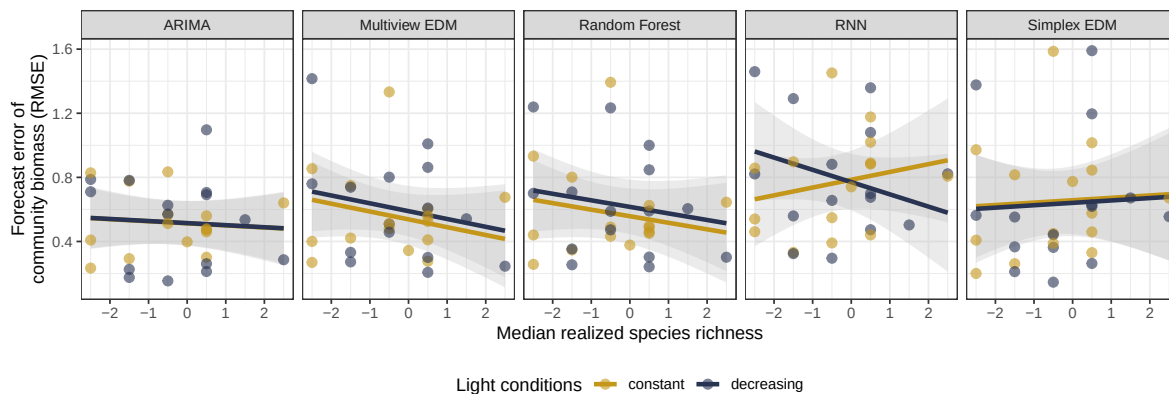


Figure 17: Effects of species richness and light on the forecast error of community biomass. Forecasts are for 30 bottles (yellow: constant light, blue: decreasing light). The panels show the results for five different forecasting methods, i.e. ARIMA (Auto-Regressive Integrated Moving Average), Multiview EDM (Empirical Dynamic Modeling), Random Forest, RNN (Recurrent Neural Networks), and Simplex EDM. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Composition and incubator were included as random intercepts in the linear mixed model.

```

complete.forecasts %>%
  dplyr::filter(Target %in% c("Mean_oxygen")) %>%
  ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.oxy,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.oxy,
            linewidth = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +

```

```

facet_grid( ~ forecast.method) +
labs(x = "Median realized species richness", y = "Forecast error of\noxygen concentration (RMSE)", col =
      "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                   aesthetics = c("color", "fill")) +

theme_bw() +
theme(legend.position = "bottom")

```

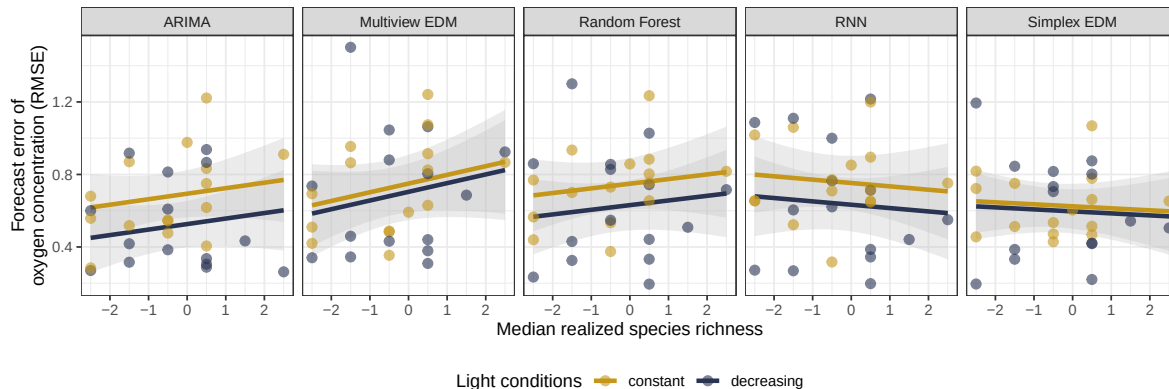


Figure 18: Effects of species richness and light on the forecast error of oxygen concentration. Forecasts are for 30 bottles (yellow: constant light, blue: decreasing light). The panels show the results for five different forecasting methods, i.e. ARIMA (Auto-Regressive Integrated Moving Average), Multiview EDM (Empirical Dynamic Modeling), Random Forest, RNN (Recurrent Neural Networks), and Simplex EDM. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Composition and incubator were included as random intercepts in the linear mixed model.

```

rownames(tab.bio) <- NULL

tab.bio$p.value <- ifelse(tab.bio$p.value < 0.0001, "<0.0001", sprintf("%.4f", round(tab.bio$p.value, 4)))

tab.bio <- tab.bio %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.bio %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Method",
      "Covariate",
      "Type",

```

Table 10: Results of the regressions investigating the effects of species richness and light on the forecast errors of the community biomass (for different forecast methods). The estimates for the random intercepts are standard deviations.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Forecast error of community biomass	Simplex EDM	Intercept	Fixed effect	0.657	20	0.108	-	-	-
		Richness		0.015	20	0.062	<i>F</i>	0.058	0.8118
		Light (decreasing)		-0.017	14	0.099	<i>F</i>	0.028	0.8698
		Composition	Random intercept	0.302	1	-	χ^2	5.108	0.0238
		Incubator		0.001	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.538	8	0.082	-	-	-
		Richness		-0.049	15	0.045	<i>F</i>	1.191	0.2921
		Light (decreasing)		0.051	3	0.094	<i>F</i>	0.298	0.6197
		Composition	Random intercept	0.170	1	-	χ^2	1.418	0.2337
		Incubator		0.027	1	-	χ^2	0.003	0.9574
	ARIMA	Intercept	Fixed effect	0.513	6	0.071	-	-	-
		Richness		-0.013	22	0.032	<i>F</i>	0.165	0.6883
		Light (decreasing)		0.001	5	0.099	<i>F</i>	0.000	0.9890
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.066	1	-	χ^2	0.144	0.7047
	RNN	Intercept	Fixed effect	0.785	19	0.093	-	-	-
		Richness		0.049	25	0.059	<i>F</i>	0.071	0.7924
		Light (decreasing)		-0.015	13	0.079	<i>F</i>	0.035	0.8554
		Richness:Light		-0.125	13	0.056	<i>F</i>	4.963	0.0442
		Composition	Random intercept	0.274	1	-	χ^2	6.706	0.0096
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.558	8	0.088	-	-	-
		Richness		-0.041	15	0.045	<i>F</i>	0.821	0.3792
		Light (decreasing)		0.058	4	0.106	<i>F</i>	0.305	0.6106
		Composition	Random intercept	0.166	1	-	χ^2	1.182	0.2769
		Incubator		0.068	1	-	χ^2	0.094	0.7594

```

"Estimate",
"DF",
"SE",
"Test",
"Value",
"\\textit{p}-value"
),
caption = "Results of the regressions investigating the effects of species
richness and light on the forecast errors of the community biomass (for
different forecast methods). The estimates for the random intercepts are
standard deviations.",
align = c("llllrrrrcrr"),
escape = F,
label = "BioTab"
) %>%
collapse_rows(
  columns = c(1, 2, 4),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 4),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

```

rownames(tab.oxy) <- NULL

tab.oxy$p.value <- ifelse(tab.oxy$p.value < 0.0001, "<0.0001",
                          sprintf("%.4f", round(tab.oxy$p.value, 4)))

tab.oxy <- tab.oxy %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.oxy %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Method",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of the regressions investigating the effects of species
richness and light on the forecast errors of mean oxygen (for different
forecast methods). The estimates for the random intercepts are standard deviations.",
    align = c("llllrrrcrr"),
    escape = F,
    label = "OxyTab"
  ) %>%
  collapse_rows(
    columns = c(1, 2, 4),
    latex_hline = 'custom',
    custom_latex_hline = c(1, 4),
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

rownames(tab.aggr.bio) <- NULL

tab.aggr.bio$p.value <- ifelse(tab.aggr.bio$p.value < 0.0001,
                              "<0.0001",
                              sprintf("%.4f", round(tab.aggr.bio$p.value, 4)))

tab.aggr.bio <- tab.aggr.bio %>%

```

Table 11: Results of the regressions investigating the effects of species richness and light on the forecast errors of mean oxygen (for different forecast methods). The estimates for the random intercepts are standard deviations.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	<i>p</i> -value
Forecast error of mean oxygen	Simplex EDM	Intercept	Fixed effect	0.624	27	0.063	-	-	-
		Richness		-0.011	27	0.032	<i>F</i>	0.121	0.7304
		Light (decreasing)		-0.028	27	0.087	<i>F</i>	0.103	0.7502
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.750	27	0.081	-	-	-
		Richness		0.048	15	0.042	<i>F</i>	1.334	0.2659
		Light (decreasing)		-0.046	14	0.108	<i>F</i>	0.178	0.6799
		Composition	Random intercept	0.075	1	-	χ^2	0.050	0.8237
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.694	27	0.067	-	-	-
		Richness		0.030	27	0.033	<i>F</i>	0.834	0.3692
		Light (decreasing)		-0.168	27	0.092	<i>F</i>	3.342	0.0786
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	RNN	Intercept	Fixed effect	0.753	27	0.076	-	-	-
		Richness		-0.018	27	0.038	<i>F</i>	0.234	0.6322
		Light (decreasing)		-0.120	27	0.105	<i>F</i>	1.300	0.2643
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.749	27	0.072	-	-	-
		Richness		0.026	27	0.036	<i>F</i>	0.506	0.4829
		Light (decreasing)		-0.118	27	0.099	<i>F</i>	1.414	0.2448
		Composition	Random intercept	0.000	1	-	χ^2	0.000	1.0000
		Incubator		0.000	1	-	χ^2	0.000	1.0000

```

dplyr::mutate(
  Estimate = round(Estimate, 3),
  SE = round(SE, 3),
  Test.value = round(Test.value, 3)
)

options(knitr.kable.NA = '-')
tab.aggr.bio %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Method",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of the regressions investigating whether the community
level measure total biomass is forecasted better than taxa based forecasts
(here: mean forecast error across all forecasted taxa). As explanatory
variables, the model included median centered richness, light treatment
and measure (i.e. total biomass forecasts versus taxa abundance forecasts),
as well as all pairwise interactions. The table reports the results for all
considered forecast methods. The estimates for the random intercepts are
standard deviations.",
    align = c("llllrrrcrr"),
    escape = F,
    label = "AggregateTabBio"
  ) %>%
  collapse_rows(
    columns = c(1, 2, 4),
    latex_hline = 'custom',
    custom_latex_hline = c(1, 4),
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

rownames(tab.aggr.oxy) <- NULL

tab.aggr.oxy$p.value <- ifelse(tab.aggr.oxy$p.value < 0.0001,
                              "<0.0001",
                              sprintf("%.4f", round(tab.aggr.oxy$p.value, 4)))

tab.aggr.oxy <- tab.aggr.oxy %>%
  dplyr::mutate(

```

Table 12: Results of the regressions investigating whether the community level measure total biomass is forecasted better than taxa based forecasts (here: mean forecast error across all forecasted taxa). As explanatory variables, the model included median centered richness, light treatment and measure (i.e. total biomass forecasts versus taxa abundance forecasts), as well as all pairwise interactions. The table reports the results for all considered forecast methods. The estimates for the random intercepts are standard deviations.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Mean forecast error of abundances Versus Forecast error of community biomass	Simplex EDM	Intercept	Fixed effect	0.715	42	0.075	-	-	-
		Richness		-0.001	27	0.033	F	0.000	0.9868
		Light (decreasing)		-0.032	27	0.090	F	0.130	0.7214
		Measure (biomass)		-0.056	29	0.072	F	0.607	0.4422
		Bottle	Random intercept	0.148	1	-	χ^2	1.322	0.2503
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.669	34	0.062	-	-	-
		Richness		0.013	26	0.040	F	1.724	0.2006
		Light (decreasing)		-0.029	26	0.080	F	0.129	0.7228
		Measure (biomass)		-0.082	29	0.045	F	3.293	0.0799
		Richness:Light		-0.100	26	0.056	F	3.210	0.0848
		Bottle	Random intercept	0.171	1	-	χ^2	7.074	0.0078
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.549	9	0.054	-	-	-
		Richness		-0.015	22	0.021	F	0.487	0.4925
		Light (decreasing)		-0.057	6	0.069	F	0.684	0.4404
		Measure (biomass)		-0.007	29	0.041	F	0.031	0.8623
		Bottle	Random intercept	0.115	1	-	χ^2	3.311	0.0688
		Incubator		0.051	1	-	χ^2	0.261	0.6095
	RNN	Intercept	Fixed effect	0.894	39	0.065	-	-	-
		Richness		0.053	26	0.040	F	0.111	0.7421
		Light (decreasing)		-0.030	26	0.080	F	0.137	0.7142
		Measure (biomass)		-0.099	29	0.059	F	2.831	0.1032
		Richness:Light		-0.124	26	0.056	F	4.888	0.0360
		Bottle	Random intercept	0.137	1	-	χ^2	1.829	0.1763
		Incubator		0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.692	36	0.066	-	-	-
		Richness		-0.038	27	0.030	F	1.562	0.2221
		Light (decreasing)		0.011	27	0.084	F	0.016	0.8997
		Measure (biomass)		-0.109	29	0.051	F	4.561	0.0413
		Bottle	Random intercept	0.181	1	-	χ^2	6.084	0.0136
		Incubator		0.000	1	-	χ^2	0.000	1.0000

```

    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
tab.aggr.oxy %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Method",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = "Results of the regressions investigating whether the community
level measure mean oxygen concentration is forecasted better than taxa based
forecasts (here: mean forecast error across all forecasted taxa). As
explanatory variables, the model included median centered richness, light
treatment and measure (i.e. oxygen concentration forecasts versus taxa
abundance forecasts), as well as all pairwise interactions. The table
reports the results for all considered forecast methods. The estimates
for the random intercepts are standard deviations.",
    align = c("llllrrrrcrr"),
    escape = F,
    label = "AggregateTabOxy"
  ) %>%
  collapse_rows(
    columns = c(1, 2, 4),
    latex_hline = 'custom',
    custom_latex_hline = c(1, 4),
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

rownames(em.tab) <- NULL

em.tab$p.value <- ifelse(em.tab$p.value < 0.0001, "<0.0001",
  sprintf("%.4f", round(em.tab$p.value, 4)))

em.tab <- em.tab %>%
  dplyr::mutate(
    estimate = round(estimate, 3),
    SE = round(SE, 3),

```

Table 13: Results of the regressions investigating whether the community level measure mean oxygen concentration is forecasted better than taxa based forecasts (here: mean forecast error across all forecasted taxa). As explanatory variables, the model included median centered richness, light treatment and measure (i.e. oxygen concentration forecasts versus taxa abundance forecasts), as well as all pairwise interactions. The table reports the results for all considered forecast methods. The estimates for the random intercepts are standard deviations.

Response	Method	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Mean forecast error of abundances Versus Forecast error of mean oxygen concentration	Simplex EDM	Intercept	Fixed effect	0.711	39	0.052	-	-	-
		Richness		-0.019	27	0.024	<i>F</i>	0.639	0.4309
		Light (decreasing)		-0.037	27	0.065	<i>F</i>	0.331	0.5701
		Measure (biomass)		-0.085	29	0.046	<i>F</i>	3.415	0.0748
		Bottle Incubator	Random intercept	0.125	1	-	χ^2	3.295	0.0695
				0.000	1	-	χ^2	0.000	1.0000
	Multiview EDM	Intercept	Fixed effect	0.673	42	0.061	-	-	-
		Richness		-0.027	51	0.034	<i>F</i>	0.160	0.6926
		Light (decreasing)		-0.039	27	0.073	<i>F</i>	0.279	0.6014
		Measure (biomass)		0.074	28	0.058	<i>F</i>	1.582	0.2189
		Richness:Measure		0.076	28	0.041	<i>F</i>	3.454	0.0736
		Bottle Incubator	Random intercept	0.127	1	-	χ^2	1.824	0.1769
				0.000	1	-	χ^2	0.000	1.0000
	ARIMA	Intercept	Fixed effect	0.620	42	0.047	-	-	-
		Richness		0.050	26	0.028	<i>F</i>	0.147	0.7050
		Light (decreasing)		-0.173	26	0.056	<i>F</i>	9.514	0.0048
		Measure (biomass)		0.072	29	0.047	<i>F</i>	2.341	0.1369
		Richness:Light		-0.085	26	0.039	<i>F</i>	4.639	0.0407
		Bottle Incubator	Random intercept	0.074	1	-	χ^2	0.559	0.4545
				0.000	1	-	χ^2	0.000	1.0000
	RNN	Intercept	Fixed effect	0.918	42	0.057	-	-	-
		Richness		0.023	26	0.034	<i>F</i>	0.940	0.3411
		Light (decreasing)		-0.092	26	0.068	<i>F</i>	1.812	0.1898
		Measure (biomass)		-0.177	29	0.056	<i>F</i>	9.921	0.0038
		Richness:Light		-0.092	26	0.048	<i>F</i>	3.701	0.0654
		Bottle Incubator	Random intercept	0.093	1	-	χ^2	0.669	0.4134
				0.000	1	-	χ^2	0.000	1.0000
	Random Forest	Intercept	Fixed effect	0.771	41	0.058	-	-	-
		Richness		0.042	26	0.035	<i>F</i>	0.080	0.7792
		Light (decreasing)		-0.116	26	0.071	<i>F</i>	2.695	0.1127
		Measure (biomass)		-0.032	29	0.056	<i>F</i>	0.323	0.5740
		Richness:Light		-0.099	26	0.049	<i>F</i>	3.977	0.0567
		Bottle Incubator	Random intercept	0.104	1	-	χ^2	0.961	0.3269
				0.000	1	-	χ^2	0.000	1.0000

```

    t.ratio = round(t.ratio, 3)
  )

options(knitr.kable.NA = '-')
em.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Contrast",
      "Method",
      "Light",
      "Estimate",
      "SE",
      "DF",
      "\\textit{t}-Value",
      "\\textit{p}-value"
    ),
    caption = "Table reporting the differences (i.e. the contrasts) between the
marginal estimated mean of the forecast errors of the community level
variables (total biomass and mean oxygen concentration) and of the mean taxa
abundance forecast error. Also reported are the results of the tests of
whether the estimated differences are different from zero.",
    align = c("lllrrrrcrr"),
    escape = F,
    label = "ContrastTab"
  ) %>%
  collapse_rows(
    columns = c(1, 2),
    latex_hline = 'custom',
    custom_latex_hline = c(1, 2),
    valign = "middle"
  ) %>%
  kable_styling(font_size = 7)

```

7.2 Relations of time series properties with forecast errors and with experimental conditions

```

simplex.forecasts.taxa <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass", Target != "Mean_oxygen") %>%
  dplyr::select(
    "bottle",
    "richness",
    "composition",
    "incubator",
    "realized_richness",
    "light_treatment",
    "Target",
    "RMSE",
    "forecast.method",

```

Table 14: Table reporting the differences (i.e. the contrasts) between the marginal estimated mean of the forecast errors of the community level variables (total biomass and mean oxygen concentration) and of the mean taxa abundance forecast error. Also reported are the results of the tests of whether the estimated differences are different from zero.

Contrast	Method	Light	Estimate	SE	DF	t-Value	p-value
Species - Biomass	Simplex EDM	constant	0.056	0.072	29	0.779	0.4422
		decreasing	0.056	0.072	29	0.779	0.4422
	Multiview EDM	constant	0.082	0.045	29	1.815	0.0799
		decreasing	0.082	0.045	29	1.815	0.0799
	ARIMA	constant	0.007	0.041	29	0.175	0.8623
		decreasing	0.007	0.041	29	0.175	0.8623
	RNN	constant	0.099	0.059	29	1.683	0.1032
		decreasing	0.099	0.059	29	1.683	0.1032
	Random Forest	constant	0.109	0.051	29	2.136	0.0413
		decreasing	0.109	0.051	29	2.136	0.0413
Species - Oxygen	Simplex EDM	constant	0.085	0.046	29	1.848	0.0748
		decreasing	0.085	0.046	29	1.848	0.0748
	Multiview EDM	constant	-0.044	0.056	28	-0.788	0.4374
		decreasing	-0.044	0.056	28	-0.788	0.4374
	ARIMA	constant	-0.072	0.047	29	-1.530	0.1369
		decreasing	-0.072	0.047	29	-1.530	0.1369
	RNN	constant	0.177	0.056	29	3.150	0.0038
		decreasing	0.177	0.056	29	3.150	0.0038
	Random Forest	constant	0.032	0.056	29	0.569	0.5740
		decreasing	0.032	0.056	29	0.569	0.5740

```

"mean_richness",
"median_richness",
"realized_richness_center",
"median_richness_center",
"mean_richness_center",
"pe5",
"cv.res",
"acf_lag5"
) %>%
dplyr::filter(!(Target %in% c("Total_biomass", "Mean_oxygen")), forecast.method ==
               "Simplex EDM") %>%
mutate(cv.res.log10 = log10(cv.res))

## acf vs RMSE

m.acf.rmse <- lmer(
  RMSE ~ acf_lag5 * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.acf.rmse.no.int <- lmer(
  RMSE ~ acf_lag5 + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)

## CV vs RMSE

```

```

m.cv.rmse <- lmer(
  RMSE ~ cv.res.log10 * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.cv.rmse.no.int <- lmer(
  RMSE ~ cv.res.log10 + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
### pe vs rmse

m.pe.rmse <- lmer(
  RMSE ~ pe5 * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.pe.rmse.no.int <- lmer(
  RMSE ~ pe5 + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)

# tab

mods <- list(m.acf.rmse.no.int, m.cv.rmse.no.int, m.pe.rmse.no.int)
CoefNams <- c(
  "Autocorrelation\n(lag = 5)",
  "log10-transformed\nCoefficient of\nvariation)",
  "Permutation\nentropy"
)
CoefNamsShort <- c("ACF (lag 5)", "log10(CV)", "PE")
InteractionNams <- c("ACF:Light", "log10(CV):Light", "PE:Light")
CvAcfPeRmseTab <- data.frame()

for (i in 1:length(mods)) {
  m <- mods[[i]]
  r <- rand(m)

  t1 <- data.frame(
    MainCovariate = CoefNams[i],
    Covariate = c(
      "Intercept",
      CoefNamsShort[i],
      "Light (decreasing)",
      "Target (taxa)",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 3),
      "Random\nintercept",
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:3]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:4]),
  )
}

```

```

SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA, NA),
Test = c(rep("\\textit{t}", 3), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:4]),
p.value = c(summary(m)$coef[, 5, drop = FALSE], r`Pr(>Chisq)`[2:4])
)

t2 <- data.frame(
  MainCovariate = CoefNams[i],
  Covariate = c(CoefNamsShort[i], "Light (decreasing)"),
  Type = "Sum of\\nsquares",
  Estimate = anova(m)[, 1],
  DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
  SE = NA,
  Test = "\\textit{F}",
  Test.value = anova(m)[, 5],
  p.value = anova(m)[, 6]
)

CvAcfPeRmseTab <- rbind(CvAcfPeRmseTab, t1, t2)
}

rownames(CvAcfPeRmseTab) <- NULL

CvAcfPeRmseTab$p.value <- ifelse(CvAcfPeRmseTab$p.value < 0.0001,
                                "<0.0001",
                                sprintf("%.4f", round(CvAcfPeRmseTab$p.value, 4)))

CvAcfPeRmseTab <- CvAcfPeRmseTab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
CvAcfPeRmseTab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Main covariate",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",

```

```

      "\\textit{p}-value"
    ),
    caption = 'Results of linear mixed models investigating the relationship
between forecast error (response variable) and time series metrics. The
models were fitted separately for the explanatory variables "autocorrelation"
(at lag 5), "log10-transformed coefficient of variation", and "permutation
entropy". This table includes both the regression tables and the analysis
of variance (Type II) tables of the models. The models included the respective
time series metrics as a covariate, alongside the light conditions and their
interaction as fixed effects. The random intercepts were included for the
bottles, incubators and the targets (i.e. forecasted taxa). Forecast error
is based on forecasts computed with Simplex EDM. The estimates for the random
intercepts are standard deviations.',
    align = c("lllrrrrcrr"),
    escape = F,
    label = "CvAcfPeRmseTab"
  ) %>%
collapse_rows(
  columns = c(1, 3),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 3),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

## acf vs exp

m.acf.exp <- lmer(
  acf_lag5 ~ median_richness_center * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.acf.exp.no.int <- lmer(
  acf_lag5 ~ median_richness_center + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)

## cv vs exp

m.cv.exp <- lmer(
  cv.res.log10 ~ median_richness_center * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.cv.exp.no.int <- lmer(
  cv.res.log10 ~ median_richness_center + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)

### pe vs rmse

m.pe.exp <- lmer(
  pe5 ~ median_richness_center * light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
  simplex.forecasts.taxa
)
m.pe.exp.no.int <- lmer(

```

Table 15: Results of linear mixed models investigating the relationship between forecast error (response variable) and time series metrics. The models were fitted separately for the explanatory variables "auto-correlation" (at lag 5), "log10-transformed coefficient of variation", and "permutation entropy". This table includes both the regression tables and the analysis of variance (Type II) tables of the models. The models included the respective time series metrics as a covariate, alongside the light conditions and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. forecasted taxa). Forecast error is based on forecasts computed with Simplex EDM. The estimates for the random intercepts are standard deviations.

Main covariate	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Autocorrelation (lag = 5)	Intercept	Fixed effect	0.972	26	0.101	t	9.650	<0.0001
	ACF (lag 5)		-0.690	249	0.111	t	-6.239	<0.0001
	Light (decreasing)		0.009	7	0.069	t	0.126	0.9038
	Target (taxa)	Random intercept	0.000	1	-	χ^2	65.009	<0.0001
	Bottle		0.269	1	-	χ^2	0.000	1.0000
	Incubator		0.061	1	-	χ^2	1.035	0.3089
	ACF (lag 5)	Sum of squares	6.399	1, 248.5	-	F	38.923	<0.0001
	Light (decreasing)		0.003	1, 6.5	-	F	0.016	0.9038
log10-transformed Coefficient of variation)	Intercept	Fixed effect	0.439	25	0.110	t	3.977	0.0005
	log10(CV)		-0.519	227	0.117	t	-4.424	<0.0001
	Light (decreasing)		-0.001	6	0.094	t	-0.014	0.9893
	Target (taxa)	Random intercept	0.083	1	-	χ^2	50.055	<0.0001
	Bottle		0.257	1	-	χ^2	0.740	0.3896
	Incubator		0.099	1	-	χ^2	2.075	0.1497
	log10(CV)	Sum of squares	3.313	1, 227.3	-	F	19.573	<0.0001
	Light (decreasing)		0.000	1, 6.3	-	F	0.000	0.9893
Permutation entropy	Intercept	Fixed effect	-0.079	147	0.219	t	-0.360	0.7193
	PE		1.162	245	0.307	t	3.789	0.0002
	Light (decreasing)		-0.029	6	0.101	t	-0.288	0.7829
	Target (taxa)	Random intercept	0.134	1	-	χ^2	47.132	<0.0001
	Bottle		0.234	1	-	χ^2	3.509	0.0610
	Incubator		0.100	1	-	χ^2	1.281	0.2577
	PE	Sum of squares	2.423	1, 245.2	-	F	14.357	0.0002
	Light (decreasing)		0.014	1, 6	-	F	0.083	0.7829

```

    pe5 ~ median_richness_center + light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator),
    simplex.forecasts.taxa
)

# tab

mods <- list(m.acf.exp.no.int, m.cv.exp.no.int, m.pe.exp.no.int)
CoefNams <- c(
  "Autocorrelation\n(lag = 5)",
  "log10-transformed\nCoefficient of\nvariation)",
  "Permutation\nentropy"
)

EXPCvAcfPeTab <- data.frame()

for (i in 1:length(mods)) {
  m <- mods[[i]]
  r <- rand(m)

  t1 <- data.frame(
    Response = CoefNams[i],
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Target (taxa)",
      "Bottle",
      "Incubator"
    ),
    Type = c(
      rep("Fixed\neffect", 3),
      "Random\nintercept",
      "Random\nintercept",
      "Random\nintercept"
    ),
    Estimate = c(fixef(m), as.data.frame(VarCorr(m))$sdcor[1:3]),
    DF = c(round(summary(m)$coef[, 3, drop = FALSE], 0), r$Df[2:4]),
    SE = c(summary(m)$coef[, 2, drop = FALSE], NA, NA, NA),
    Test = c(rep("\\textit{t}", 3), "\\chi^2$", "\\chi^2$", "\\chi^2$"),
    Test.value = c(summary(m)$coef[, 4, drop = FALSE], r$LRT[2:4]),
    p.value = c(summary(m)$coef[, 5, drop = FALSE], r$`Pr(>Chisq)`[2:4])
  )

  t2 <- data.frame(
    Response = CoefNams[i],
    Covariate = c("Richness", "Light (decreasing)"),
    Type = "Sum of\nsquares",
    Estimate = anova(m)[, 1],
    DF = paste(anova(m)[, 3], round(anova(m)[, 4], 1), sep = ", "),
    SE = NA,
    Test = "\\textit{F}",
    Test.value = anova(m)[, 5],
    p.value = anova(m)[, 6]
  )
}

```

```

)

EXPCvAcfPeTab <- rbind(EXPCvAcfPeTab, t1, t2)
}

rownames(EXPCvAcfPeTab) <- NULL

EXPCvAcfPeTab$p.value <- ifelse(EXPCvAcfPeTab$p.value < 0.0001,
                                "<0.0001",
                                sprintf("%.4f", round(EXPCvAcfPeTab$p.value, 4)))

EXPCvAcfPeTab <- EXPCvAcfPeTab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SE = round(SE, 3),
    Test.value = round(Test.value, 3)
  )

options(knitr.kable.NA = '-')
EXPCvAcfPeTab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Type",
      "Estimate",
      "DF",
      "SE",
      "Test",
      "Value",
      "\\textit{p}-value"
    ),
    caption = 'Results of linear mixed models investigating the the effect of
species richness and light on time series metrics (response variables). The
models were fitted separately for the response variables "autocorrelation"
(at lag 5), "log10-transformed coefficient of variation", and "permutation
entropy". This table includes both the regression tables and the analysis of
variance (Type II) tables of the models. The models included median centered
richness, light treatment and their interaction as fixed effects. The random
intercepts were included for the bottles, incubators and the targets (i.e.
forecasted taxa). Forecast error is based on forecasts computed with Simplex
EDM. The estimates for the random intercepts are standard deviations.',
    align = c("lllrrrrcrr"),
    escape = F,
    label = "EXPCvAcfPeTab"
  ) %>%

```

Table 16: Results of linear mixed models investigating the the effect of species richness and light on time series metrics (response variables). The models were fitted separately for the response variables "autocorrelation" (at lag 5), "log10-transformed coefficient of variation", and "permutation entropy". This table includes both the regression tables and the analysis of variance (Type II) tables of the models. The models included median centered richness, light treatment and their interaction as fixed effects. The random intercepts were included for the bottles, incubators and the targets (i.e. forecasted taxa). Forecast error is based on forecasts computed with Simplex EDM. The estimates for the random intercepts are standard deviations.

Response	Covariate	Type	Estimate	DF	SE	Test	Value	p-value
Autocorrelation (lag = 5)	Intercept	Fixed effect	0.431	16	0.054	<i>t</i>	8.017	<0.0001
	Richness		0.004	22	0.016	<i>t</i>	0.281	0.7811
	Light (decreasing)		0.127	5	0.046	<i>t</i>	2.769	0.0363
	Target (taxa)	Random intercept	0.087	1	-	χ^2	54.562	<0.0001
	Bottle		0.152	1	-	χ^2	7.027	0.0080
	Incubator		0.025	1	-	χ^2	0.050	0.8228
	Richness	Sum of squares	0.004	1, 21.8	-	<i>F</i>	0.079	0.7811
	Light (decreasing)		0.367	1, 5.4	-	<i>F</i>	7.669	0.0363
log10-transformed Coefficient of variation)	Intercept	Fixed effect	-0.429	12	0.064	<i>t</i>	-6.723	<0.0001
	Richness		-0.007	239	0.011	<i>t</i>	-0.680	0.4972
	Light (decreasing)		0.133	236	0.027	<i>t</i>	4.841	<0.0001
	Target (taxa)	Random intercept	0.000	1	-	χ^2	90.879	<0.0001
	Bottle		0.221	1	-	χ^2	0.000	1.0000
	Incubator		0.000	1	-	χ^2	0.000	1.0000
	Richness	Sum of squares	0.022	1, 238.8	-	<i>F</i>	0.462	0.4972
	Light (decreasing)		1.107	1, 236	-	<i>F</i>	23.440	<0.0001
Permutation entropy	Intercept	Fixed effect	0.642	14	0.023	<i>t</i>	27.923	<0.0001
	Richness		0.016	238	0.004	<i>t</i>	3.782	0.0002
	Light (decreasing)		-0.033	6	0.019	<i>t</i>	-1.756	0.1290
	Target (taxa)	Random intercept	0.000	1	-	χ^2	73.226	<0.0001
	Bottle		0.068	1	-	χ^2	0.000	1.0000
	Incubator		0.022	1	-	χ^2	3.945	0.0470
	Richness	Sum of squares	0.100	1, 238.1	-	<i>F</i>	14.302	0.0002
	Light (decreasing)		0.022	1, 6.1	-	<i>F</i>	3.082	0.1290

```
collapse_rows(
  columns = c(1, 3),
  latex_hline = 'custom',
  custom_latex_hline = c(1, 3),
  valign = "middle"
) %>%
kable_styling(font_size = 7)
```

7.3 Structural equation model

```
semDD <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass", Target != "Mean_oxygen") %>%
  ungroup() %>%
  mutate(light_treatment_num = ifelse(light_treatment == "constant", 0, 1))

semDD <- data.frame(semDD)

## create random effect variable
semDD$group <- paste0(semDD$Target, "_",
  semDD$bottle, "_",
```

```

semDD$incubator)

semDD.simplex <- semDD %>%
  dplyr::filter(forecast.method == "Simplex EDM")

semDD.rf <- semDD %>%
  dplyr::filter(forecast.method == "Random Forest")

semDD.mv <- semDD %>%
  dplyr::filter(forecast.method == "Multiview EDM")

semDD.rnn <- semDD %>%
  dplyr::filter(forecast.method == "RNN")

semDD.ar <- semDD %>%
  dplyr::filter(forecast.method == "ARIMA")

model <-
  "
  RMSE ~ median_richness + cv.res + pe5 + acf_lag5
  cv.res ~ median_richness
  pe5 ~ median_richness
  acf_lag5 ~ median_richness
  cv.res ~~ pe5
  "

# semDD.simplex, semDD.rf, semDD.mv, semDD.rnn, semDD.ar
out <- sem(model, data = semDD.simplex, group = "light_treatment")

### Survey Design Object
des.card <- svydesign(ids = ~ group,
  nest = T,
  data = semDD.simplex)

### Post-Processing
fit.out.survey <- lavaan.survey(out, des.card, estimator = "MLM")

SEMsum <- summary(fit.out.survey, standardized = T, rsquare = T)

SEMtab <- SEMsum$pe %>%
  dplyr::filter(rhs != "") %>%
  dplyr::mutate(
    Light = ifelse(group == 1, "Constant light", "Decreasing light"),
    lhs = case_when(
      lhs == "RMSE" ~ "Forecast error",
      lhs == "cv.res" ~ "Coefficient of variation",
      lhs == "pe5" ~ "Permutation entropy",
      lhs == "acf_lag5" ~ "Autocorrelation (lag 5)",
      lhs == "median_richness" ~ "Species richness"
    ),
    rhs = case_when(
      rhs == "RMSE" ~ "Forecast error",
      rhs == "cv.res" ~ "Coefficient of variation",

```

```

    rhs == "pe5" ~ "Permutation entropy",
    rhs == "acf_lag5" ~ "Autocorrelation (lag 5)",
    rhs == "median_richness" ~ "Species richness"
  )
)

SEMtabR2 <- SEMtab %>%
  dplyr::filter(op == "r2") %>%
  dplyr::select(Light, Variable = lhs, est)

SEMtab <- SEMtab %>%
  dplyr::filter(op != "r2",
    lhs != rhs) %>%
  dplyr::mutate(op = ifelse(op == "~", "$\\sim$", "$\\sim$$\\sim$")) %>%
  dplyr::mutate(term = paste0(lhs, op, rhs)) %>%
  dplyr::select(Light, term, est, se, z, pvalue, std.all)

rownames(SEMtab) <- NULL

SEMtab$pvalue <- ifelse(SEMtab$pvalue < 0.0001, "<0.0001",
  sprintf("%.4f", round(SEMtab$pvalue, 4)))

SEMtab <- SEMtab %>%
  dplyr::mutate(
    est = round(est, 3),
    se = round(se, 3),
    z = round(z, 3),
    std.all = round(std.all, 3)
  )

options(knitr.kable.NA = '-')

SEMtab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Light conditions",
      "Term",
      "Estimate",
      "SE",
      "\\textit{z}-value",
      "\\textit{p}-value",
      "Stand. est."
    ),
    caption = 'Table of path coefficients of the structural equation model.
    The last column reports the standardized coefficients.',
    align = c("llrrrrr"),
    escape = F,
    label = "SEMtab"
  ) %>%
  collapse_rows(

```

Table 17: Table of path coefficients of the structural equation model. The last column reports the standardized coefficients.

Light conditions	Term	Estimate	SE	z-value	p-value	Stand. est.
Constant light	Forecast error~Species richness	0.015	0.031	0.469	0.6391	0.035
	Forecast error~Coefficient of variation	-0.602	0.177	-3.412	0.0006	-0.416
	Forecast error~Permutation entropy	0.085	0.659	0.128	0.8978	0.017
	Forecast error~Autocorrelation (lag 5)	-0.695	0.112	-6.230	<0.0001	-0.368
	Coefficient of variation~Species richness	0.005	0.022	0.243	0.8084	0.018
	Permutation entropy~Species richness	0.010	0.007	1.459	0.1446	0.117
	Autocorrelation (lag 5)~Species richness	-0.018	0.019	-0.948	0.3429	-0.081
Decreasing light	Coefficient of variation~Permutation entropy	-0.032	0.008	-4.110	<0.0001	-0.784
	Forecast error~Species richness	-0.077	0.035	-2.224	0.0261	-0.220
	Forecast error~Coefficient of variation	-0.095	0.173	-0.548	0.5834	-0.072
	Forecast error~Permutation entropy	0.430	0.569	0.756	0.4495	0.092
	Forecast error~Autocorrelation (lag 5)	-0.350	0.143	-2.455	0.0141	-0.185
	Coefficient of variation~Species richness	-0.030	0.026	-1.150	0.2500	-0.111
	Permutation entropy~Species richness	0.014	0.007	2.186	0.0288	0.191
	Autocorrelation (lag 5)~Species richness	0.010	0.017	0.592	0.5536	0.053
	Coefficient of variation~Permutation entropy	-0.022	0.006	-3.968	<0.0001	-0.628

```

columns = 1,
latex_hline = 'custom',
custom_latex_hline = 1,
valign = "middle"
) %>%
kable_styling(font_size = 7)

rownames(SEMtabR2) <- NULL

SEMtabR2 <- SEMtabR2 %>%
  dplyr::mutate(est = round(est, 4))

options(knitr.kable.NA = '-')

SEMtabR2 %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 4,
    col.names = c("Light conditions", "Variable", "R-squared"),
    caption = 'Achieved coefficients of determination (R-squared)
for the structural equation model.',
    align = c("llr"),
    escape = F,
    label = "SEMtabR2"
) %>%
collapse_rows(
  columns = 1,
  latex_hline = 'custom',
  custom_latex_hline = 1,
  valign = "middle"
) %>%
kable_styling(font_size = 8)

```

Table 18: Achieved coefficients of determination (R-squared) for the structural equation model.

Light conditions	Variable	R-squared
Constant light	Forecast error	0.3222
	Coefficient of variation	0.0003
	Permutation entropy	0.0137
	Autocorrelation (lag 5)	0.0066
Decreasing light	Forecast error	0.0972
	Coefficient of variation	0.0124
	Permutation entropy	0.0365
	Autocorrelation (lag 5)	0.0028

8 Robustness analyses

8.1 Effects of chosen richness variable and forecast method

```

formula <-
  as.formula(
    paste0(
      "RMSE ~ ",
      "richness_var",
      "*light_treatment + (1|Target) + (1|bottle) + (1|incubator)"
    )
  )
rich_vars2 <-
  list(
    "realized_richness_center",
    "median_richness_center",
    "mean_richness_center",
    "richness"
  )
fore.methods <-
  list("Multiview EDM", "Simplex EDM", "RNN", "ARIMA", "Random Forest")

tab <-
  robustnessFunction(rich_vars2, fore.methods, complete.forecasts,
    formula, phases = F)

residuals <- do.call("rbind", map(tab, 2))
tab.robust.main <- do.call("rbind", map(tab, 1))

tab.robust.main <- tab.robust.main %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(

```

```

    term == "(Intercept)" ~ "Intercept",
    term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
    term == "richness_var" ~ "Species richness",
    term == "richness_var:light_treatmentdecreasing" ~
      "Interaction: light conditions and species richness"
  ),
  term = factor(
    term,
    levels = c(
      "Intercept",
      "Light conditions (decreasing)",
      "Species richness",
      "Interaction: light conditions and species richness"
    )
  )
)

tab.robust.main %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(position = position_dodge(width = 0.4)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Forecast method", x = "Used species richness variable", y =
    "Coefficient estimates + 95% CI") +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold")
  )
)

```

8.2 Effect of used detrending method

```

dd_list$densities.int.detrend.complete %>%
  dplyr::filter(!(
    species %in% c(
      "Algae",
      "Debris",
      "Didinium nasutum",
      "Digested algae",
      "Small unidentified",
      "Colpidium vacuoles"
    )
  ),
  ForecastTarget == T) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, density.seg, col = species)) +

```

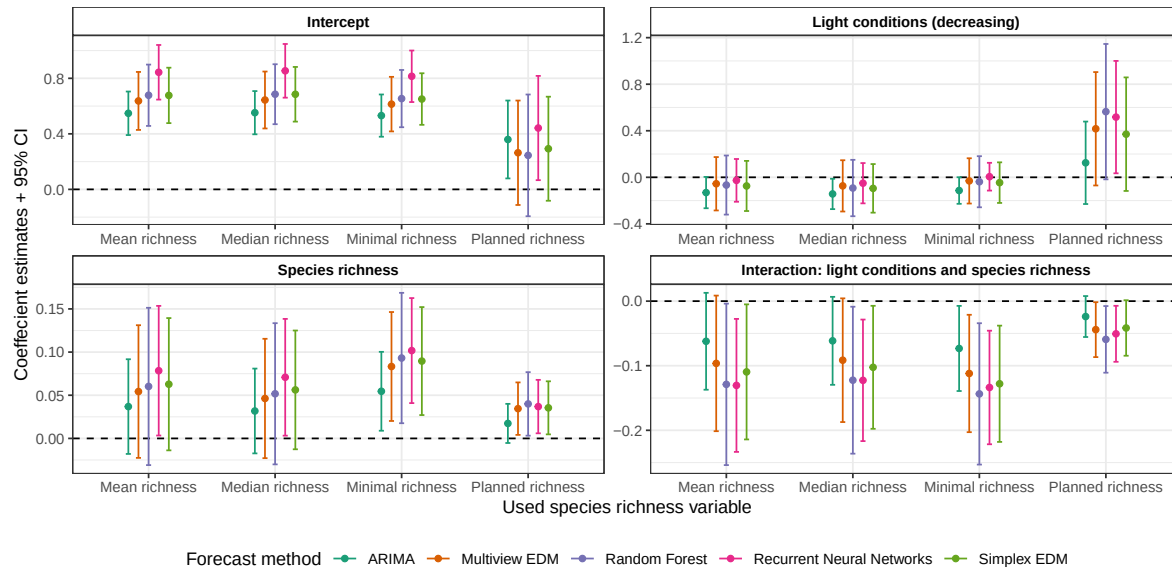


Figure 19: Robustness analysis figure regarding the chosen species richness variable and the forecast method. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

```
geom_line(linewidth = 0.6) +
facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
theme_bw() +
theme(
  legend.position = c(0.75, 0.055),
  panel.grid = element_blank(),
  legend.text = element_text(size = 12),
  axis.text = element_text(size = 11),
  axis.title = element_text(size = 16),
  strip.text = element_text(size = 12),
  # legend.key.size = unit(3, "line"),
  # strip.text = element_text(colour = "white"),
) +
scale_color_manual(values = unname(kelly()[2:16])) +
labs(col = "", y = "Standardized density",
  x = "Sampling day") +
guides(col = guide_legend(ncol = 3, byrow = TRUE))
```

```
detrend.compare.complete <-
full_join(
  seg.complete.forecasts %>%
    dplyr::mutate(detrend = "segmented"),
  complete.forecasts %>%
    dplyr::mutate(detrend = "simple")
)
```

```
detrend.compare.complete2 <-
```



Figure 20: *Time series detrending based on the segmented temporal regression.* The *transformed* biotic density time series that were used as forecast targets, shown in the different bottles (i.e. the panels). The time series are coloured by forecast target. The background colour of the subpanel strips indicates whether the bottles were in a constant (orange) or in a decreasing (blue) light environment.

```

full_join(
  complete.forecasts %>%
    dplyr::select(
      bottle,
      Target,
      richness,
      composition,
      realized_richness,
      light_treatment,
      RMSE,
      forecast.method
    ),
  seg.complete.forecasts %>%
    dplyr::select(
      bottle,
      Target,
      richness,
      composition,
      realized_richness,
      light_treatment,
      RMSE,
      forecast.method
    ) %>%
    dplyr::rename(RMSE.seg = RMSE)
)

```

```

tab <-
  robustnessFunction(rich_vars2,
                    fore.methods,
                    seg.complete.forecasts,
                    formula,
                    phases = F)

tab.detrending <- do.call("rbind", map(tab, 1)) %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(

```

```

    "Intercept",
    "Light conditions (decreasing)",
    "Species richness",
    "Interaction: light conditions and species richness"
  )
)
)

tab.detrending %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(position = position_dodge(width = 0.4)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Forecast method", x = "Used species richness variable", y =
    "Coefficient estimates + 95% CI") +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold")
  )
)

```

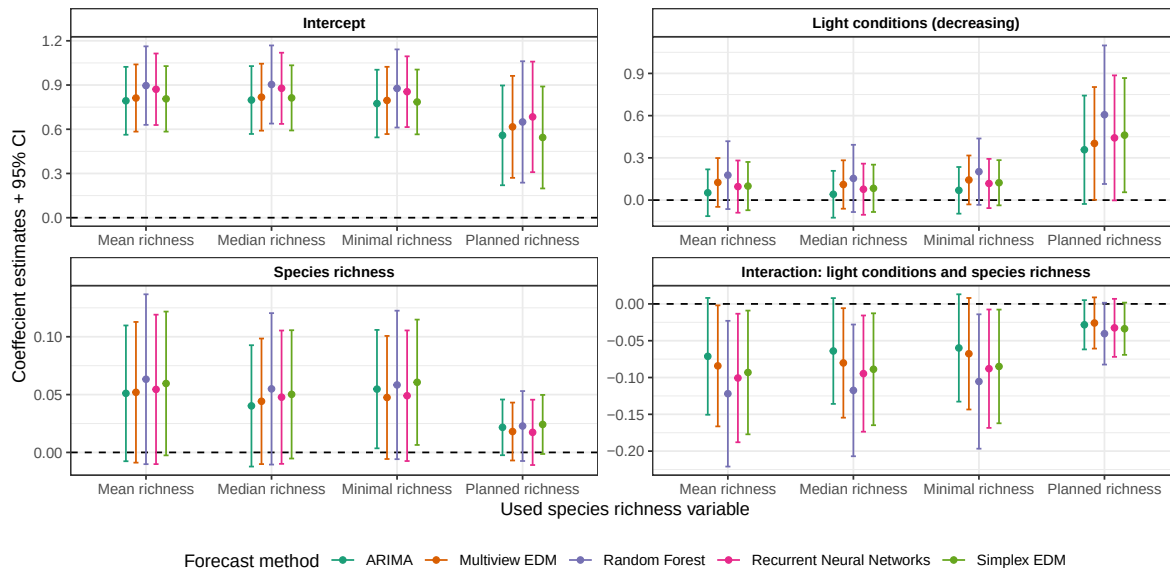


Figure 21: Robustness analysis figure for when the time series were detrended using the **segmented regression** approach. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

8.3 Species Effects on results

```
targets_excl <- unique(complete.forecasts$Target)
targets_excl <-
  targets_excl[!(targets_excl %in% c("Total_biomass", "Mean_oxygen"))]

tab <-
  robustnessFunctionSpecies(rich_vars2,
                           "Simplex EDM",
                           complete.forecasts,
                           formula,
                           targets_excl,
                           phases = F)

tab.speciesEffects <- do.call("rbind", tab) %>%
  mutate(
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(
        "Intercept",
        "Light conditions (decreasing)",
        "Species richness",
        "Interaction: light conditions and species richness"
      )
    )
  )

tab.speciesEffects %>%
  ggplot(aes(richness_var, estimate, col = ExcludedTarget)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.5)
  ) +
  geom_point(position = position_dodge(width = 0.5)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Excluded\\ntaxa", x = "Used species richness variable",
```

```

y = "Coefficient estimates + 95% CI" +
# scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour"))+
scale_color_manual(values = unname(kelly()[2:15])) +
theme_bw() +
theme(
  legend.position = "bottom",
  strip.background = element_rect(fill = "white"),
  strip.text = element_text(color = "black", face = "bold")
)

```

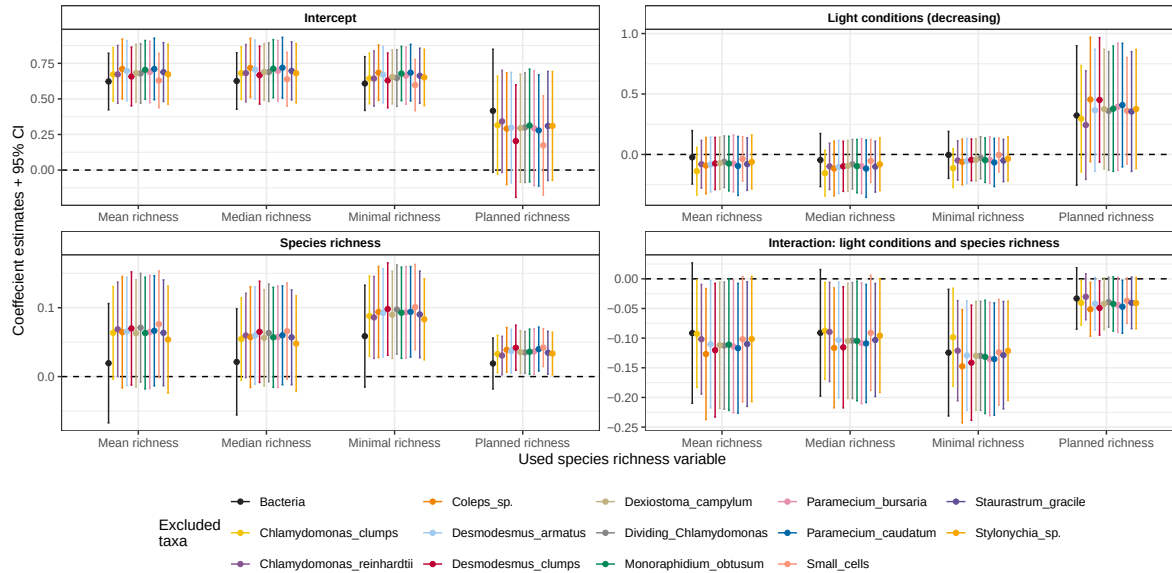


Figure 22: Robustness analysis figure regarding the sensitivity of the results on specific forecasted taxa. The figure depicts the coefficient estimates for regression between the target abundance forecast errors (response) and the covariates "species richness" and "light conditions" (as specified by the panel labels) for when the specified taxa were excluded (i.e. not forecasted). The dots are the estimates and the corresponding error bars are the 95% CI. The colour specifies the taxa that was excluded and the x-axis specifies the used species richness variable. The forecasts are based on the Simplex EDM approach.

8.4 Standardization using historical data only

```

inSample.complete.forecasts %>%
  rename(RMSEhist = RMSE, kHist = k) %>%
  full_join(complete.forecasts) %>%
  dplyr::filter(forecast.method == "Simplex EDM") %>%
  ggplot(aes(RMSE, RMSEhist)) +
  geom_abline(intercept = 0, slope = 1) +
  geom_smooth(method = "lm", col = "darkgrey") +
  geom_point(aes(col = bottle, shape = light_treatment)) +
  facet_wrap(~ Target, scales = "free") +
  theme_bw() +
  labs(x = "RMSE (Time series processed based on entire data)",
       y = "RMSE (Time series processed based on historic data)",
       shape = "Light conditions",
       col = "Bottle")

```

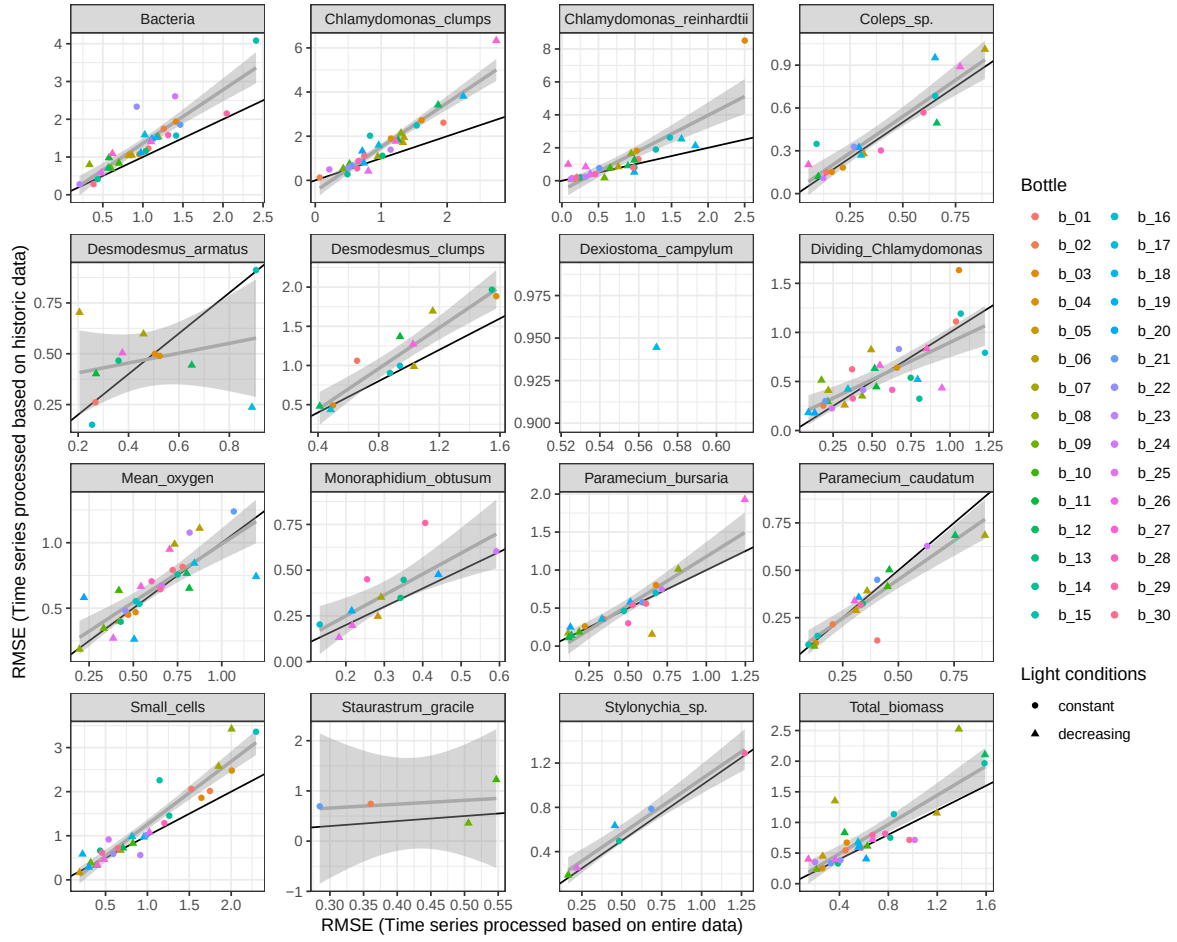


Figure 23: Scatter plot of the RMSE values for when the time series were processed (i.e. detrended, standardized) based on the entire data available against the RMSE values for when the time series were processed using only historic data. RMSE is based on Simplex EDM. Each forecast target is shown in a separate panel, with the colour of the dots indicating the experimental bottle and the shape the light conditions. The black line indicates the identity line, while the grey line (estimate) and the corresponding shaded area (95% CI) are the results of a simple linear regression (fixed effects only).

```

formula <- as.formula(
  paste0(
    "RMSE ~ ",
    "richness_var",
    "*light_treatment + (1|Target) + (1|bottle) + (1|incubator)"
  )
)
rich_vars2 <- list(
  "realized_richness_center",
  "median_richness_center",
  "mean_richness_center",
  "richness"
)
fore.methods <- list("Multiview EDM", "Simplex EDM", "RNN", "ARIMA", "Random Forest")

tab <- robustnessFunction(rich_vars2,
  fore.methods,
  inSample.complete.forecasts,
  formula,
  phases = F)

residuals <- do.call("rbind", map(tab, 2))
tab.robust.main <- do.call("rbind", map(tab, 1))

tab.robust.main <- tab.robust.main %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(
        "Intercept",
        "Light conditions (decreasing)",
        "Species richness",
        "Interaction: light conditions and species richness"
      )
    )
  )
)

```

```

tab.robust.main %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(position = position_dodge(width = 0.4)) +
  facet_wrap(~ term, scales = "free") +
  labs(col = "Forecast method",
       x = "Used species richness variable",
       y = "Coefficient estimates + 95% CI",
       title = "Robustness analyses: standardization using historical data only") +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold"))

```

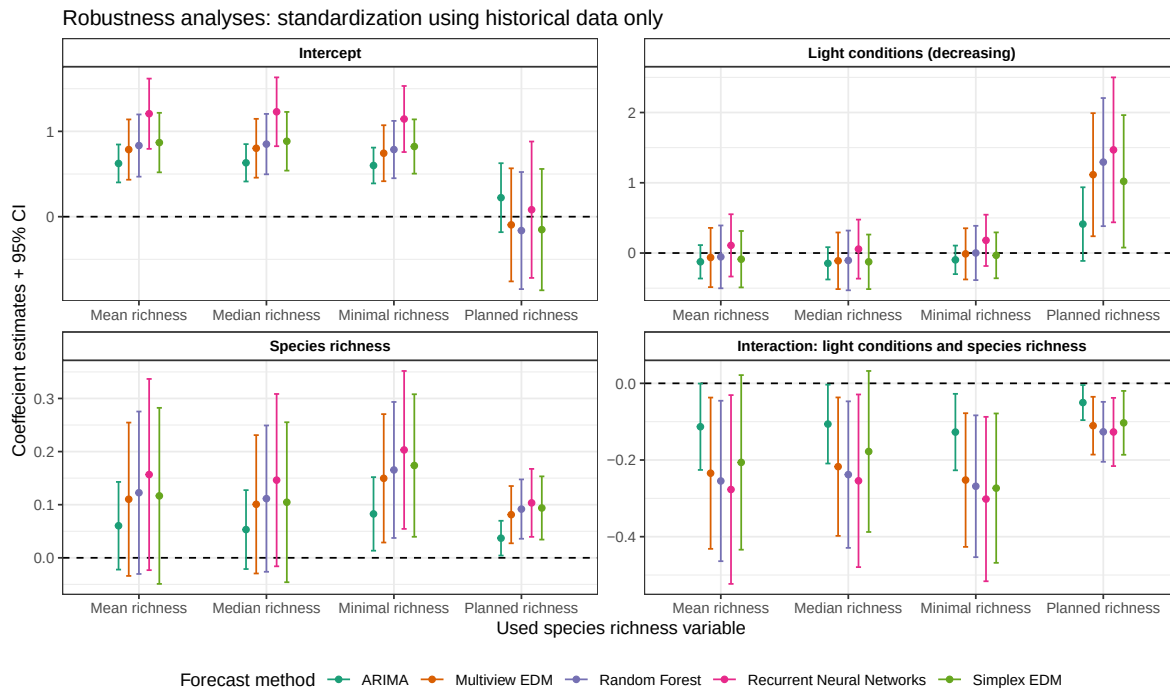


Figure 24: Robustness analysis figure investigating the effects of the standardization using only historical data (instead of the entire time series) on the main results. The relation between biodiversity, light and forecast error is shown for different species richness variables and the different forecast methods. The figure depicts the coefficient estimates for regressions between the taxa abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

8.5 Forecast error evaluation on original and transformed scale

```
RMSEBackT.complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM") %>%
  ggplot(aes(RMSE, RMSE_BackT)) +
  geom_abline(intercept = 0, slope = 1) +
  geom_smooth(method = "lm", col = "darkgrey") +
  geom_point(aes(col = bottle, shape = light_treatment)) +
  facet_wrap(~ Target, scales = "free") +
  theme_bw() +
  labs(x = "RMSE (transformed scale)",
       y = "RMSE (back-transformed forecasts)",
       shape = "Light conditions",
       col = "Bottle")
```

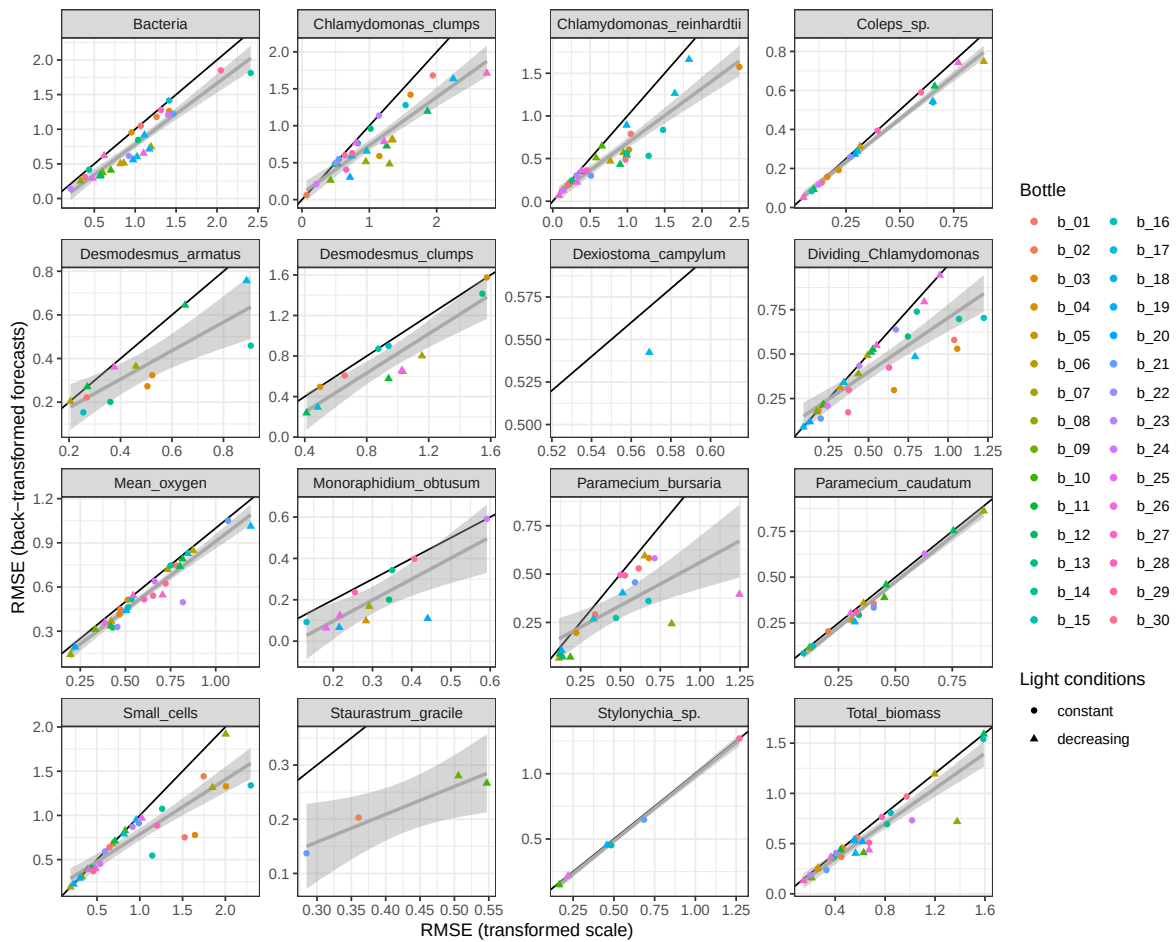


Figure 25: Scatter plot of the RMSE values for the forecasts left on the transformed scale against the RMSE values for the back-transformed forecasts (i.e. the original scale). RMSE is based on Simplex EDM. Each forecast target is shown in a separate panel, with the colour of the dots indicating the experimental bottle and the shape the light conditions. The black line indicates the identity line, while the grey line (estimate) and the corresponding shaded area (95% CI) are the results of a simple linear regression (fixed effects only).

```
formula <- as.formula(paste0("RMSE_BackT ~ ", "richness_var",
                             "*light_treatment + (1 | Target) + (1 | bottle) + (1 | incubator)"))
```

```

rich_vars2 <- list(
  "realized_richness_center",
  "median_richness_center",
  "mean_richness_center",
  "richness"
)
fore.methods <- list("Multiview EDM", "Simplex EDM", "RNN", "ARIMA", "Random Forest")

tab <- robustnessFunction(rich_vars2,
  fore.methods,
  RMSEBackT.complete.forecasts,
  formula,
  phases = F)

residuals <- do.call("rbind", map(tab, 2))
tab.robust.main <- do.call("rbind", map(tab, 1))

tab.robust.main <- tab.robust.main %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "mean_richness_center" ~ "Mean richness",
      richness_var == "median_richness_center" ~
        "Median richness",
      richness_var == "realized_richness_center" ~
        "Minimal richness",
      richness_var == "richness" ~ "Planned richness"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions (decreasing)",
      term == "richness_var" ~ "Species richness",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction:light conditions and species richness"
    ),
    term = factor(
      term,
      levels = c(
        "Intercept",
        "Light conditions (decreasing)",
        "Species richness",
        "Interaction:light conditions and species richness"
      )
    )
  )

tab.robust.main %>%
  ggplot(aes(richness_var, estimate, col = method)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  )

```

```

) +
geom_point(position = position_dodge(width = 0.4)) +
facet_wrap(~ term, scales = "free") +
labs(col = "Forecast method",
      x = "Used species richness variable",
      y = "Coefficient estimates + 95% CI",
      title = "Robustness analyses:error evaluation on original scale") +
scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
theme_bw() +
theme(
  legend.position = "bottom",
  strip.background = element_rect(fill = "white"),
  strip.text = element_text(color = "black", face = "bold")
)

```

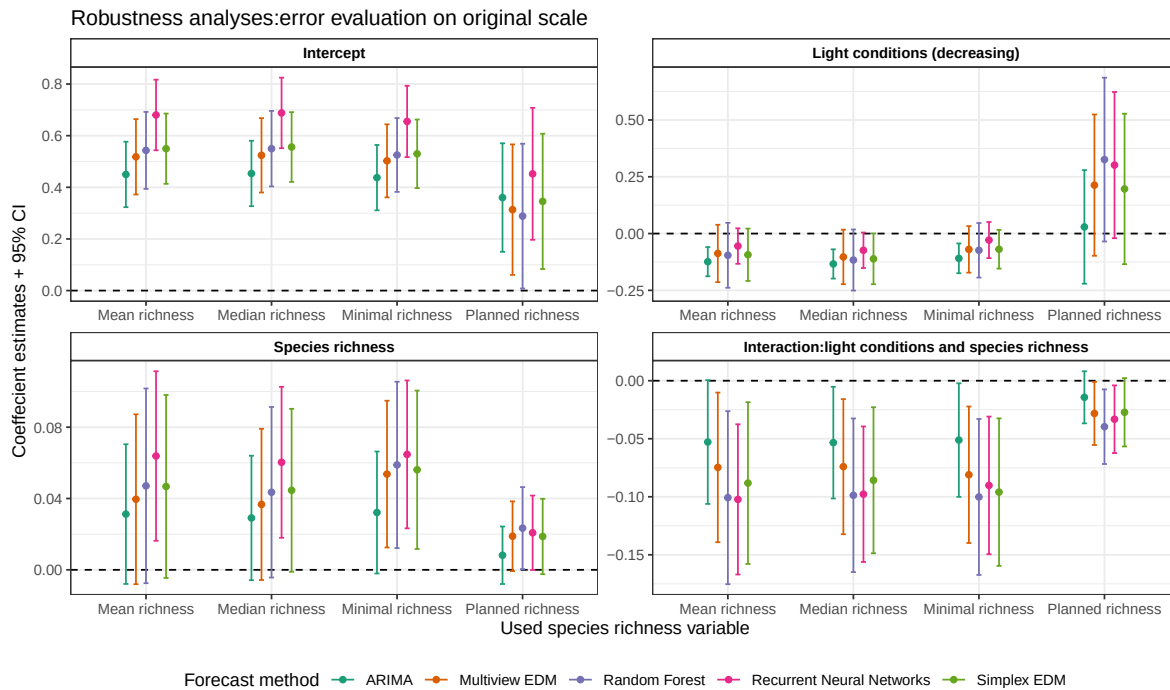


Figure 26: Robustness analysis figure investigating the effects of calculating the RMSE based on back-transformed forecasts (i.e. on the original scale instead of the transformed scale) on the main results. The relation between biodiversity, light and forecast error is shown for different species richness variables and the different forecast methods. The figure depicts the coefficient estimates for regressions between the taxa abundance forecast errors (response) and the covariates "species richness" and "light conditions", as specified by the panel labels. The dots are the estimates and the corresponding error-bars are the 95% CI. The colour specifies the used forecast method and the x-axis specifies the used species richness variable.

9 Further figures mentioned in the discussion

9.1 Insurance effect of diversity and its consequences for forecastability

```
stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))

# Only colour strips in x-direction
strip <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter((
    species %in% c(
      "Coleps sp.",
      "Paramecium bursaria",
      "Colpidium striatum",
      "Loxocephalus sp.",
      "Paramecium caudatum",
      "Stylonychia sp.",
      "Dexiostoma campylum",
      "Euplotes daidaleos"
    )
  )) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
    {
      -1
    }, ")"))),
    x = "Sampling day") +
  guides(col = guide_legend(ncol = 3, byrow = TRUE))

stripColdd <- dd_list$densities %>%
  ungroup() %>%
  dplyr::select(bottle, light_treatment) %>%
  distinct() %>%
  mutate(stripCol = ifelse(light_treatment == "constant", "#f5d682", "#bdc5de"))
```

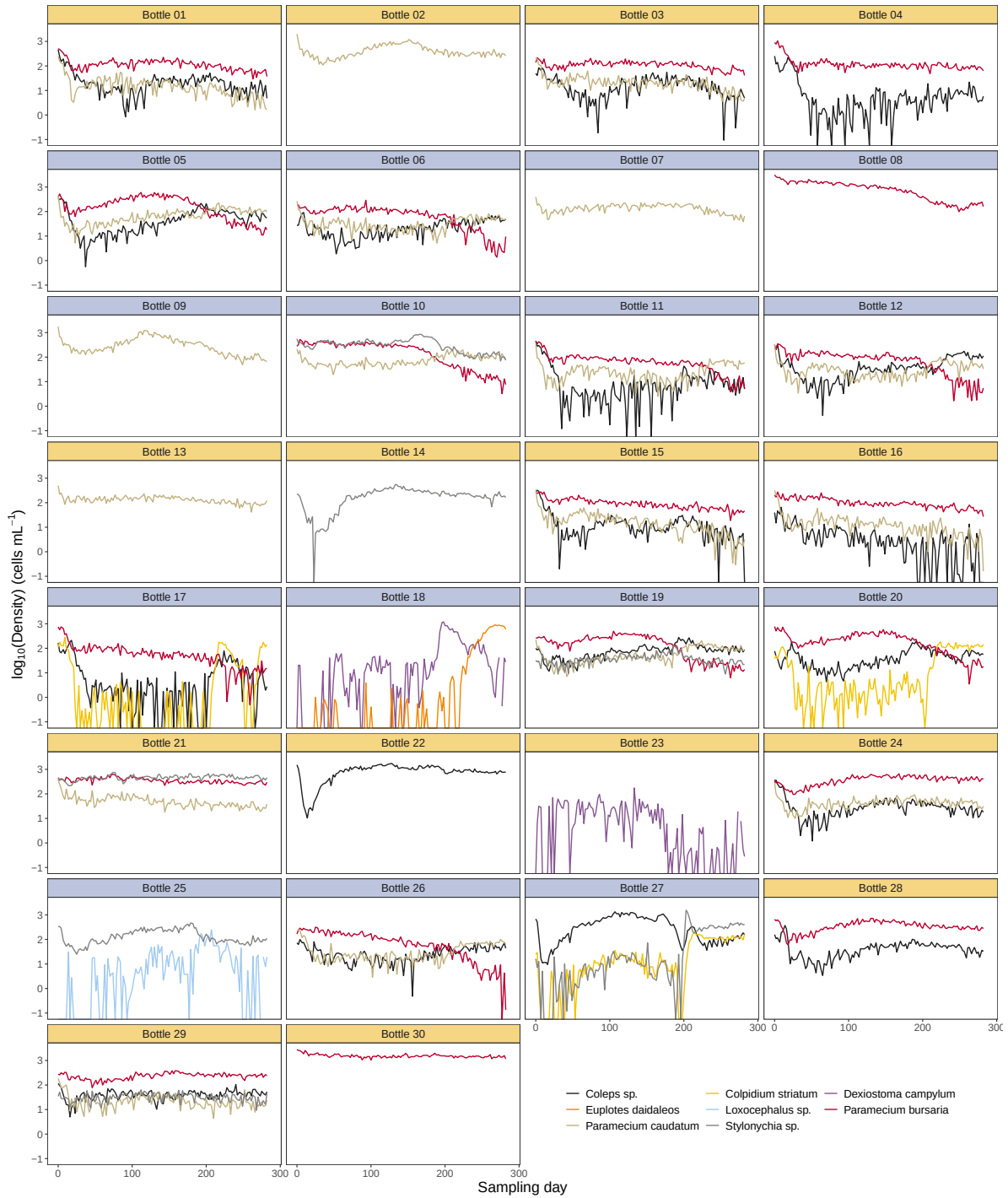


Figure 27: Time series of ciliate densities in the 30 experimental communities. Time series are coloured by ciliate species. The background colour of the subpanel strips indicates whether bottles were maintained in constant light (orange) or in decreasing light (blue).

```

# Only colour strips in x-direction
strip <-
  strip_themed(background_x = elem_list_rect(fill = stripColdd$stripCol))

dd_list$densities %>%
  dplyr::filter((
    species %in% c(
      "Chlamydomonas clumps",
      "Chlamydomonas reinhardtii",
      "Cosmarium botrytis",
      "Desmodesmus armatus",
      "Desmodesmus clumps",
      "Dividing Chlamydomonas",
      "Small cells",
      "Staurostrum gracile",
      "Monoraphidium obtusum",
      "Staurostrum polytrichum"
    )
  )) %>%
  mutate(bottle = gsub("b_", "Bottle ", bottle)) %>%
  ggplot(aes(day, log10(density), col = species)) +
  geom_line(linewidth = 0.6) +
  facet_wrap2( ~ bottle, ncol = 4, strip = strip) +
  theme_bw() +
  theme(
    legend.position = c(0.75, 0.055),
    panel.grid = element_blank(),
    legend.text = element_text(size = 12),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 16),
    strip.text = element_text(size = 12),
  ) +
  scale_color_manual(values = unname(kelly()[2:16])) +
  labs(col = "", y = expression(paste("log"[10], "(Density) (cells ", mL ^
    {
      -1
    }, ")")),
    x = "Sampling day") +
  guides(col = guide_legend(ncol = 3, byrow = TRUE))

```

9.2 Coefficient of variation of aggregate ecosystem properties

BIOMASS CV VERSUS EXPERIMENTAL CONDITIONS

```

complete.forecasts.pe.cv2.biomass <- complete.forecasts %>%
  ungroup() %>%
  dplyr::filter(Target == "Total_biomass", forecast.method == "Simplex EDM") %>%
  dplyr::select(
    bottle,
    median_richness,
    richness,
  )

```

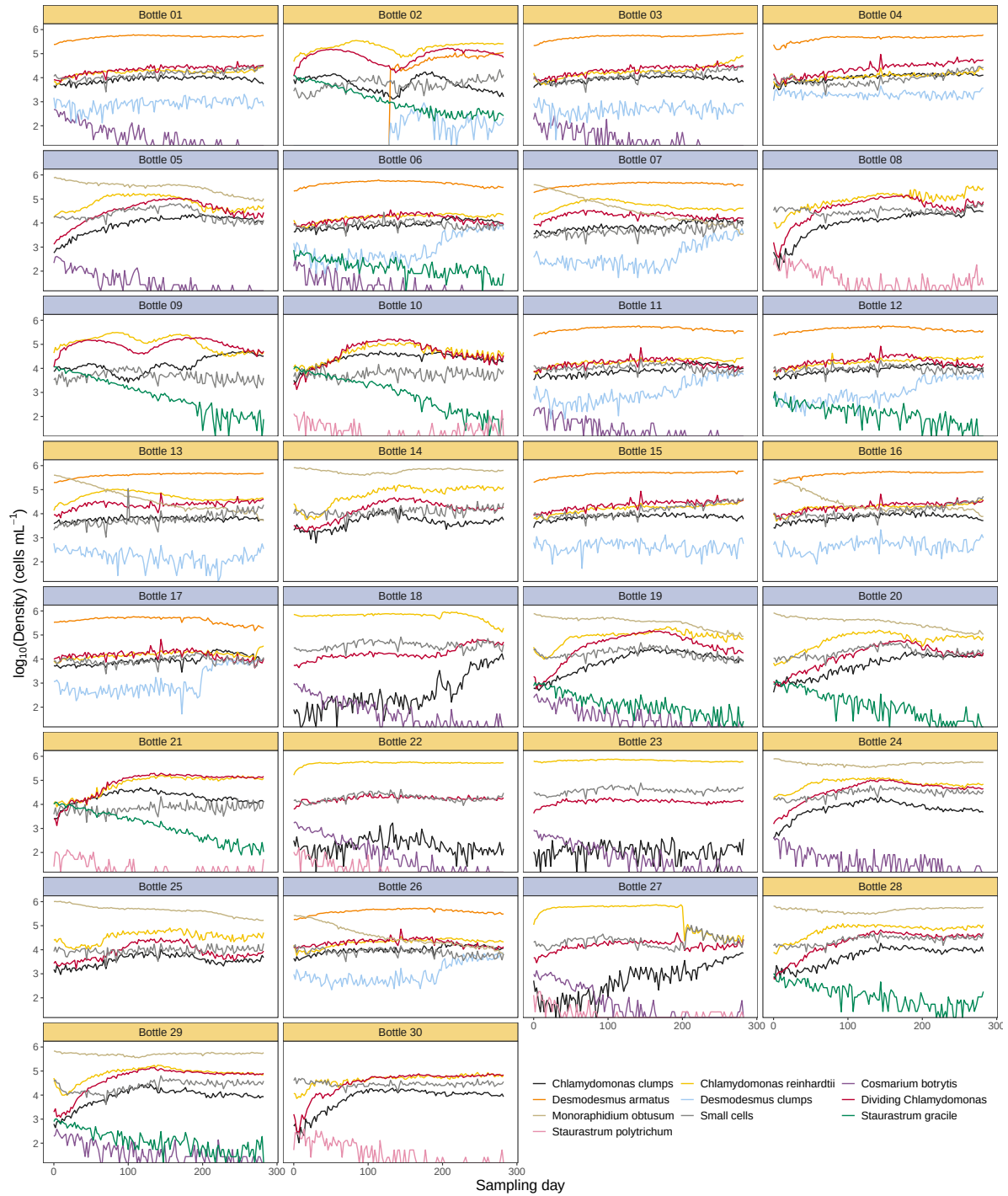


Figure 28: Time series of algae densities in the 30 experimental communities. Time series are coloured by algal species. The background colour of the subpanel strips indicates whether bottles were maintained in constant light (orange) or in decreasing light (blue).

```

    light_treatment,
    composition,
    cv.res,
    median_richness_center,
    incubator
  ) %>%
  dplyr::mutate(log10.cv.res = log10(cv.res))

m.cv.bio <-
  lmer(
    log10.cv.res ~ median_richness_center * light_treatment +
      (1 | composition) + (1 | incubator),
    complete.forecasts.pe.cv2.biomass
  )

newdat.m.cv.bio <- with(
  complete.forecasts.pe.cv2.biomass,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment)
  )
)

newdat.m.cv.bio <-
  conf_interval(m.cv.bio,
    newdat.m.cv.bio,
    "log10.cv.res",
    "log10(Coefficient of variation)")

### OXYGEN CV VERSUS EXPERIMENTAL CONDITIONS

complete.forecasts.pe.cv2.oxygen <- complete.forecasts %>%
  ungroup() %>%
  dplyr::filter(Target == "Mean_oxygen", forecast.method == "Simplex EDM") %>%
  dplyr::select(
    bottle,
    median_richness,
    richness,
    light_treatment,
    composition,
    cv.res,
    median_richness_center,
    incubator
  ) %>%
  dplyr::mutate(log10.cv.res = log10(cv.res))

```

```

m.cv.oxy <-
  lmer(
    log10.cv.res ~ median_richness_center * light_treatment +
      (1 | composition) + (1 | incubator),
    complete.forecasts.pe.cv2.oxygen
  )

newdat.m.cv.oxy <- with(
  complete.forecasts.pe.cv2.oxygen,
  expand.grid(
    median_richness_center = seq(
      min(median_richness_center),
      max(median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(light_treatment)
  )
)

newdat.m.cv.oxy <-
  conf_interval(m.cv.oxy,
    newdat.m.cv.oxy,
    "log10.cv.res",
    "log10(Coefficient of variation)")

#### PLOT

complete.forecasts.pe.cv2.biomass %>%
  ggplot(aes(median_richness_center, log10.cv.res, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.cv.bio,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,
    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.cv.bio,
    size = 1.3,
    aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Centered median realized species richness",
    y = "log10(Coefficient of variation of total biomass)",
    col = "Light conditions") +
  scale_color_manual(values = c("#C4961A", "#293352"),
    aesthetics = c("color", "fill")) +

```

```

theme_bw() +

complete.forecasts.pe.cv2.oxygen %>%
ggplot(aes(median_richness_center, log10.cv.res, col = light_treatment)) +
geom_ribbon(
  data = newdat.m.cv.oxy,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.m.cv.oxy,
  size = 1.3,
  aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
  y = "log10(Coefficient of variation of mean oxygen)",
  col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill")) +
theme_bw() +

plot_layout(guides = "collect") + plot_annotation(tag_levels = "a")

```

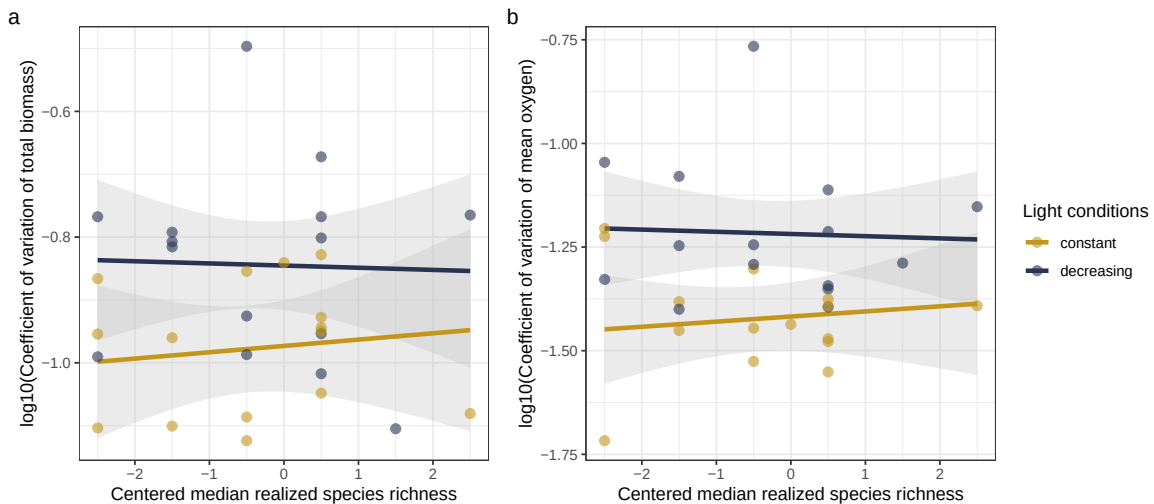


Figure 29: Coefficient of variation of (a) total community biomass, and (b) oxygen concentration. Lines display the fit of the corresponding linear mixed model, shaded areas denote the 95% confidence intervals. Incubator and composition were included as random effects in the model.

```
# compare average cv of species biomass time series with cv of total biomass
# calculate average cv.res of species biomass (of species used as forecast targets)
```

```
density.complete.pe.cv <- dd_list$densities.int.detrend.complete %>%
  group_by(
    species,
    bottle,
    composition,
    ForecastTarget,
    richness,
    incubator,
    method,
    light_treatment,
    realized_richness,
    mean_richness,
    median_richness
  ) %>%
  summarize(
    density.pe.5 = PE(
      x = density,
      weighted = T,
      word_length = 5,
      tau = 1,
      tie_method = "noise",
      noise_amount = 10
    ),
    biomass.pe.5 = ifelse(
      all(is.na(biomass)),
      NA,
      PE(
        x = biomass,
        weighted = T,
        word_length = 5,
        tau = 1,
        tie_method = "noise",
        noise_amount = 10
      )
    ),
    biomass.cv.res = sd(biomass.detrended.unstandardized, na.rm =
      T) / mean(biomass.original, na.rm = T),
    density.cv.res = sd(density.detrended.unstandardized, na.rm =
      T) / mean(density.original, na.rm = T),
    biomass.acf_lag5 = stats::acf(
      biomass,
      lag.max = 5,
      plot = F,
      na.action = na.pass
    )$acf[6],
    density.acf_lag5 = stats::acf(
      density,
      lag.max = 5,
      plot = F,
      na.action = na.pass
    )$acf[6]
```

```

    )$acf[6]
  )

mean.cv.biomass <- density.complete.pe.cv %>%
  dplyr::filter(ForecastTarget == T) %>%
  group_by(bottle,
            median_richness,
            richness,
            light_treatment,
            composition) %>%
  dplyr::summarize(cv.res = mean(biomass.cv.res, na.rm = T)) %>%
  dplyr::mutate(level = "population")

temp.data <- complete.forecasts.pe.cv2.biomass
temp.data$level <- "community"

mean.cv.biomass <- full_join(mean.cv.biomass, temp.data)

mean.cv.biomass %>%
  ggplot(aes(
    x = level,
    y = cv.res,
    colour = as.factor(median_richness)
  )) +
  geom_line(aes(group = bottle)) +
  scale_color_manual(values = unname(okabe())) +
  theme_bw() +
  labs(x = "Level of aggregation", y = "Coefficient of variation",
       col = "Median richness") +
  scale_x_discrete(expand = c(0.2, 0))

```

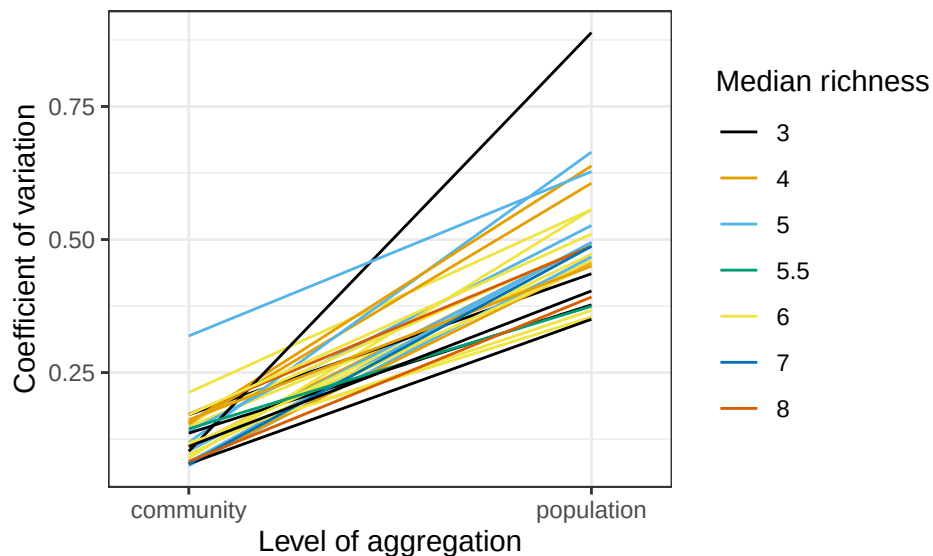


Figure 30: Coefficient of variation (CV) of community and population biomass. For each bottle, we calculated the CV of the time series of total community biomass, and the average CV of the biomass time series of all species in a bottle. Lines connect the two values of a given bottle, and are coloured by median richness.

10 Other diversity indices

```
vars <- c(
  "Coleps sp.",
  "Paramecium bursaria",
  "Paramecium caudatum",
  "Stylonychia sp.",
  "Colpidium striatum",
  "Euplotes daidaleos",
  "Dexiostoma campylum",
  "Loxocephalus sp.",
  "Chlamydomonas reinhardtii",
  "Cosmarium botrytis",
  "Desmodesmus armatus",
  "Staurastrum gracile",
  "Monoraphidium obtusum",
  "Staurastrum polytrichum",
  "Bacteria"
)

Shannon <- dd_list$densities.int.detrend.complete %>%
  dplyr::filter(
    species %in% vars,
    (method == "FlowCam" &
      density.original >= 100) |
    (method == "Video" &
      density.original >= 10) | (method == "Flowcytometer")
  ) %>%
  mutate(
    FunctionalGroup = case_when(
      species %in% c("Paramecium bursaria", "Euplotes daidaleos") ~ "Mixotrophs",
      species %in% c(
        "Cosmarium botrytis",
        "Desmodesmus armatus",
        "Staurastrum gracile",
        "Monoraphidium obtusum",
        "Staurastrum polytrichum"
      ) ~ "InedibleAlgae",
      species %in% c(
        "Coleps sp.", "Paramecium caudatum", "Stylonychia sp."
      ) ~ "Omnivores",
      species %in% c(
        "Colpidium striatum", "Dexiostoma campylum", "Loxocephalus sp."
      ) ~ "Bacterivores",
      species %in% c("Chlamydomonas reinhardtii") ~ "EdibleAlgae",
      species %in% c("Bacteria") ~ "Bacteria"
    ),
    TrophicLevel = ifelse(
      FunctionalGroup %in% c("Mixotrophs", "Omnivores", "Bacterivores"),
      "Consumers",
      "Producers"
    )
  ) %>%
```

```

group_by(bottle, day, light_treatment) %>%
mutate(
  day_richness = n(),
  NumberFunctionalGroups = length(unique(FunctionalGroup)),
  NumberTrophicLevels = length(unique(TrophicLevel)),
  speciesProp = density.original / sum(density.original),
  speciesPropBio = biomass.original / sum(biomass.original)
) %>%
group_by(bottle, day, light_treatment) %>%
summarise(
  NumberFunctionalGroups = unique(NumberFunctionalGroups),
  NumberTrophicLevels = unique(NumberTrophicLevels),
  ShannonDiv = -sum(speciesProp * log(speciesProp)),
  ShannonDivBio = -sum(speciesPropBio * log(speciesPropBio)),
  day_richness = unique(day_richness),
  ShannonEven = ShannonDiv / log(day_richness),
  ShannonEvenBio = ShannonDivBio / log(day_richness)
) %>%
group_by(bottle, light_treatment) %>%
summarise(
  Richness = median(day_richness),
  NumberFunctionalGroups = median(NumberFunctionalGroups),
  NumberTrophicLevels = as.factor(median(NumberTrophicLevels)),
  ShannonAbundance = median(ShannonDiv),
  EvennessAbundance = median(ShannonEven),
  ShannonBiomass = median(ShannonDivBio),
  EvennessBiomass = median(ShannonEvenBio)
)

# group_by(bottle) %>%
# dplyr::mutate(mean_richness = mean(day_richness, na.rm = T),
#               median_richness = median(day_richness, na.rm = T))

```

```

Shannon %>%
rename(
  "Median richness" = Richness,
  "Abundance evenness" = EvennessAbundance,
  "Biomass evenness" = EvennessBiomass,
  "Functional groups" = NumberFunctionalGroups
) %>%
ggpairs(
  mapping = aes(col = light_treatment, fill = light_treatment),
  columns = c(
    "Median richness",
    "Abundance evenness",
    "Biomass evenness",
    "Functional groups"
  ),
  lower = list(continuous = lowerfun, alpha = 0.5),
  diag = list(continuous = my_hist),
  legend = 1,
) +
scale_color_manual(values = c("#C4961A", "#293352"),
  aesthetics = c("color", "fill")) +

```

```
labs(col = "Light conditions", fill = "Light conditions") +
theme(legend.position = "bottom")
```

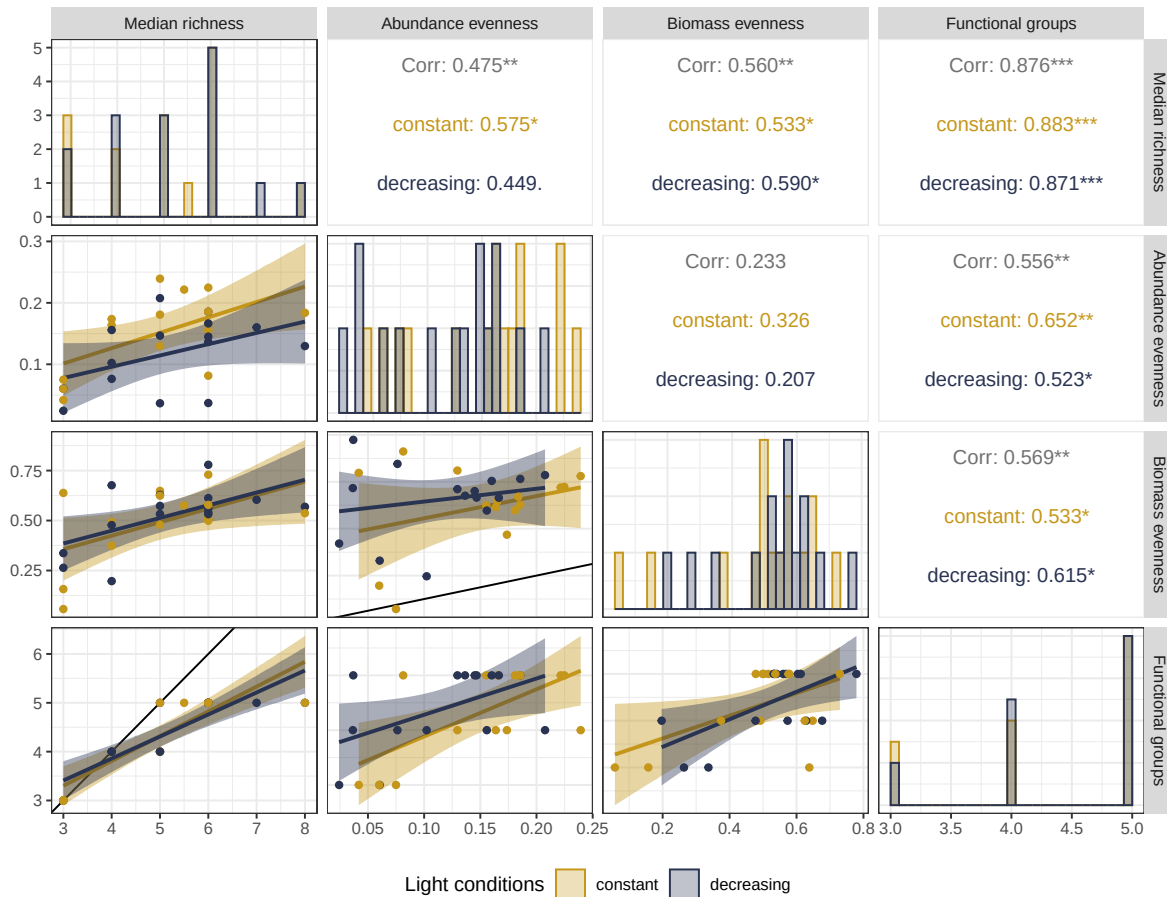


Figure 31: Correlations and distributions of different diversity indices (i.e. the median species richness, the median Shannon evenness of the species abundances and biomasses, and the median number of functional groups), colour-coded for the light conditions. The panels in the diagonal show the bar-plots of the richness variables. The panels below the diagonal show the pairwise scatter-plots as indicated by the facet labels (the black line is the identity line, the coloured lines and the corresponding shaded areas show the fit and the 95% CI of a linear regression). The panels above the diagonal show the corresponding correlation estimates.

```
Shannon <- Shannon %>%
  ungroup() %>%
  mutate(
    ShannonAbundance_center = ShannonAbundance - mean(ShannonAbundance),
    EvennessAbundance_center = EvennessAbundance - mean(EvennessAbundance),
    ShannonBiomass_center = ShannonBiomass - mean(ShannonBiomass),
    EvennessBiomass_center = EvennessBiomass - mean(EvennessBiomass),
    NumberFunctionalGroups_center = NumberFunctionalGroups - median(NumberFunctionalGroups)
  )

complete.forecasts.shannon <- full_join(complete.forecasts,
  Shannon %>% dplyr::select(-Richness))
```

```

formula <-
  as.formula(
    paste0(
      "RMSE ~ ",
      "richness_var",
      "*light_treatment + (1|Target) + (1|bottle) + (1|incubator)"
    )
  )

community_vars <- list(
  "median_richness_center",
  "EvennessAbundance_center",
  "EvennessBiomass_center",
  "NumberFunctionalGroups_center"
)

fore.methods <- list("Simplex EDM")

tab.shannon <-
  robustnessFunction(community_vars,
    fore.methods,
    complete.forecasts.shannon,
    formula,
    phases = F)

residuals.shannon <- do.call("rbind", map(tab.shannon, 2))
tab.shannon <- do.call("rbind", map(tab.shannon, 1))

tab.shannon <- tab.shannon %>%
  mutate(
    method = ifelse(method == "RNN", "Recurrent Neural Networks", method),
    richness_var = case_when(
      richness_var == "median_richness_center" ~ "Median richness",
      richness_var == "EvennessAbundance_center" ~
        "Abundance evenness",
      richness_var == "EvennessBiomass_center" ~
        "Biomass evenness",
      richness_var == "NumberFunctionalGroups_center" ~
        "Functional groups"
    ),
    term = case_when(
      term == "(Intercept)" ~ "Intercept",
      term == "light_treatmentdecreasing" ~ "Light conditions\n(decreasing)",
      term == "richness_var" ~ "Diversity\nindex",
      term == "richness_var:light_treatmentdecreasing" ~
        "Interaction: light conditions \nand diversity index"
    ),
    term = factor(
      term,
      levels = c(
        "Intercept",
        "Light conditions\n(decreasing)",
        "Diversity\nindex",

```

```

    "Interaction: light conditions \nand diversity index"
  )
)
)

tab.shannon %>%
  ggplot(aes(term, estimate, col = richness_var)) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_errorbar(
    aes(ymin = conf.low, ymax = conf.high),
    width = 0.2,
    position = position_dodge(width = 0.4)
  ) +
  geom_point(
    aes(fill = richness_var, shape = richness_var),
    position = position_dodge(width = 0.4),
    col = "black"
  ) +
  labs(
    x = "Model coefficient",
    y = "Coefficient estimates + 95% CI",
    col = "Diversity index\nused in model",
    fill = "Diversity index\nused in model",
    shape = "Diversity index\nused in model"
  ) +
  scale_color_brewer(palette = "Dark2", aesthetics = c("fill", "colour")) +
  scale_shape_manual(values = 21:24) +
  standard_theme2() +
  theme(
    legend.position = "right",
    strip.background = element_rect(fill = "white"),
    strip.text = element_text(color = "black", face = "bold")
  )
)

```

11 Versions of R and of the R-packages used

R version: R version 4.3.2 (2023-10-31).

Base packages: grid, parallel, stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: r2glmm 0.1.2, effectsize 0.8.8, lavaan.survey 1.1.3, lavaan 0.6-18, survey 4.4-2, survival 3.7-0, LEEF.measurement.bemovi 1.0.0, bemovi 1.0, flowViz 1.66.0, flowCore 2.8.0, keras 2.15.0, reticulate 1.37.0, slider 0.3.1, broom.mixed 0.2.9.5, GGally 2.2.1, ggh4x 0.2.8, lmerTest 3.1-3, lme4 1.1-35.5, Matrix 1.6-5, caret 6.0-94, lattice 0.22-6, here 1.0.1, pals 1.9, kableExtra 1.4.0, rEDM 1.15.4, doParallel 1.0.17, iterators 1.0.14, foreach 1.5.2, forecast 8.23.0, pracma 2.4.4, data.table 1.15.4, patchwork 1.2.0, randomForest 4.7-1.1, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.5.1, tidyverse 2.0.0, ggstance 0.3.7, emmeans 1.10.2, knitr 1.48.

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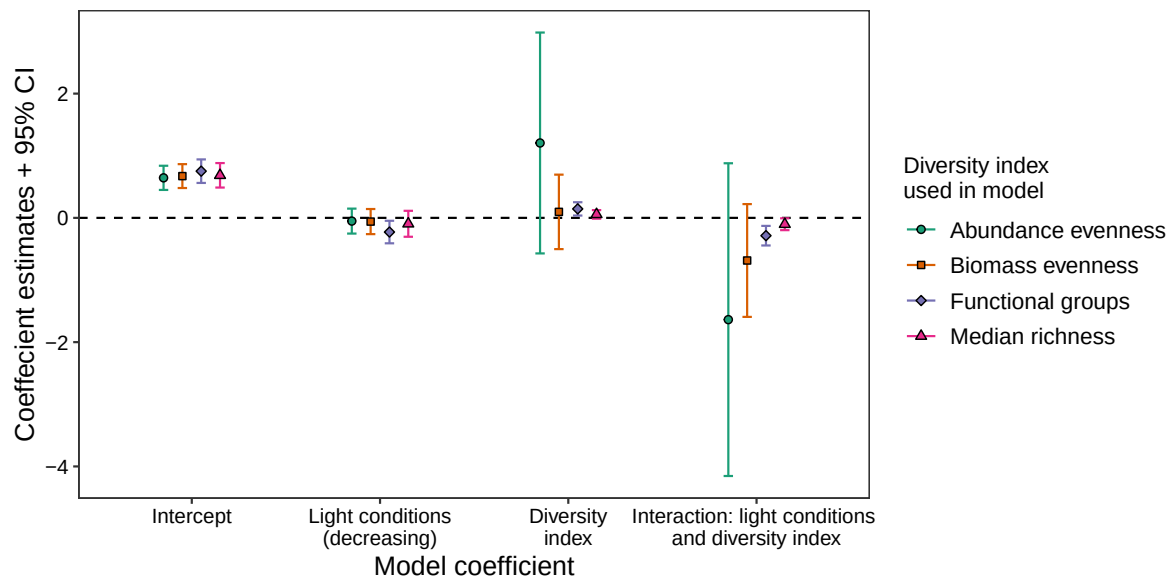


Figure 32: Results of forecast error analyses using different diversity indices (i.e. the median species richness, the median Shannon evenness of the species abundances and biomasses, and the median number of functional groups). The figure depicts the coefficient estimates for regressions between the taxa abundance forecast errors (response) and the covariates "diversity index" and "light conditions". The dots are the estimates and the corresponding error-bars are the 95% CI. The colour and shape of the dots specify the used diversity index and the x-axis specifies model coefficient.